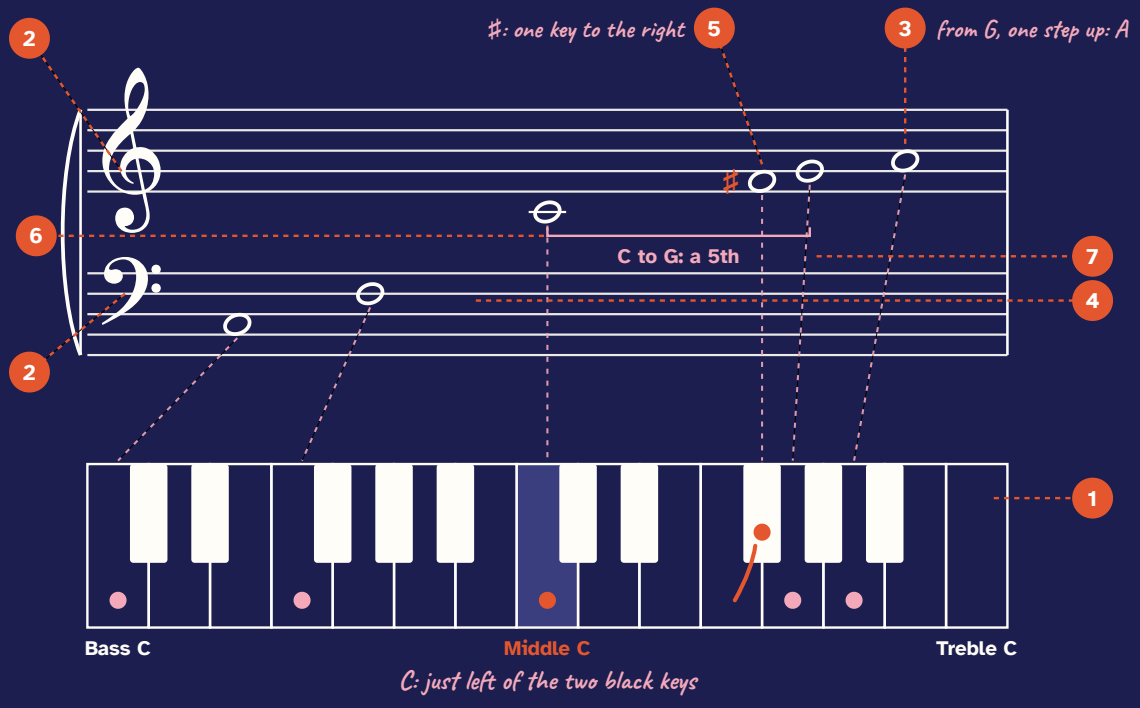
PART I · WHICH NOTE

Pitch

High or low? That is the first question every note answers. Start with the keys under your fingers, find where they live on the staff, see how sharps and flats move them, then measure how far apart two notes are. Everything in this part is in this one picture.



1 **The keyboard** p. 22
two and three black keys; C is left of the two

2 **The staff and the clefs** p. 26
the treble clef names G, the bass clef F

3 **Reading the notes** p. 29
start from a clef note, then step or skip

4 **The staff and the keyboard** p. 32
every line and space is one white key

5 **Sharps, flats and naturals** p. 35
a sharp moves one key right, a flat one key left

6 **Counting intervals** p. 38
count the notes, both ends: C to G is a 5th

7 **Major, minor and perfect** p. 41
count the keys between: 6 make a perfect 5th

★ **Tips: when a note reads wrong** p. 44

PART I · PITCH · 1

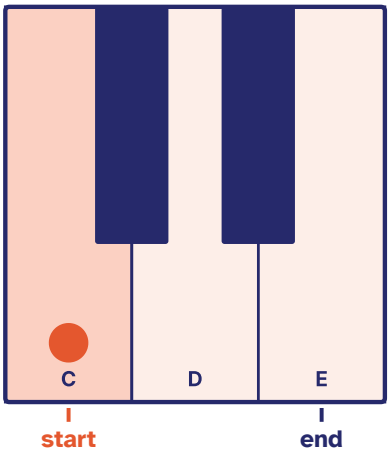
The keyboard

Left is **low**, right is **high**. The black keys come in groups of **two** and **three**. Each group has a first white key, just to its left: **C** for the two, **F** for the three. Find the group, find its first key, then walk on.

THE SECRET

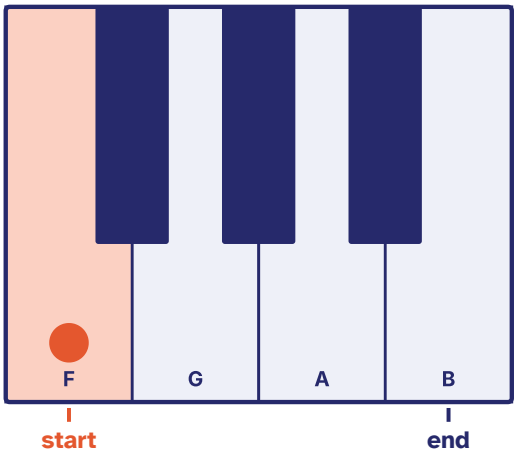
Two black keys: start on C. Three black keys: start on F. Then walk on.

2 black keys: C D E



on the piano E and F touch

3 black keys: F G A B



Pulled apart: C starts the two, F starts the three · E and B end them

THE RULE

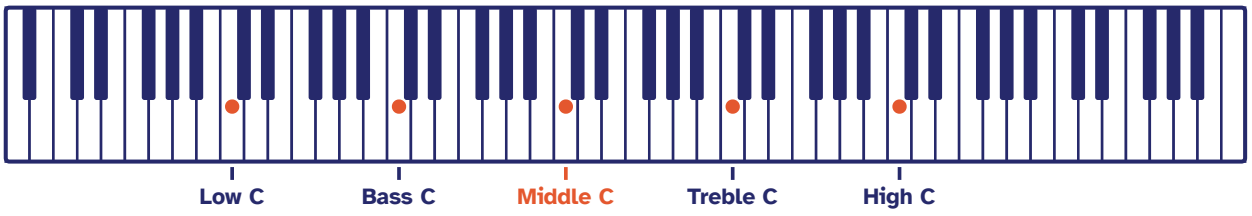
Find the group, then its first key. Looking for D or E? Find two black keys and start on C. Looking for G, A or B? Find three black keys and start on F. You never have to count from far away.

Seven letters, two families

The white keys use seven letters, **A B C D E F G**, then start again, like the days of the week. They live in two families: **C D E** round the two black keys, **F G A B** round the three.

The five Cs have names

The C nearest the middle is **Middle C**. The Cs either side are **Bass C** and **Treble C**; further out, **Low C** and **High C**. From one C to the next is an **octave**: C D E F G A B C, eight notes, both Cs included.

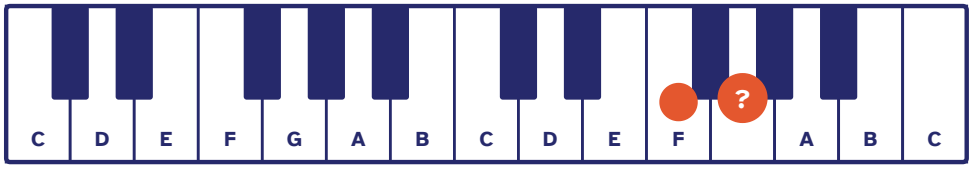


The whole piano, all 88 keys: the five named Cs sit in the middle

PART I · PITCH · 1 · CONTINUED

EXAMPLE 1

Find G

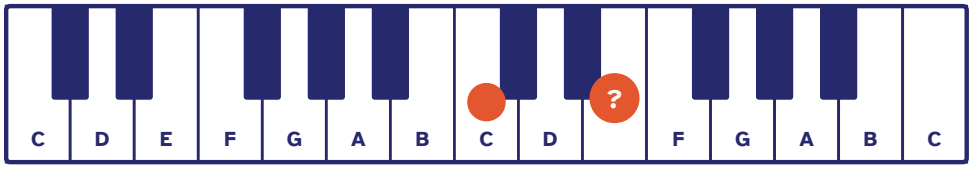


1. **Which family?** G is in F G A B: look for **three** black keys.
2. **Find its first key:** F, just left of the three (the red dot).
3. **Walk on one key.** That is the **?**.

The **?** is **G**: the key after F. No need to start from C and count five.

EXAMPLE 2

Find E



1. **Which family?** E is in C D E: look for **two** black keys.
2. **Find its first key:** C, just left of the two (the red dot).
3. **Walk on:** C, D … the third key is the **?**.

The **?** is **E**: the last of the C D E family, just right of the two black keys.

EXAMPLE 3

Find B



1. **Which family?** B is in F G A B: three black keys.
2. **Start on F** (the red dot).
3. **Walk on:** F, G, A … the last key of the family is the **?**.

The **?** is **B**: the last of F G A B, just right of the three black keys.

GAME

Which family?

1	2	3	4	5	6
D	A	F	E	B	G

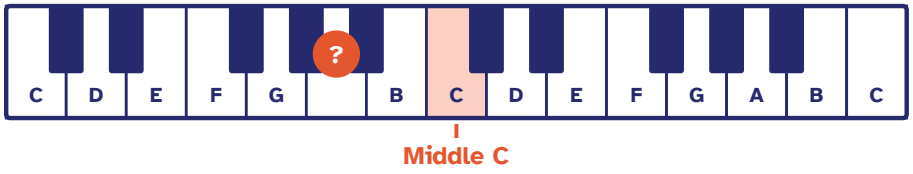
Does each white key live with the **two** black keys (start on C) or the **three** (start on F)?

Answer: 1 two · 2 three · 3 three (the first) · 4 two (the last) · 5 three (the last) · 6 three.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 4

Find the A below Middle C

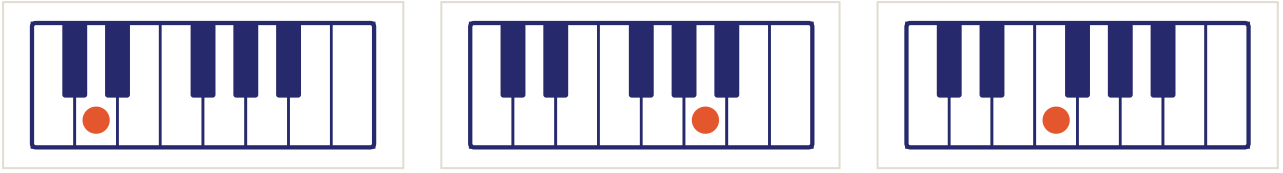


1. **Which family?** A is in F G A B.
2. **From Middle C, look left** for the nearest three black keys, and start on their F.
3. **Walk on:** F, G … the third key is the **?**.

The **?** is **A**: F, G, A. Find the family first, and every key is at most three steps away.

GAME

Name the key



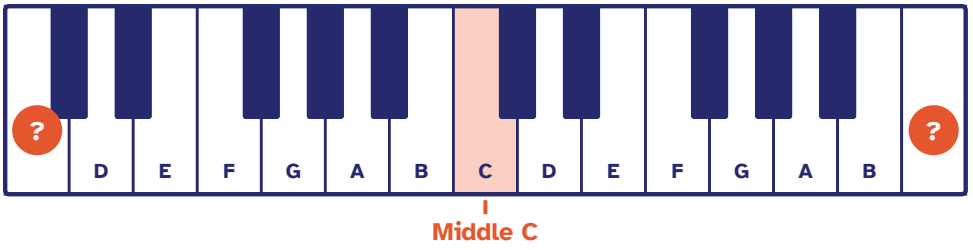
1 2 3

Find the group, then its first key, then walk on. Name each red key.

Answer: 1 D (C, then D) · 2 A (F, G, A) · 3 F (the first of the three).

EXAMPLE 5

Find Bass C and Treble C



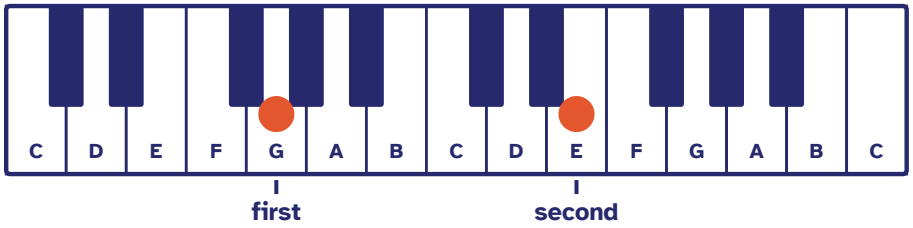
1. **Start on Middle C.**
2. **Walk left** to the next pair of black keys. The C just left of it is one **?**.
3. **Walk right** the same way, to the next pair: the other **?**.

Left **?**: **Bass C**. Right **?**: **Treble C**. Each is one octave, eight notes, from Middle C.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 6

Higher or lower?



1. **Find the two red keys.**
2. **Which one is further right?**
3. **Right means higher:** play both and listen.

The second note is **higher**: it is to the right. Left is always lower, right always higher.

TRY IT · 10 SECONDS

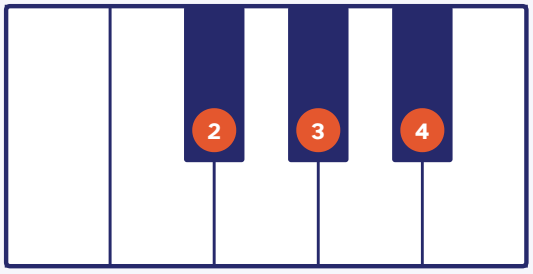


Play every C, then every F.

Bottom to top, saying the names: Low C, Bass C, Middle C, Treble C, High C. Then every F: left of the three.

YOU KNOW THIS

Hot Cross Buns, on three black keys



Right hand, fingers **2 3 4** on the three black keys. Play from the top: **4 3 2**, 4 3 2, then 2 2 2 2, 3 3 3 3, 4 3 2. A whole tune, and not one white key.

Try it. Find another group of three, higher up, and play it again there.

THINK OF IT The black keys are bumps you can feel in the dark. Close your eyes and slide your hand along: two bumps, and the key to their left is C; three bumps, and the key to their left is F.

TRAP

- × Counting every key from C: find the nearest group and start from its C or F.
- × Mixing up the families: C is left of the **two** black keys, F left of the **three**.

SEE ALSO

The staff and the clefs p. 26 · The staff and the keyboard p. 32

PRACTISE IT

Workbook Pitch 1 · Online: Library › Keyboard

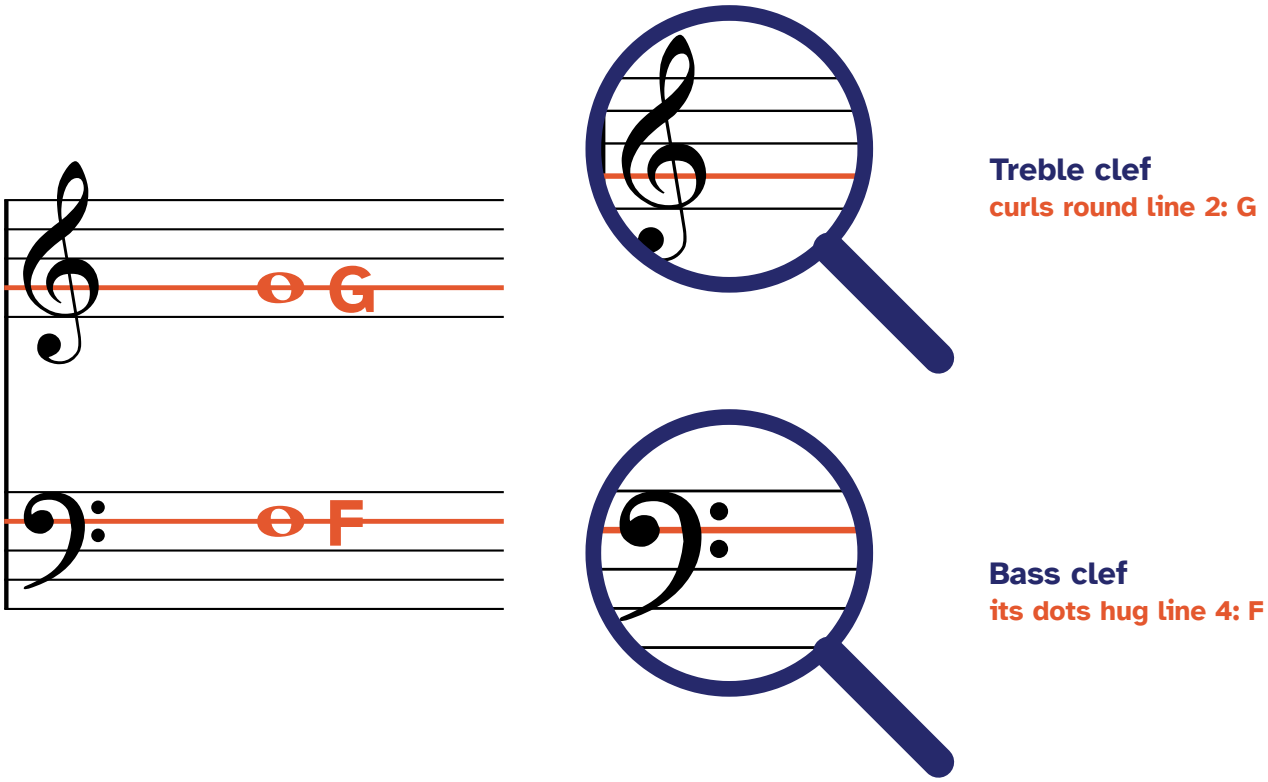
PART I · PITCH · 2

The staff and the clefs

Music is written on a **staff**: five lines and four spaces, counted from the **bottom**. Higher on the staff means higher on the keyboard. The **clef** at the start is the key to the code: it names one line, and every other note follows from it.

THE SECRET

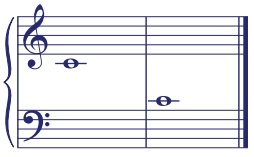
The treble clef curls round G; the bass clef's two dots hug F.



Look closely: the treble clef curls round G · the bass clef's dots hug F

The grand staff

Piano music uses two staves joined by a brace: the **treble** staff on top, usually for the right hand, and the **bass** staff below, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.

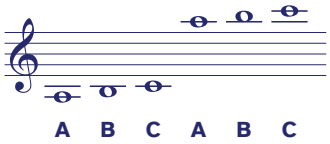


Middle C in the treble staff (left) and in the bass staff (right): the same key.

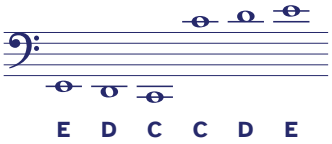
Ledger lines: the staff carried on

Notes too high or too low for the staff sit on short extra lines called **ledger lines**, spaced exactly like the staff lines. Read them by stepping on from the nearest staff line: line, space, line, space.

PART I · PITCH · 2 · CONTINUED



Treble: A, B, C below; A, B, C above.



Bass: E, D, C below; C, D, E above.

8va: play it an octave higher

Lots of ledger lines are hard to read. So a very high passage is written on the staff with **8va** and a dotted line above it: play those notes **one octave higher**, eight notes up, until the dotted line stops. **8vb** under the staff means an octave lower.

THINK OF IT A clef is a pair of glasses: the same five lines read as different notes, depending on which pair you put on.

EXAMPLE 1

One line, two notes

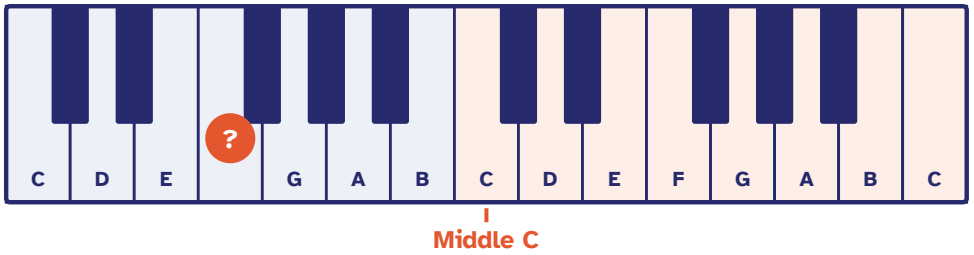


1. **Look at the red notes:** both sit on the bottom line.
2. **Now look at the clefs:** treble on the left, bass on the right.
3. **Read each from its clef note:** treble G is line 2, so step down (F in the space) to line 1; bass F is line 4, so step down line by line.

Treble bottom line: **E**. Bass bottom line: **G**. Same line, different note: always read the **clef** first.

EXAMPLE 2

Which staff is this key written on?



1. **Find Middle C.**
2. **Is the ? to its left or right?** Left is lower: the blue half.
3. **Low keys go on the bass staff**, high keys on the treble staff.

The **?** is below Middle C, so it is written on the **bass staff**, usually for the left hand.

PART I · PITCH · 2 · CONTINUED

EXAMPLE 3

Read a high note on ledger lines



1. **The top line** of the treble staff is F.
2. **Step up** one note at a time: G sits on top of the staff, A on the first ledger line.
3. **Keep stepping** to the red note: B above it, then the second ledger line.

Two ledger lines above the treble staff: **High C**, two octaves above Middle C.

GAME

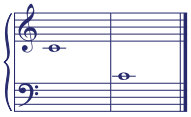
Treble or bass?

1	2	3	4
the E above Middle C	the lowest key on the piano	Middle C	High C

Which staff would each note be written on?

Answer: 1 treble · 2 bass, with ledger lines · 3 either: it sits between the two · 4 treble, two ledger lines up.

TRY IT · 10 SECONDS
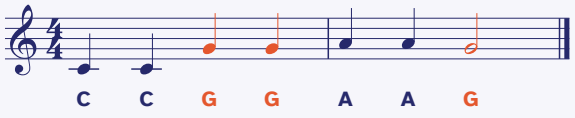


Point, then play.

Point to the line the treble clef curls round and play that G. Point to the line between the bass clef's dots and play that F. Middle C sits between them.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The treble clef curls round **line 2: G**, the very note Twinkle leaps up to. Middle C hangs on its own ledger line below.

Try it. Point to the G line, then play C C G G A A G.

TRAP

- × Reading bass-clef notes as if they were treble: check the clef on every line.
- × Reading ledger lines from the wrong end: start at the staff and step outwards.

SEE ALSO
Reading the notes p. 29 · The staff and the keyboard p. 32

PRACTISE IT
Workbook Pitch 2 · Online: Library › Staff and clefs

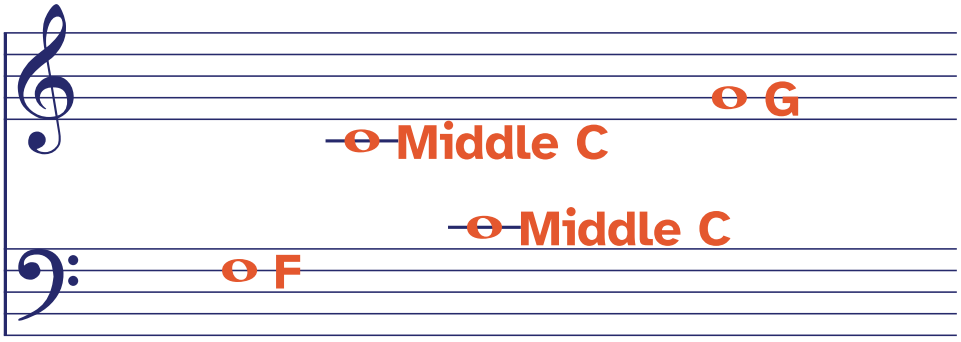
PART I · PITCH · 3

Reading the notes

Every line and space has one letter. You do not have to learn them all at once: start from a note you always know, then **step** or **skip** to the one you want.

THE SECRET

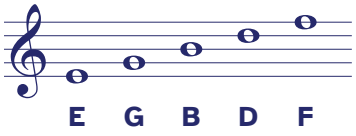
Start from a note you know (bass F, Middle C, treble G), then step or skip.



Notes you always know: treble G, bass F, and Middle C in either clef

THE RULE

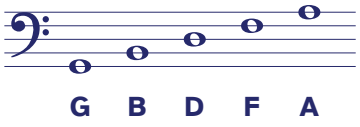
Step: to the very next line or space (one letter on). Skip: line to the next line, or space to the next space (two letters on).



Treble lines, E G B D F: *Every Good Boy Deserves Fun*



Treble spaces, F A C E: they spell *FACE*



Bass lines, G B D F A: *Great Big Dragons Fly Away*

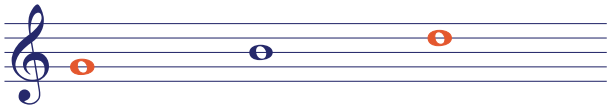


Bass spaces, A C E G: *All Cows Eat Grass*

PART I · PITCH · 3 · CONTINUED

EXAMPLE 1

Read a note you are not sure of

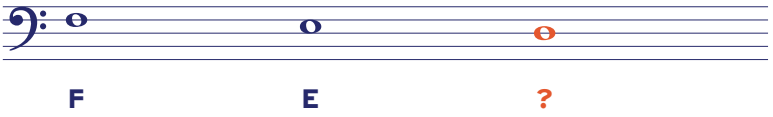


1. **The clef:** treble, so the clef note is G, on line 2.
2. **From G, skip up** line to line: one letter jumped each time.
3. **Count the skips** up to the other red note, on line 4: G, then B, then …

The red note is **D**: G, B, D, two skips up from the clef's G.

EXAMPLE 2

Read a bass note from the clef

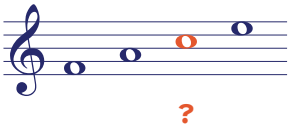


1. **The bass clef's dots** sit either side of line 4: that line is F.
2. **Step down** to the space below.
3. **One more step down** to the red note on line 3.

F, E, **D**: bass line 3 is D, two steps down from the clef's F.

EXAMPLE 3

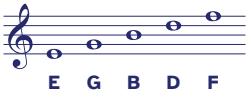
Find Treble C on the staff



1. **These notes sit in the spaces.** The bottom space is F.
2. **Step up through the spaces:** F, A … they spell a word.
3. **The red note is space 3.** Which letter is it?

F A **C** E: space 3 is **C**, Treble C, the same C you found on the keyboard, one octave above Middle C.

TRY IT · 10 SECONDS



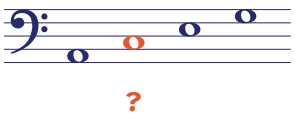
Say the treble lines up and down.

"E G B D F", then "F D B G E", out loud, as fast as you can without a slip. Then the spaces: "F A C E".

PART I · PITCH · 3 · CONTINUED

EXAMPLE 4

Find Bass C on the staff



1. **These notes sit in the bass spaces.** The bottom space is A: *All.*
2. **Say the next word:** *All Cows …*
3. **The red note is space 2.** Which letter?

A **C** E G: space 2 of the bass staff is **C**, Bass C, one octave below Middle C.

GAME

Name it from the clef note
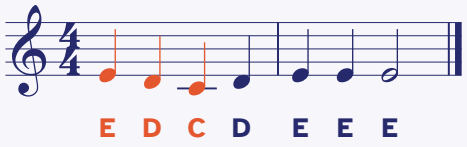
1	2	3	4
treble line 3	treble space 1	bass line 3	bass space 3

Start from the clef note (treble G on line 2, bass F on line 4), then step or skip.

Answer: 1 B (skip up from G) · 2 F (step down from G) · 3 D (two steps down from F) · 4 E (step down from F).

YOU KNOW THIS

Mary Had a Little Lamb



E D C: **line, space, ledger line**, one step down each time. Read the shape and the tune walks down with it.

Try it. Read the notes aloud before you play them.

THINK OF IT The clef notes are lamp posts on a dark road: walk from the nearest one and you are never lost.

TRAP

- × Naming bass notes with treble names: in the bass, the bottom line is G, not E.
- × Mixing up step and skip: a step changes line to space; a skip stays on lines (or on spaces).

SEE ALSO

The staff and the clefs p. 26 · The staff and the keyboard p. 32

PRACTISE IT

Workbook Pitch 2 · Online: Library › Staff and clefs

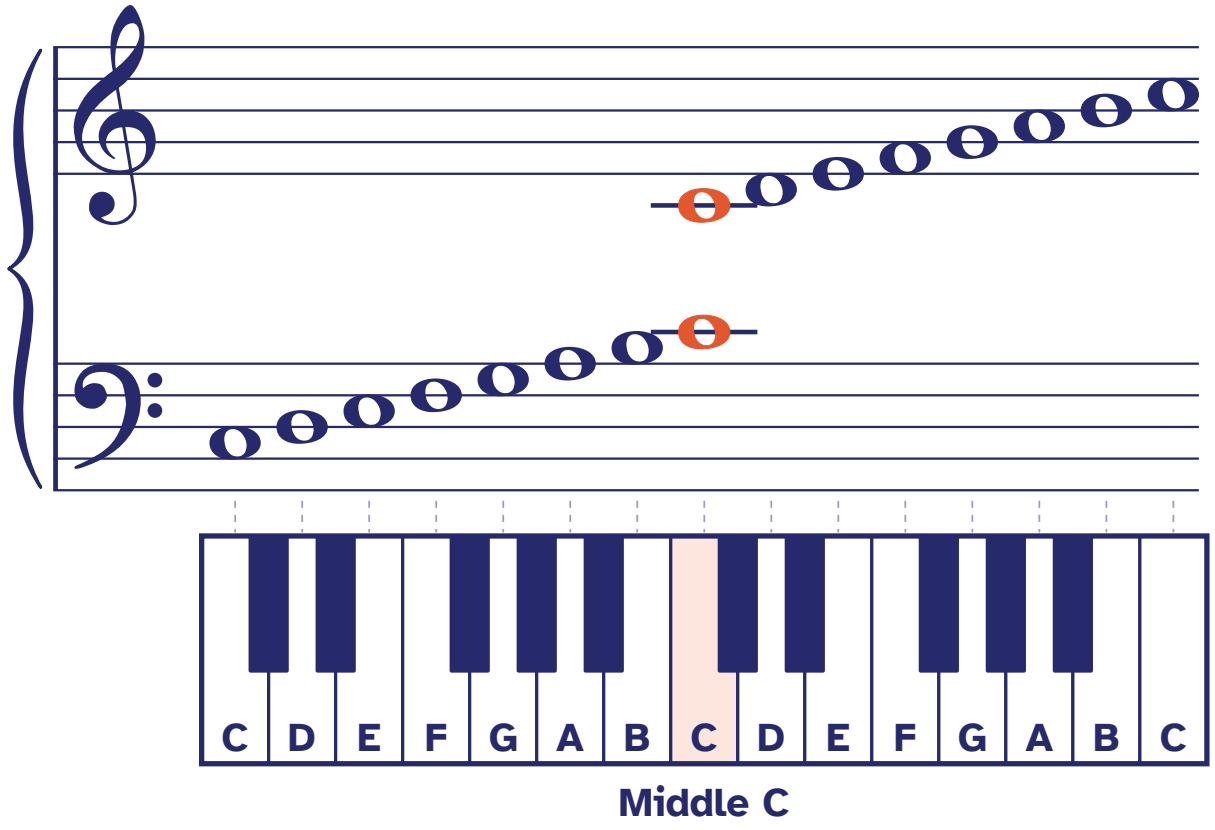
PART I · PITCH · 4

The staff and the keyboard

Each line and each space of the grand staff is one white key. Climb the staff and you climb the keyboard: here are two octaves, from Bass C to Treble C, with Middle C in red where the two staves meet.

THE SECRET

Every line and every space is one white key, in order.



Middle C can be written in either staff: on a ledger line below the treble staff, or above the bass staff. Same key; the hand that plays it chooses.

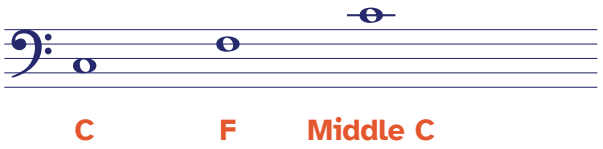
THE RULE

One step on the staff (line to space, or space to line) is the very next white key. The black keys sit in between; how they are written is the next chapter (p. 35).

PART I · PITCH · 4 · CONTINUED

EXAMPLE 1

From the page to the key

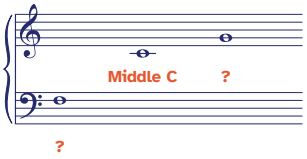


1. **Name the note:** line 4 of the bass staff, between the clef's dots, is F.
2. **On the piano,** find the three black keys just below Middle C.
3. **F is the first key** of that family: play it with the left hand.

The bass-clef F is the **F just below Middle C**: left of the three black keys under it.

EXAMPLE 2

Mirror notes either side of Middle C

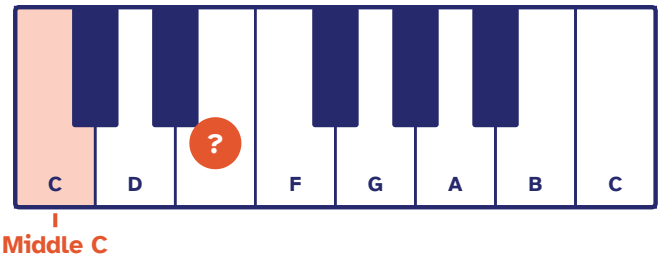


1. **Treble G:** the three black keys just above Middle C; G is the second key, after F.
2. **Bass F:** the three black keys just below Middle C; F is the first key.
3. **Now count each from Middle C:** C D E F G up, C B A G F down. Which number?

Both are **note 5**: treble G and bass F are mirror images, the same distance either side of Middle C.

EXAMPLE 3

From the key to the page



1. **Find the ? key:** count up from Middle C, C 1, D 2 ...
2. **Do the same on the staff:** Middle C on its ledger line, D in the space under the staff ...
3. **Where do you land?**

The **?** is **E**, note 3, and it sits on the **bottom line** of the treble staff.

PART I · PITCH · 4 · CONTINUED

GAME

Count the notes

1	2	3
Middle C up to A	Middle C down to E	G up to the next G

Count the notes, both ends included. The staff climbs the same way: one line or space per note.

Answer: 1 six (C D E F G A) · 2 six (C B A G F E) · 3 eight: an octave.

TRY IT · 10 SECONDS

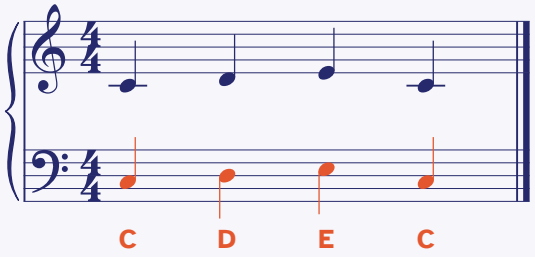


Play Middle C, then walk up.

Say each white key as you play it, C D E F G A B C, and picture each one climbing the staff: line, space, line, space.

YOU KNOW THIS

Frère Jacques, in both hands



The left hand plays the **same letters**, C D E C, one octave lower, written in the bass clef.

Try it. Play it right hand, then left hand, then both together.

THINK OF IT The staff is a ladder laid beside the keyboard: each rung, a line or a space, is the next white key.

TRAP

- × Skipping a key: every line and every space is used, one white key each.
- × Counting from 0: the note you start on is note 1.

SEE ALSO

The keyboard p. 22 · Reading the notes p. 29

PRACTISE IT

Workbook Pitch 1–2 · Online: Library › Staff and clefs

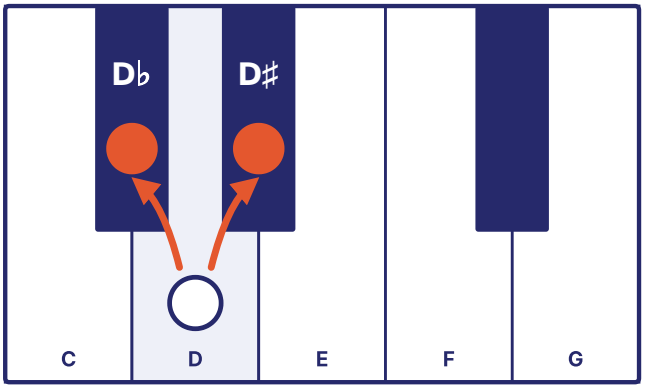
PART I · PITCH · 5

Sharps, flats and naturals

A sign in front of a note moves it to the very next key. **Sharp**: one key to the **right**. **Flat**: one key to the **left**. Usually that next key is black, which is how the black keys get their names.

THE SECRET

Sharp: the very next key to the right. Flat: the very next key to the left.

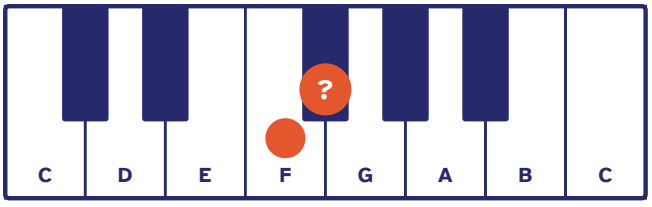


♯ one key right · ♭ one key left

SIGN	NAME	WHAT IT DOES	FIRST MET
♯	sharp	one key to the right (a semitone higher)	Level 0
♭	flat	one key to the left (a semitone lower)	Level 0
♮	natural	cancels a sharp or flat: back to the white key	Level ½

EXAMPLE 1

Find F♯



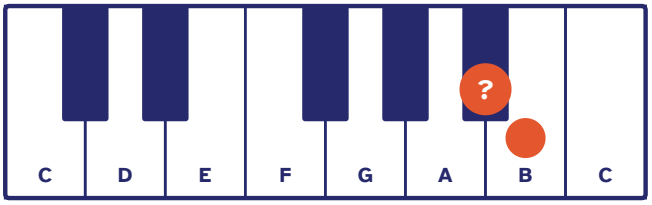
1. **Find F** (the red dot): just left of the three black keys.
2. **Sharp means right:** move to the very next key.
3. **Is it black or white?** Put your finger on it.

The **?** is **F♯**: the first black key of the group of three.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 2

Find B♭
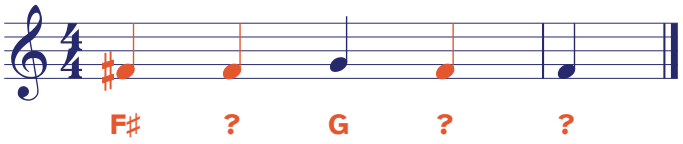


1. **Find B** (the red dot): just right of the three black keys.
2. **Flat means left:** move to the very next key.
3. **Which black key is it?**

The **?** is **B♭**: the top black key of the group of three.

EXAMPLE 3

How long does the sharp last?

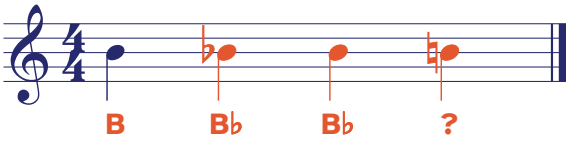


1. **The first note** has a sharp: F♯.
2. **Later in the same bar** there is no sign. Is it still sharp?
3. **After the bar line**: is it still sharp?

Still **F♯** until the bar line; after it, plain **F** again. A sign lasts **to the end of its bar**.

EXAMPLE 4

A natural cancels the flat
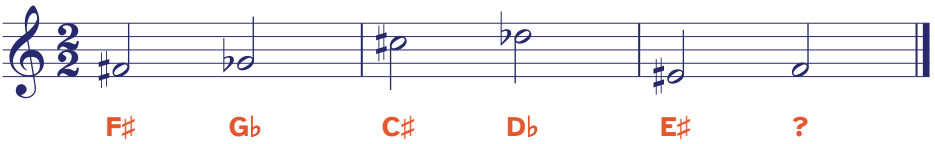


1. **B**: the white key.
2. **B♭**: one key left, and it stays flat for the rest of the bar.
3. **The last note** has a natural sign: what does it do?

The natural **cancels** the flat: back to the white key B, until the bar line.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 5

One key, two names



1 **The first bar:** F♯, then G♭. Find both on the keyboard.

2 **Are they different keys?**

3 **The last bar:** E♯. Which key is one right of E?

F♯ and G♭ are the **same black key**; E♯ is the white key **F**. Same sound, different names: **enharmonic**.

GAME

Which key is it?
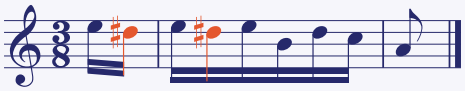
1	2	3	4
C♯	E♭	B♯	F♭

Sharp: one key to the right. Flat: one key to the left. Which key do you press?

Answer: 1 the black key right of C · 2 the black key left of E · 3 the white key C · 4 the white key E.

YOU KNOW THIS

Für Elise (Beethoven)



E, D♯, E, D♯: the very next key down and back. The sharp is written again in the new bar, because the bar line wiped it out.

Try it. Play E and D♯ back and forth: the black key just left of E.

THINK OF IT A sharp or flat is a sticky note on one line of the staff: it stays stuck until the bar line, then it falls off.

TRAP

× Carrying a sharp or flat past the bar line: after it, the note is plain again.

SEE ALSO
Keys and key signatures p. 78

PRACTISE IT
Workbook Pitch 3 · Online: Library › Signs

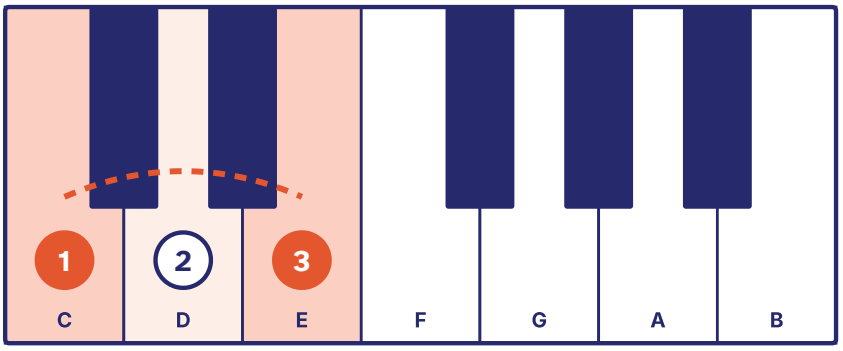
PART I · PITCH · 6

Counting intervals

How far apart are two notes? That distance is an **interval**. To give it a number, **count the notes**: the first note is 1, and you count every letter up to the second, both ends included.

THE SECRET

Count the notes, both ends included: C D E is 3 notes, a 3rd.

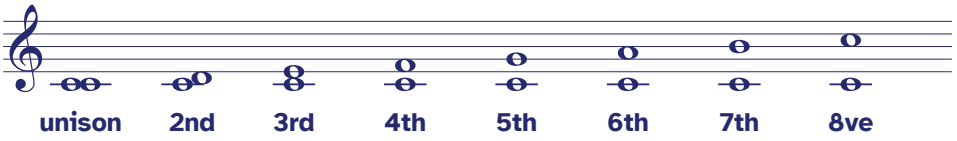


C to E: count the notes, 1 2 3: a 3rd

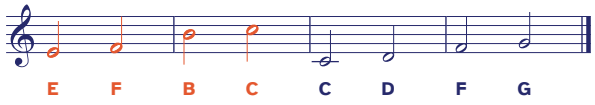
THE RULE

Count the notes for the number: start at 1 on the first note. C to E is C D E: 3 notes, a 3rd. Sharps and flats never change the number.

Interval numbers from C



The smallest steps: semitones and tones



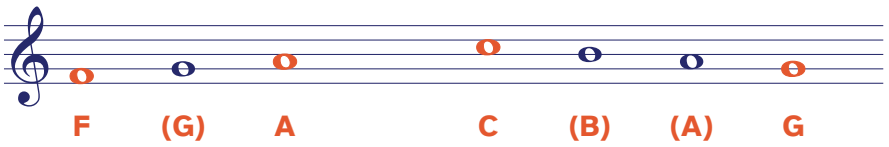
E–F and B–C: no black key between, a semitone. C–D and F–G: a black key between, a tone.

From any key to the **very next key**, black or white, is a **semitone**: the smallest step on the piano. Two semitones make a **tone**. On the white keys most steps are tones, but **E–F** and **B–C** have no black key between them: they are semitones.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 1

How far apart are these notes?

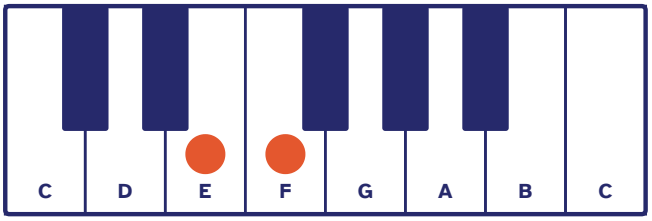


1. **Call the first note 1:** here F is 1.
2. **Count every letter** up to the second note.
3. **Going down works the same:** start at 1 on the top note and count down.

F G A is 3 notes: a **3rd**. C B A G is 4 notes: a **4th**.

EXAMPLE 2

Tone or semitone?

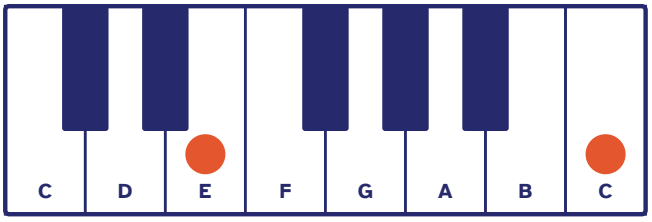


1. **Look between the red keys:** is there a black key?
2. **No black key** means they are next-door keys.
3. **Now try C and D**: is there a black key between?

E–F is a **semitone** (next door). C–D has C♯ between: a **tone**.

EXAMPLE 3

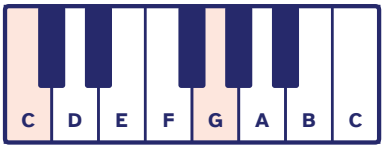
Name an interval on the keyboard



1. **Find both red keys:** E, and the C above it.
2. **Count the notes** from E: E is 1 …
3. **Stop on the C.** What number is it?

E F G A B C is 6 notes: a **6th**.

TRY IT · 10 SECONDS



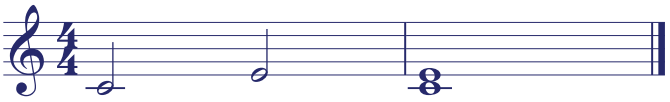
Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 4

Melodic or harmonic?



1. **The first bar:** C, then E, one after the other.
2. **The second bar:** C and E at the same time.
3. **Count the notes** in each.

Both are a **3rd**: **melodic** when played one after the other, **harmonic** when played together.

GAME

Count the notes
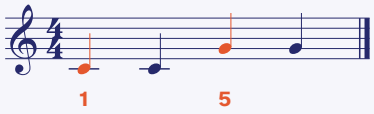
1	2	3	4
C up to G	D up to F	E up to the next E	G up to A

Count every note, both ends included. What is each interval called?

Answer: 1 a 5th (C D E F G) · 2 a 3rd · 3 an octave (8) · 4 a 2nd.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from the first “twinkle” to the second is C up to G: C D E F G, **five notes, a 5th**.

Try it. Sing the first two notes of any song, then count the notes between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: C to E is 3 notes, not 2 gaps.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 41 · Major scales p. 72

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

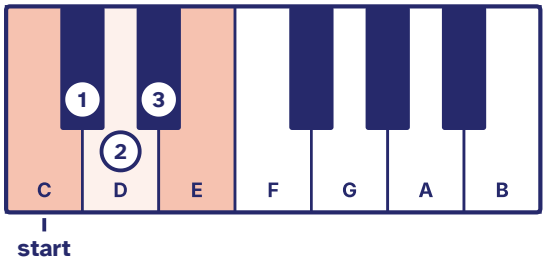
PART I · PITCH · 7

Major, minor and perfect

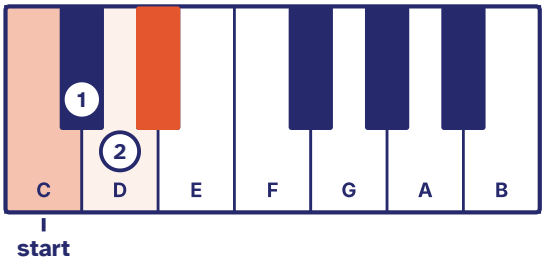
C to E and C to E♭ are both 3rds: three notes each. But they sound different, bright and dark. To tell them apart, **count the keys in between**, black and white: the keys you do not play.

THE SECRET

Count the keys between: 3 make a major 3rd, 2 a minor 3rd.



C to **E**: 3 keys between. A **major** 3rd: bright.



C to **E♭**: 2 keys between, one fewer. A **minor** 3rd: dark.

One key fewer between: a major 3rd becomes minor

THE RULE

Count the notes for the number (a 3rd). Count the keys in between for the size (major or minor): black and white, not the two you play.

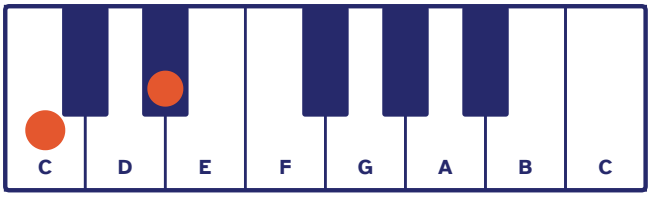
NUMBER	KEYS BETWEEN: SMALLER	MAIN SIZE
3rd	2 (minor)	3 (major)
4th		4 (perfect)
5th		6 (perfect)
6th	7 (minor)	8 (major)
octave		11 (perfect)

3rds and 6ths come in two sizes: **major**, and **minor**, one key fewer between.

4ths, 5ths and octaves have one main size, called **perfect**. They are never major or minor.

EXAMPLE 1

Major or minor 3rd?



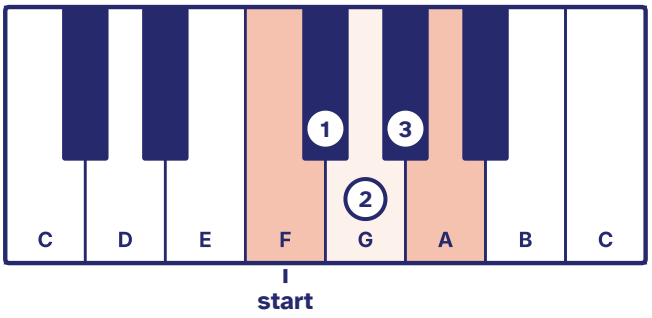
1. **Count the notes:** C D E, three letters, so it is a 3rd.
2. **Count the keys in between** C and the red black key, E♭: C♯, D …
3. **A major 3rd has 3 between.** How many here?

C to E♭ has **2 keys between**: a **minor 3rd**.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 2

Name F up to A

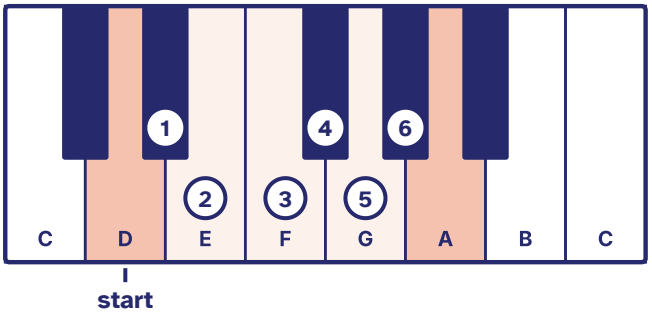


1. **Count the notes:** F G A is 3, so it is a 3rd.
2. **Count the keys in between:** they are numbered.
3. **3 between, or 2?**

F to A has 3 keys between: a **major 3rd**, the same size as C to E.

EXAMPLE 3

Name D up to A



1. **The number:** D E F G A, five notes, a 5th.
2. **The size:** count the keys in between: the numbered keys.
3. **A 5th with 6 between** is its main size. What is that called?

D up to A is a **perfect 5th**: 6 keys between.

GAME

Perfect, major or minor?

1	2	3	4
C–G	D–F	C–E	E–A

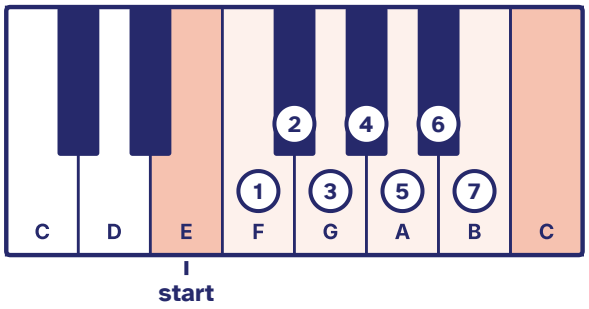
Count the notes for the number, then the keys in between for the size.

Answer: 1 perfect 5th (6 between) · 2 minor 3rd (2) · 3 major 3rd (3) · 4 perfect 4th (4).

PART I · PITCH · 7 · CONTINUED

EXAMPLE 4

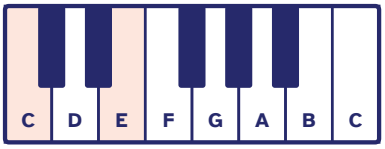
Name E up to C



1. **The number:** E F G A B C, six notes, a 6th.
2. **The size:** count the keys in between.
3. **A major 6th** has 8 between. Is this one more or one fewer?

E up to C has 7 keys between, one fewer than major: a **minor 6th**.

TRY IT · 10 SECONDS
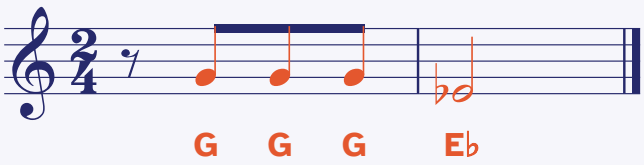


Play C–E, then C–E♭.

Bright, then dark. One key makes the difference: the major 3rd has three keys between, the minor 3rd two.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to **E♭** is three notes, a 3rd, with 3 keys between (F♯, F, E): a **major 3rd**.

Try it. Count the keys between G and E♭ on the keyboard.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key between, black and white.
- × Counting the two notes you play: count only the keys in between.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO
Counting intervals p. 38 · Chords and triads p. 89

PRACTISE IT
Workbook Pitch 4 · Online: Library › Intervals

PART I · PITCH · TIPS

When a note reads wrong

Most wrong notes come from a handful of slips. Name the slip first, then fix only that note: playing the whole line again rarely finds it.

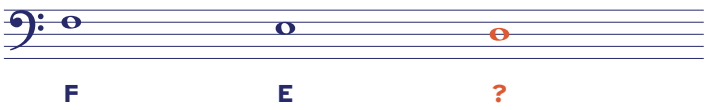
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
The left hand reads like the right	**the clef**: bass lines are G B D F A, not E G B D F	p. 26
You land an octave too high or too low	**the Cs**: find Middle C first, then walk out to Bass C or Treble C	p. 22
Notes above or below the staff slow you down	**the ledger lines**: step on from the nearest staff line	p. 26
A sharp or flat vanishes, or stays too long	**the bar line**: a sign lasts only to the end of its bar	p. 36
You play the black key on the wrong side	**the direction**: sharp goes right, flat goes left	p. 35
Two notes sound too close or too far apart	**the interval**: count the notes, then the keys between	p. 41

EXAMPLE

Spot the slip



Sam played B. Wrong!

1. **Check the clef first:** this is the bass clef, not the treble.
2. **Start from the clef note:** the dots hug line 4, F.
3. **Step down** to the red note on line 3: F, E ...

The note is **D**. Sam read the bass staff with treble names (treble line 3 is B): the most common slip of all.

HOW TO

Three checks before you read a new line

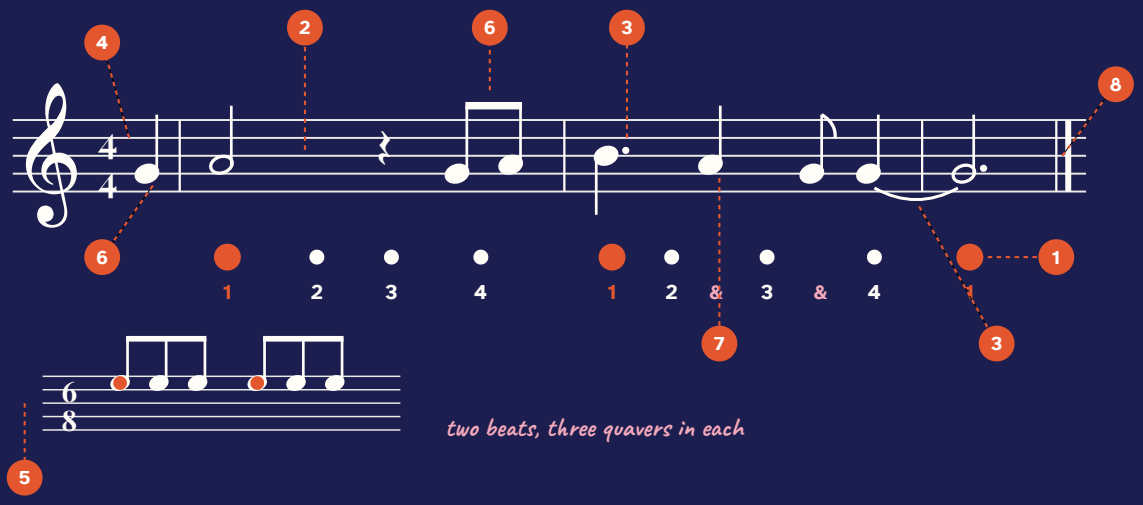
1. **The clefs:** which staff is treble, which is bass?
2. **The clef notes:** find treble G, bass F and Middle C, and read the other notes from them.
3. **The signs:** mark each sharp, flat or natural, and see where its bar ends.

THINK OF IT Like reading a map: find the big landmarks first, then the streets around them. A note far from every clef note is the one most often misread.

PART II · WHEN, AND FOR HOW LONG

Time

Music never stands still. Underneath runs a steady **pulse**, packed into **bars**, and every note and rest gets its fair share. Everything in this part is in these two bars.



1 **Pulse, tempo and the bar** p. 46
the steady beat underneath; beat 1 is strongest

2 **Notes and rests** p. 49
the shape says how long; rests are counted too

3 **How long is a note?** p. 52
a dot adds half; a tie glues two notes into one

4 **Time signatures** p. 55
four seats in every bar, a crotchet in each

5 **Compound and irregular time** p. 58
beats in threes; bars of five or seven

6 **Rhythm: groups and upbeats** p. 61
beams show the beats; starting before bar 1

7 **Rhythm: off the beat** p. 64
a note between the beats that holds on

8 **Repeats and road signs** p. 67
repeat signs, D.C., D.S., Fine and Coda

★ **Tips: when the rhythm slips** p. 70

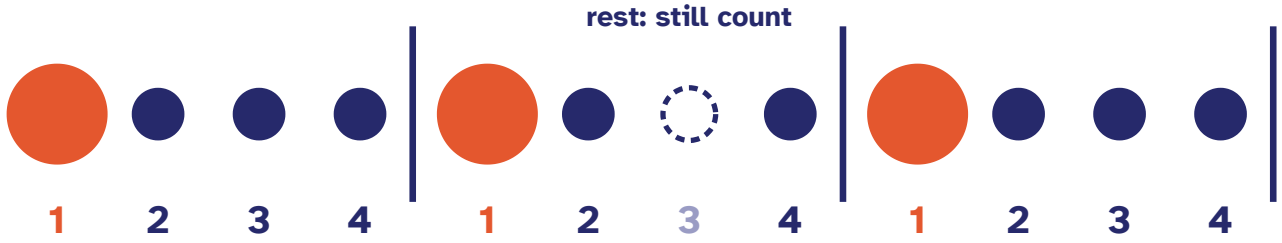
PART II · TIME · 1

Pulse, tempo and the bar

Put your hand on your heart: that steady beat is a **pulse**, and almost every piece has one. Its speed is the **tempo**. Beats travel in groups called **bars**, and beat 1 is the strongest.

THE SECRET

The beat never stops, not even in a rest. Feel it before you play.



Beat 1 of every bar is the strongest

Tempo

Tempo is measured in beats per minute. A **metronome mark** at the top of the music shows it: ♩ **= 60** means 60 crotchet beats a minute, one beat a second. ♩ **= 120** is twice as fast. Tempo words such as *Andante* and *Allegro* come with Expression, later in the book.

Strong and weak beats

BEATS	PATTERN	FEELS LIKE
2	**Strong** weak	a march
3	**Strong** weak weak	a waltz
4	**Strong** weak **medium** weak	most songs

Bars and bar lines

SIGN	NAME	MEANING
	bar line	divides the music into bars
	double bar line	marks the end of a section, or a change of time signature or tempo
	final bar line	marks the end of the piece
	repeat signs	play the passage between them again; if there is no start sign, go back to the beginning

PART II · TIME · 1 · CONTINUED

The beat in tunes you know

TUNE	BEATS IN A BAR	PATTERN
Twinkle, Twinkle, Little Star	4	**Strong** weak **medium** weak
Happy Birthday	3	**Strong** weak weak
The Blue Danube (a waltz)	3	**Strong** weak weak
Radetzky March	2	**Strong** weak

Counting

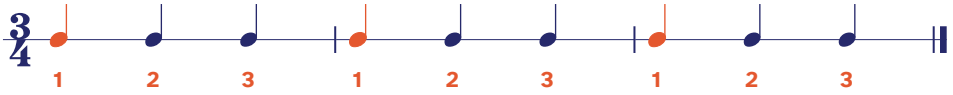


Three bars of 3/4 (the sign at the start: three beats in each bar, p. 55). Every beat is counted, even inside a long note: the red number is where you play, the grey numbers are the beats you hold.

Count aloud at one speed, a little louder on beat 1. With a metronome, start slower than you think and speed up only when the passage is steady.

EXAMPLE 1

Is this a march or a waltz?



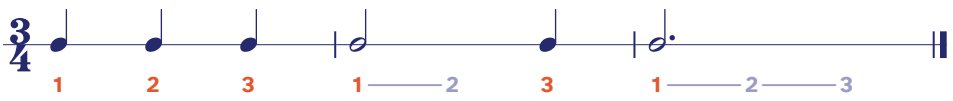
1. **Tap a steady beat** as you listen.
2. **Which beats feel heavier?** The red notes help.
3. **Count from one heavy beat to the next.** How many?

Heavy every three: 1 2 3, 1 2 3. A **waltz**. Every two would be a march.

EXAMPLE 2

Feel the beat before you play

♩ = 96



1. **Read the mark:** ♩ = 96, 96 crotchet beats a minute.
2. **Listen first:** set the metronome and let one full bar go by, counting “1 2 3” in your head.
3. **Step in** on the next beat 1, the way you step onto a moving escalator.

The first note lands **on the beat**, not a guess.

PART II · TIME · 1 · CONTINUED

GAME

How many beats in a bar?

1 2 3 4

The big dot is the strong beat. Count the beats in each bar: 2, 3 or 4?

Answer: 1 three (a waltz) · 2 four · 3 two (a march) · 4 three, twice.

TRY IT · 10 SECONDS



Hand on your heart. Tap the beat.

Say “1 2 3 4” with your taps, 1 a little louder. Keep going for ten seconds without slowing down.
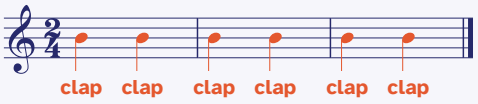
HOW TO

Find the beat of a new piece

1. **Read the time signature:** how many beats in each bar.
2. **Tap the beat** on your knee before you play, counting one bar aloud.
3. **Start slower** than the marked speed, and keep it steady to the end.

YOU KNOW THIS

Radetzky March (Johann Strauss I)



At the New Year concert the audience claps along on **every beat**. That steady clap is the pulse; the orchestra’s tune moves freely over it.

Try it. Put on any march and clap only the beat, not the tune.

THINK OF IT The beat is the stage and the notes are the actors: the stage is there before anyone walks on. So step in the way you step onto an escalator or take a plate off a sushi train: watch it move for a bar, then join in.

TRAP

- × Speeding up in easy bars and slowing in hard ones: the pulse never changes.
- × Counting only the notes you play: the beat goes on through long notes and rests.

SEE ALSO
Time signatures p. 55 · Tempo words: Expression (later)

PRACTISE IT
Workbook Level 0 · Pulse · Online: Metronome

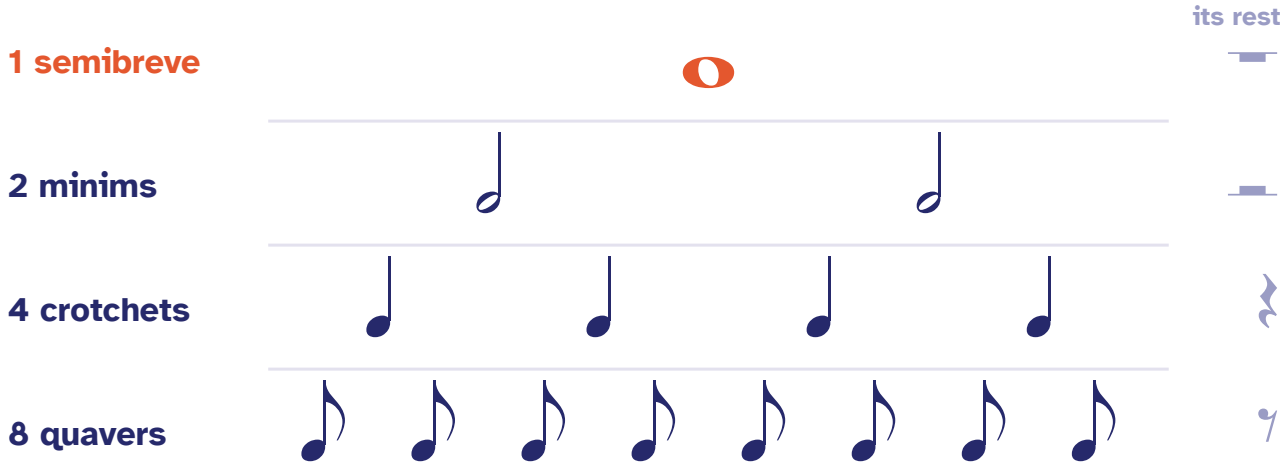
PART II · TIME · 2

Notes and rests

You can see how long a note lasts from its shape: each plain note is **half** the one above it. And every note has a twin **rest**, a silence you count just as carefully.

THE SECRET

Each note is half the one above it, and every note has a rest twin.



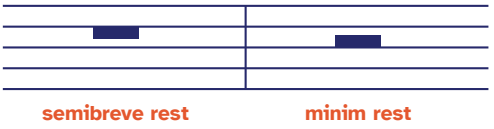
Each note splits into two of the next

Counting quavers: say the beat's number, then "&" halfway to the next beat: "1 & 2 & 3 & 4 &".

Head, stem, flag

The **head** sits on a line or in a space. A **stem** goes up on the right of the head or down on the left. **Flags** on the stem halve the value each time; groups of quavers are joined by **beams** instead of flags.

Which rest is which?

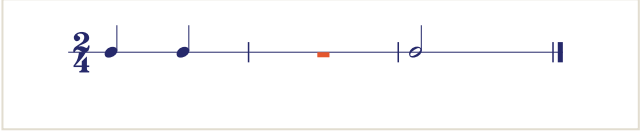


The semibreve rest **hangs** below the fourth line; the minim rest **sits** on the middle line.

THE SEMIBREVE REST'S SPECIAL POWER

Most rests last a fixed number of beats. The **semibreve rest** can also mean **a whole bar of silence** in any time signature: 4 beats in 4/4, but 3 in 3/4 and 2 in 2/4. It is then centred in the bar. No other rest can do this.





PART II · TIME · 2 · CONTINUED

NOTE	REST	NAME	ALSO CALLED	BEATS*	FIRST MET
		semibreve	whole note	**4**	Level 0
		**minim**	half note	**2**	Level 0
		**crotchet**	quarter note	**1**	Level 0
		**quaver**	eighth note	**½**	Level 0
		**semiquaver**	sixteenth note	**¼**	Level 2

Why whole, half and quarter?

Think of each beat as a pizza. A bar of 4/4 is four pizzas. The note that fills the whole bar is the **whole note**; the others are a half, a quarter, an eighth and a sixteenth of it. Two quavers beamed together fill one pizza. A rest is the same share left empty: shaded with lines.

semibreve		whole note	4 pizzas · 4 beats
semibreve rest		silence	4 empty pizzas · 4 beats
minim		half note	2 pizzas · 2 beats
minim rest		silence	2 empty pizzas · 2 beats
crotchet		quarter note	1 pizza · 1 beat
crotchet rest		silence	1 empty pizza · 1 beat
quaver		eighth note	half a pizza · ½ beat
quaver rest		silence	half an empty pizza · ½ beat
semiquaver		sixteenth note	a quarter of a pizza · ¼ beat
semiquaver rest		silence	a quarter of an empty pizza · ¼ beat

TRY IT · 10 SECONDS



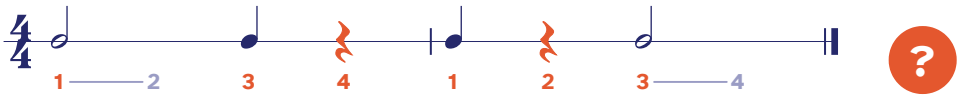
Clap once and hold for 2: a minim.

Now clap four crotchets (1 2 3 4), then one semibreve (clap, and hold all four beats). Say the counts out loud.

PART II · TIME · 2 · CONTINUED

EXAMPLE 1

How long is each sound, and each silence?

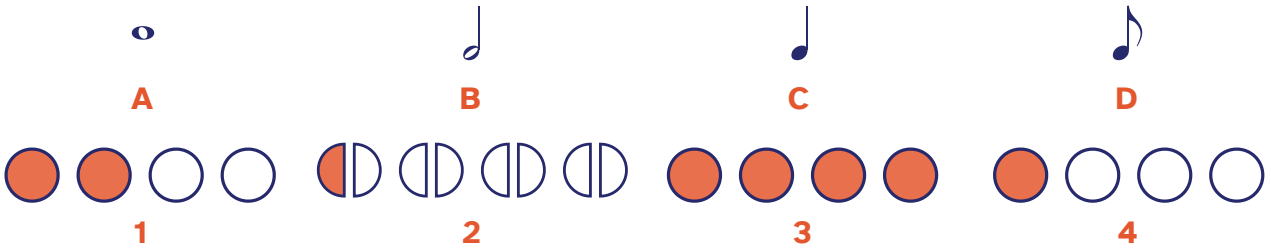


1. **Name each shape** in the bar.
2. **How many beats** does each one last?
3. **Add them up,** rests included. What is the total?

Minim 2, crotchet 1, crotchet rest 1: every bar adds up to **4 beats**, silences included.

GAME

Pizza match

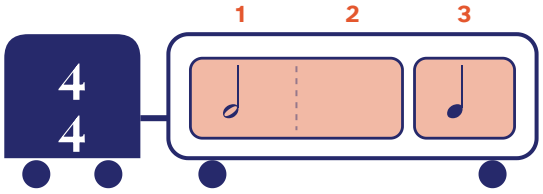


Match each note (A to D) with its pizzas (1 to 4). One pizza is one beat.

Answer: A 3, B 1, C 4, D 2.

EXAMPLE 2

Fill a bar of 4/4: what goes in the last seat?
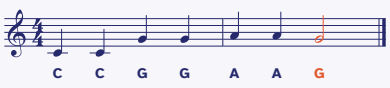


1. **How many seats** does the minim take?
2. **And the crotchet?**
3. **How many seats are left** for a silence, and which rest fills one seat?

2 + 1 = 3, one seat left: a **crotchet rest**. The bar is full.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



Six **crotchets**, then a **minim** at the end of the line: one beat, one beat … and two beats to breathe.

SEE ALSO
Pulse and the bar p. 46 · How long is a note? p. 52

PRACTISE IT
Workbook Level 0 · Note values · Online: Library › Notes

PART II · TIME · 3

How long is a note?

Need a note longer than its shape allows? Two signs stretch it: the **dot**, which adds half, and the **tie**, which glues two notes into one sound.

THE SECRET
A dot adds half. A tie glues two notes into one.
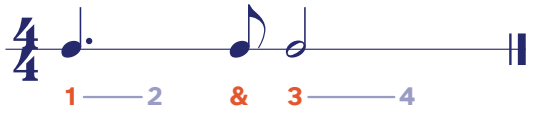


2 beats **2 + 1 = 3 beats**

The dot adds half

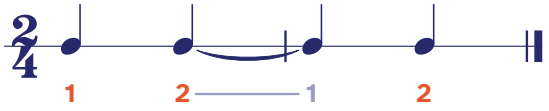
The dot

A dot adds **half the note's value**: a dotted minim lasts 2 + 1 = 3 beats; a dotted crotchet 1 + ½ = 1½.



The tie

A curve joining two notes **of the same pitch** makes one longer sound: play the first, hold through the second. Ties can cross a bar line, where a dot cannot.



The dot, in pizzas

The dot is a helper that adds **half of whatever its note is worth**. Dark slices: the note. Lighter: what the dot adds.

dotted minim		2 + **1** (half of 2) = **3 beats**
dotted crotchet		1 + **½** (half of 1) = **1½ beats**
dotted quaver		½ + **¼** (half of ½) = **¾ beat**
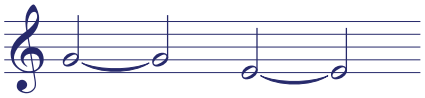
EXTRA · GOING FURTHER

The double dot

A second dot adds half of the first dot: a double-dotted minim lasts 2 + 1 + ½ = **3½ beats**. Rare before the higher Levels.

PART II · TIME · 3 · CONTINUED

Tie or slur?



Tie: the same pitch. Play once and hold: G for 4 beats, then E for 4.

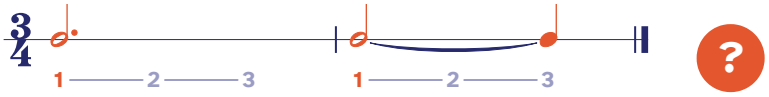


Slur: different pitches. Play every note, smoothly joined (legato).

TWO-MINUTE DRILL Clap a dotted minim ("1 2 3"), a dotted crotchet and quaver ("1 & 2"), then two tied crotchets ("1 2"). Say the counts aloud and never re-clap a tied note.

EXAMPLE 1

Dot or tie: which is longer?
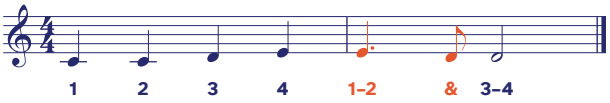


1. **The dotted minim:** a dot adds half. How many beats?
2. **The tied notes:** add the two notes together.
3. **Compare** the two totals.

Both last **3 beats**: the same length. A tie can also cross a bar line, which a dot cannot.

EXAMPLE 2

A dotted crotchet you already know: Ode to Joy



1. **Bars 3–4 of Beethoven's Ode to Joy.** Sing it: "C C D E, E — D D".
2. **The red E is a dotted crotchet:** 1 + ½ = 1½ beats, held through "1 & 2".
3. **The quaver D** falls on the "&" of beat 2; the last D takes beats 3 and 4.

Long–short, then long: the dot is why the tune **leans** before it lands.

EXAMPLE 3

Which is longer: a dotted minim or a minim tied to a quaver?



1. **Dotted minim:** add half of 2. How many pizzas?
2. **Minim tied to a quaver:** add the two.
3. **Compare** the totals.

3 against 2½: the **dotted minim** is longer, by half a beat.

PART II · TIME · 3 · CONTINUED

GAME

How many beats?



With a crotchet as one beat: how long does each one last?

Answer: 1 three (2 + 1) · 2 one and a half (1 + ½) · 3 three (2 tied to 1).

TRY IT · 10 SECONDS

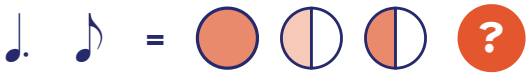


Clap once, hold for three.

Say “1 2 3” while you hold: that is a dotted minim. Then a dotted crotchet: clap on “1”, hold through “& 2”, and let go on the “&” after 2.

EXAMPLE 4

How many beats: a dotted crotchet and a quaver?

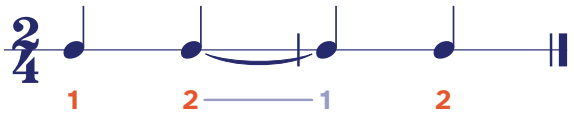


1. **The dotted crotchet:** 1 beat plus half of 1.
2. **The quaver:** half a beat.
3. **Add them.** Where does the quaver fall?

1½ + ½ = **2 beats**. The quaver lands on the “&” of beat 2: “1 (&) 2 **&**”.

EXAMPLE 5

How long do you hold a tie?



1. **Play the first note** and count its beats.
2. **Keep holding** through the tie. Do you play the second note again?
3. **Add the beats** of both notes.

One sound, never played twice: a tie **adds the beats** of both notes.

TRAP

- × Putting the dot on the wrong note: a dot adds half of **its own** note.
- × Tying two different pitches: that curve is a slur, not a tie.

SEE ALSO

Time signatures p. 55 · Rhythm p. 61

PRACTISE IT

Workbook Level 0 · Note values · Online: Library › Notes and rests

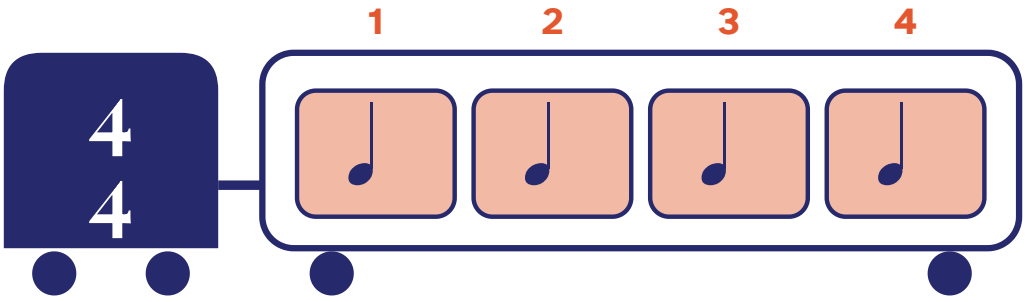
PART II · TIME · 4

Time signatures

Two numbers, like a train ticket: the **top number** says how many seats (counts) each bar has; the **bottom number** says which note fills one seat.

THE SECRET

Top number: how many beats. Bottom number (in simple time): which note gets one beat.



Top: how many seats · bottom: which note sits in one

The bottom number

BOTTOM	NOTE	ONE COUNT IS A	FIRST MET
2	𝅗𝅥	minim	Level 3
4	♩	crotchet	Level 0
8	♪	quaver	Level 1
16	𝅘𝅥𝅯	semiquaver	*Extra*

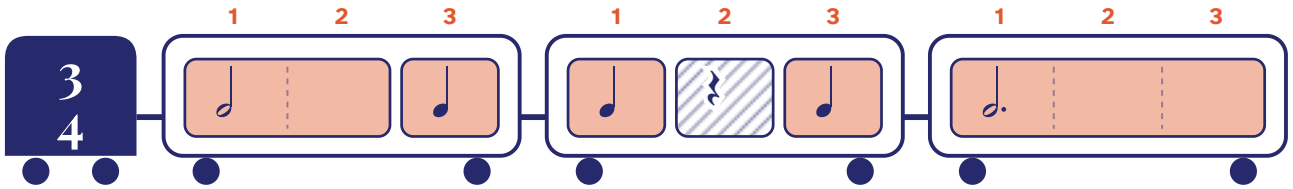
Simple time: each beat divides in two

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
2/4	2 crotchets	Level 0
3/4	3 crotchets	Level 0
4/4 𝄴	4 crotchets (the C sign: common time)	Level 0
2/2 𝄵	2 minims (the cut C: cut common time)	Level 3
3/2	3 minims	Level 3
4/2	4 minims	*Extra*
3/8	3 quavers	Level 3

PART II · TIME · 4 · CONTINUED

Think of a train

In 3/4 every carriage (bar) has exactly **3 seats**; each seat is one crotchet beat. Notes book seats: a minim takes 2, a crotchet 1. A rest books a seat and leaves it **empty** (shaded). Every seat in every carriage must be booked.



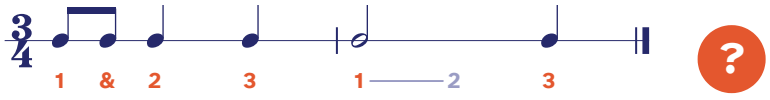
minim + crotchet · crotchet, crotchet rest, crotchet · dotted minim: 3 seats each

Changing time signature

A new time signature can appear in the middle of a piece, right after a bar line; it holds until the next one. A time signature is written only once, not on every line.

EXAMPLE 1

What does 3/4 mean?



1. **The top number:** how many counts in each bar?
2. **The bottom number:** 4 stands for a crotchet. Which note gets one count?
3. **Check a bar:** does it add up?

3/4: **three crotchet beats** in every bar (two quavers + crotchet + crotchet = 3).

GAME

Which ticket?

1	2	3	4
3 crotchets	4 crotchets	3 quavers	2 minims

Each bar holds these notes. Write its time signature.

Answer: 1 3/4 · 2 4/4 · 3 3/8 · 4 2/2 (or 4/4).

TRY IT · 10 SECONDS

4
4

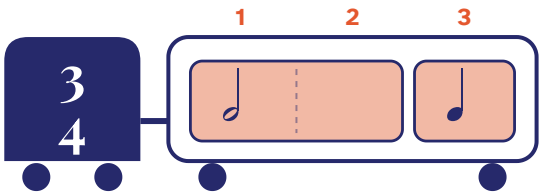
Open a piece you are playing.

Find the time signature, then count the seats in bar 1: do the notes add up to the top number? Try bar 2 as well.

PART II · TIME · 4 · CONTINUED

EXAMPLE 2

How do you count this bar of 3/4?

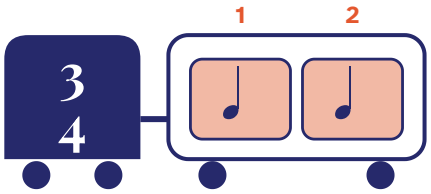


1. **How many seats** does 3/4 have?
2. **The minim**: which seats does it book?
3. **The crotchet**: which seat is left for it?

Count **1 2 3**: the minim holds through 1 and 2, the crotchet takes 3.

EXAMPLE 3

Where does the rest go?
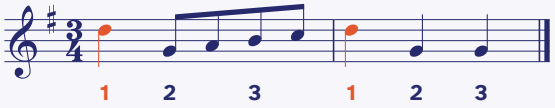


1. **How many seats** in 3/4?
2. **The two crotchets** fill which seats?
3. **Which seat is empty,** and what goes there?

Seat 3 is empty: a **crotchet rest**. Count it: 1 2 (3).

YOU KNOW THIS

Minuet (from the Notebook for Anna Magdalena Bach)



Every bar holds **three crotchet beats**: 1 2 3, 1 2 3, the step of the dance. That is 3/4.

Try it. Count 1 2 3 aloud while someone plays it.

TRAP

- × Reading the bottom number as a count: it names the note that gets one beat.
- × Reading 6/8 as six beats: at most speeds it moves in two.
- × Mixing up 3/4 and 6/8: both hold six quavers, grouped differently (p. 58).

SEE ALSO

Pulse and the bar p. 46 · Compound time p. 58

PRACTISE IT

Workbook Level 0 · Time signatures · Online: Library › Time signatures

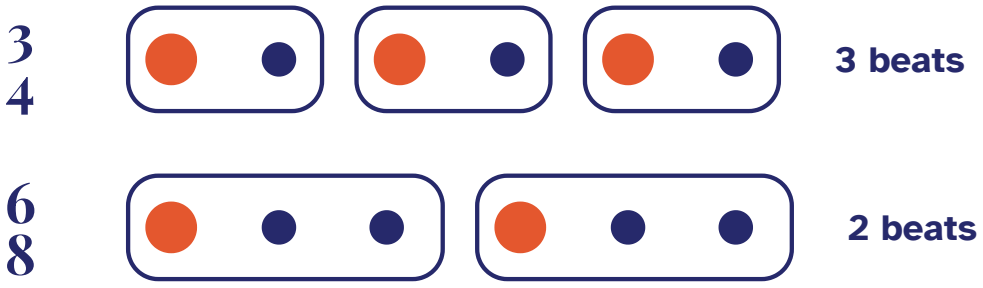
PART II · TIME · 5

Compound and irregular time

Compound time is a skipping rope: each dotted beat splits into **three**. Irregular time keeps you guessing, with **five or seven** counts in uneven groups. (9/4, 12/4 and /16 signatures are rare.)

THE SECRET

Dotted beats split in three: 6/8 is two big beats.



Same six quavers, new groups

SIMPLE OR COMPOUND?

In **simple time** each beat divides into **two**. In **compound time** each beat is a dotted note and divides into **three**.

Quick test: a top number of 6, 9 or 12 means compound. Divide it by 3 to find the beats: 6/8 has 2, 9/8 has 3, 12/8 has 4.

Compound time

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
6/8	2 dotted crotchets	Level 1
9/8	3 dotted crotchets	Level 3
12/8	4 dotted crotchets	Level 3
6/4	2 dotted minims	Level 4

Counting 6/8



Slowly, count all six quavers. At speed, feel two beats, on 1 and 4: “1 & a, 2 & a”.

PART II · TIME · 5 · CONTINUED

EXTRA · GOING FURTHER

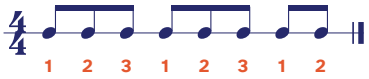
Irregular time

Five or seven counts, grouped unevenly. Met from Level 5.

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
5/4	5 crotchets: 3 + 2 or 2 + 3	Level 5
7/4	7 crotchets: 4 + 3 or 3 + 4	*Extra*
5/8	5 quavers: 3 + 2 or 2 + 3	*Extra*
7/8	7 quavers: 2 + 2 + 3 or 3 + 2 + 2	*Extra*

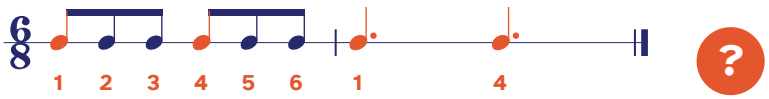
UNEVEN GROUPS IN 4/4

Many dances keep 4/4 but group the eight quavers 3 + 3 + 2: “1 2 3, 1 2 3, 1 2”.



EXAMPLE 1

How many beats in a bar of 6/8?



1. **Slowly, count all six quavers.**
2. **Which two counts** feel strong? (the red notes)
3. **Faster, count only the strong ones.** How many?

6/8 has **two beats** in a bar (1 and 4), each split into three quavers.

GAME

Two beats or three?

1	2	3	4
6/8	3/4	9/8	2/4

How many beats in a bar, and does each beat split in two or in three?

Answer: 1 two, in threes · 2 three, in twos · 3 three, in threes · 4 two, in twos.

TRY IT · 10 SECONDS

6/8

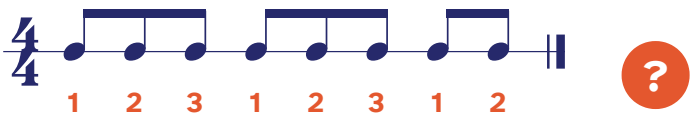
Sway left, sway right.

Say “1 2 3, 4 5 6” and sway once on 1 and once on 4. That is 6/8: six quavers, two big beats.

PART II · TIME · 5 · CONTINUED

EXAMPLE 2

Is this still 4/4?



1. **Count eight quavers** in the bar.
2. **Group them 3 + 3 + 2:** where do the accents fall?
3. **Keep your foot on the beat.** How many foot taps?

Accents on 1, 4 and 7, but four foot taps: the bar is still **4/4**, only the accents are uneven.

EXAMPLE 3

How many beats in 9/8?



1. **The top number, 9:** does it divide by 3?
2. **How many groups of three** fit in 9?
3. **What is each beat**, in quavers?

9 ÷ 3 = **three beats**, each a dotted crotchet: “1 2 3, 4 5 6, 7 8 9”.
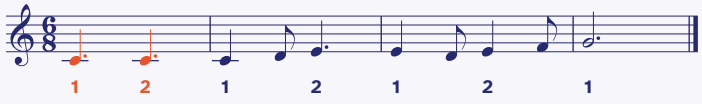
HOW TO

Feel compound time

1. **Say “pine-ap-ple”** on every beat and step only on “pine”.
2. **Clap the three syllables** while you step.

YOU KNOW THIS

Row, Row, Row Your Boat



Two big beats in every bar, each a **dotted crotchet**: “Row, row”. Each big beat is worth three quavers: “row your” is long-short, a crotchet and a quaver, filling one big beat. That is 6/8.

Try it. Sway twice per bar while you sing it.

THINK OF IT Like a skipping rope: two big swings, each with three little steps inside.

TRAP

- × Counting 6/8 as six slow beats at speed: feel two big ones.
- × Mixing up 3/4 and 6/8: the same six quavers, grouped differently.

SEE ALSO

Time signatures p. 55 · Rhythm: groups and upbeats p. 61

PRACTISE IT

Workbook Levels 1–4 · Compound time · Online: Rhythm clock

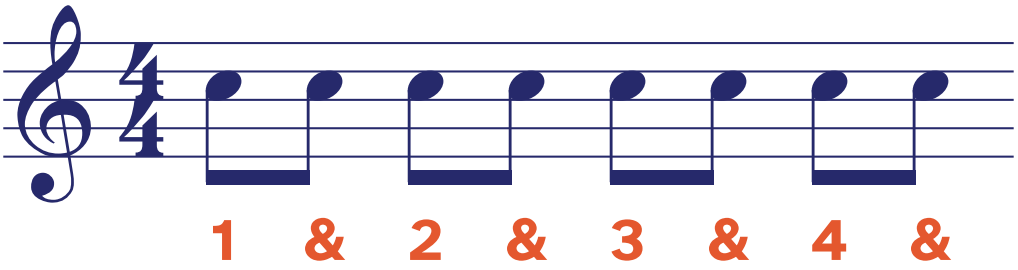
PART II · TIME · 6

Rhythm: groups and upbeats

Say your own name: long and short sounds, that's **rhythm**. The pulse marches steadily on; the rhythm dances around it. How the notes are grouped on the page shows where the beats fall.

THE SECRET

Beams group notes into beats; an upbeat borrows from the last bar.



One beam, one beat

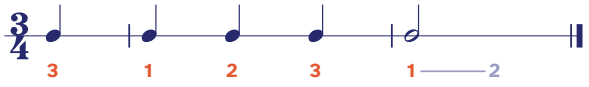
Grouping with beams

Beams join the notes of **one beat**, so the beats stay visible. In 4/4 a beam may cover beats 1–2 or 3–4, but never cross the middle of the bar (except in a deliberate grouping such as 3 + 3 + 2). In compound time, quavers are beamed in threes.

× A beam across beats 2 and 3 would hide the middle of the bar.
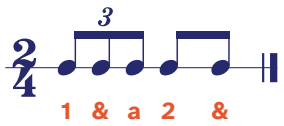
The upbeat (anacrusis)

A piece can begin **before** the first full bar, with one or more notes called an upbeat or anacrusis. The last bar is then short by the same amount, so the two together make one full bar. Count the missing beats in silence before you start.



Count "1 2" silently, then play on 3.
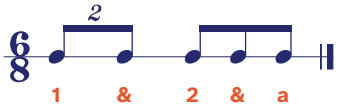
Triplets



Triplet: three exactly equal notes in the time of two. First met: Level 2.

EXTRA · GOING FURTHER

Duplet



Two notes in the time of three, in compound time: the triplet turned inside out.

PART II · TIME · 6 · CONTINUED



Two quavers: one beat cut in 2



A **triplet**: one beat cut in 3



Four semiquavers: one beat cut in 4

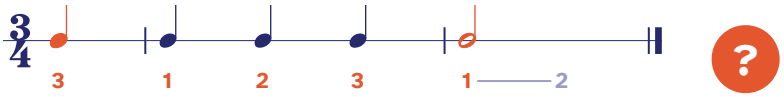
Counting smaller values



Crotchet, two quavers, four semiquavers and a triplet (“trip-o-let”): one beat each.

EXAMPLE 1

Does the upbeat balance the last bar?



1. **The upbeat:** how many beats before the first full bar?
2. **The last bar:** how many beats?
3. **Add them:** do they make a whole bar of 3/4?

1 + 2 = 3: the upbeat and the last bar **make one full bar** together.

GAME

May the beam join them?

1	2	3
beats 1 and 2	beats 2 and 3	beats 3 and 4

In 4/4, quavers are beamed so the middle of the bar stays visible. Which pairs may share one beam?

Answer: 1 yes · 2 no, that hides the middle of the bar · 3 yes.

TRY IT · 10 SECONDS



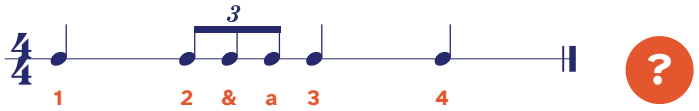
Clap “1 & 2 & 3 & 4 &”.

Now clap only the numbers, then only the &s, while your foot keeps the beat. The beat never moves; your claps do.

PART II · TIME · 6 · CONTINUED

EXAMPLE 2

Where is the triplet, and how do you count it?

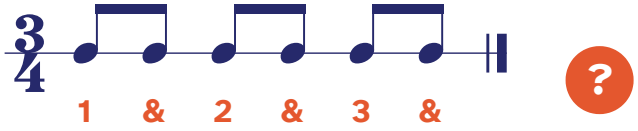


1. **Count the bar:** four crotchet beats.
2. **Find the little 3.** Which beat do the three quavers share?
3. **Say it** with "trip-o-let" on that beat.

Beat 2: "1, **trip-o-let**, 3, 4". Three notes squeeze into one beat; the beat never moves.

EXAMPLE 3

Why are six quavers beamed in twos in 3/4?



1. **How many beats** does 3/4 have, and what kind?
2. **How many quavers** fit in one beat?
3. **So how many** go under each beam?

Three crotchet beats, two quavers each: **2 + 2 + 2**. Beamed 3 + 3 they would look like 6/8.
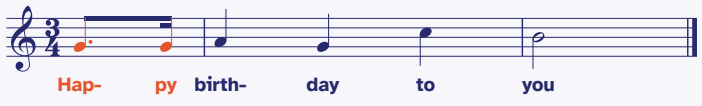
HOW TO

Practise triplets

1. **Say "trip-o-let"** on every beat, evenly.
2. **Clap the three syllables** while your foot keeps the beat.
3. **Switch** between two quavers ("1 &") and a triplet ("trip-o-let") every bar.

YOU KNOW THIS

Happy Birthday



"Hap-py" comes **before** the first strong beat: an upbeat. The strong beat lands on "birth".

Try it. Sing it and clap only on "birth", "day" and "you": you feel where bar 1 starts.

THINK OF IT An upbeat is the breath before you speak: "and-ONE". You lift on the upbeat so the strong beat can land.

SEE ALSO
Notes and rests p. 49 · Rhythm: off the beat p. 64

PRACTISE IT
Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

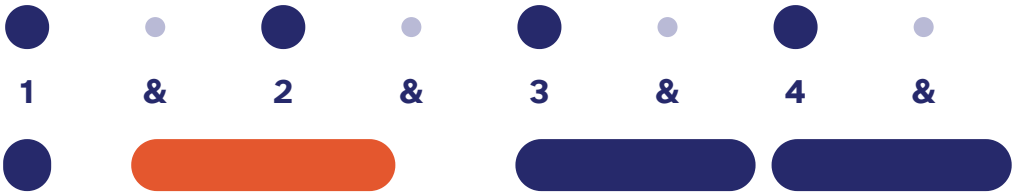
PART II · TIME · 7

Rhythm: off the beat

Now the fun part: rhythm that teases the beat. It only works if the beat underneath stays rock-steady.

THE SECRET

Keep the beat rock-steady and let the rhythm tease it.



starts on “&”, held over beat 2

A note between the beats

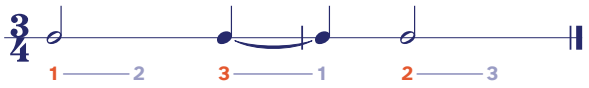
Syncopation



The crotchet starts on “&” and is held through beat 2.

Level 4. Feel beat 2 even though nothing is played on it.

Hemiola



Two bars of 3/4 felt as three minims.

Level 4. Common at the ends of phrases in Baroque dances.

Swing Level ½

Common from the first pieces: jazz, blues and many easy modern pieces are marked *swing*. Pairs of quavers are written evenly but played **long–short**, like the first and last notes of a triplet.



Written



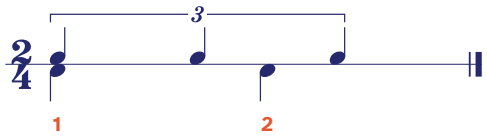
Played

PART II · TIME · 7 · CONTINUED

EXTRA · GOING FURTHER

Polyrhythm and rubato

Polyrhythm (Level 7): two rhythms at once.

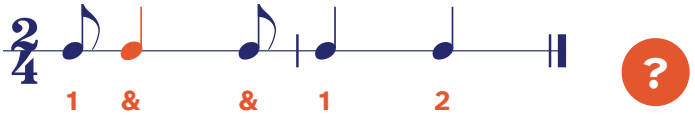


Three against two: a triplet above, two crotchets below, starting together.

Rubato: Time is stretched and then given back, for expression: the music leans into a phrase and catches up after it. The pulse always returns. Marked *rubato*, or simply part of the style in much music from the 1800s.

EXAMPLE 1

Where does the long note start?



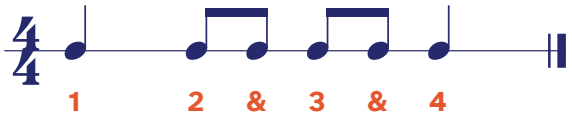
1. **Tap the beats:** 1, 2.
2. **Find the red crotchet.** Does it start on a beat, or on the "&"?
3. **Hold it:** through which beat?

It starts on the "&" and holds through beat 2: the long note starts **off the beat**. That is syncopation.

EXAMPLE 2

Which notes swing?

Swing



1. **Find the quaver pairs.** On which beats?
2. **Play each pair long-short:** "doo-ba".
3. **What about the crotchets?**

Only the **pairs** (beats 2 and 3) swing; the crotchets and the beat underneath stay even.

TRY IT · 10 SECONDS



Foot on the beat, hands on the &.

Tap a steady beat with your foot. Clap only between the taps. If the foot starts to follow the hands, slow down.

PART II · TIME · 7 · CONTINUED

GAME

On the beat or off it?

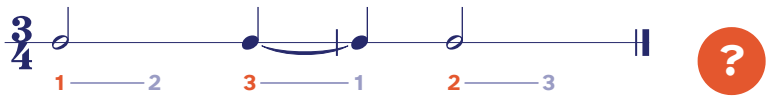
1	2	3	4
starts on 1	starts on &	starts on 2	starts on the & of 3

A note that starts on a number is on the beat; on an "&" it is off the beat. Which are off?

Answer: 2 and 4 are off the beat; 1 and 3 are on it.

EXAMPLE 3

Two bars of 3, or one big bar?
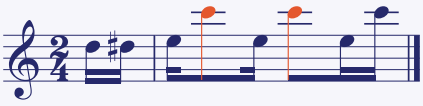


1. **Count as written:** "1 2 3, 1 2 3".
2. **Look at the notes:** how many beats does each minim last?
3. **Feel each minim** as one big beat. How many big beats?

Three minims of two beats: the six beats are felt as **three big ones**. That is a hemiola.

YOU KNOW THIS

The Entertainer (Scott Joplin)



The high **C** keeps landing just after the beat, not on it: syncopation, the bounce of ragtime.

Try it. Tap the beat with your foot and hum the tune: hear the C slip between the taps.

THINK OF IT Syncopation is a skip in your walk: a step that lands between the beats. Swing is walking with a spring in your step, long-short, long-short. Rubato is borrowing time like pocket money: whatever you spend, you pay back.

TRAP

- × Rushing the short notes after a long one: keep counting inside the long note.
- × Playing a triplet unevenly: three equal notes, not two quick and one long.
- × Letting syncopation drag the pulse: tap the beat with your foot.

SEE ALSO

Rhythm: groups and upbeats p. 61 · Words for how to play: Expression (later)

PRACTISE IT

Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

PART II · TIME · 8

Repeats and road signs

Music often says “play that again”. Instead of writing the same bars twice, it uses **repeat signs** and **road signs**. Read them like a map: they tell you which bar comes next.

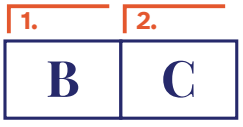
THE SECRET

Follow the map: jump back at the sign, stop at Fine, or jump to the Coda.



You play: A → B → A → B → C

Two dots facing back mean: go back and play it again

SIGN	NAME	WHAT YOU DO
	**repeat signs**	play the bars between them twice (with only the end sign: go back to the start)
	**1st and 2nd time bars**	first time play the 1. bar; after the repeat skip it and play the 2. bar
D.C.	**da capo**	“from the head”: go back to the very beginning
	**D.S.** (dal segno)	“from the sign”: go back to the segno
Fine	**fine**	“the end”: after a D.C. or D.S., stop here
	**coda**	“tail”: at “To Coda”, jump to the coda sign and play the ending

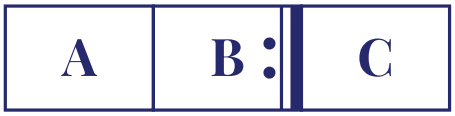
THE RULE

Follow the map in order. At a jump (D.C., D.S.) go back; then the first time you meet Fine, stop; or at To Coda, jump to the coda.

PART II · TIME · 8 · CONTINUED

EXAMPLE 1

In what order do you play these bars?



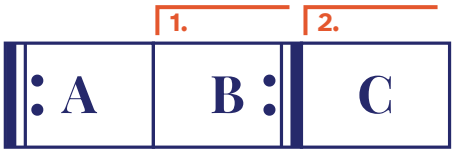
You play: ? ? ? ? ?

1. **Play A and B.**
2. **B ends with a repeat sign** facing back. There is no start sign: where do you go back to?
3. **Play it again,** then carry on.

Back to the very start: **A B A B C**.

EXAMPLE 2

First and second time



You play: ? ? ? ?

1. **First time through:** play A, then the bar marked 1.
2. **The repeat sign** sends you back to A.
3. **Second time:** which bar do you skip, and which do you play?

A B, A C: **A B A C**. The 1. bar is skipped the second time.

GAME

How many bars do you play?

1	2	3
|: A B :|	A |: B C :| D	A B (Fine) C D.C.

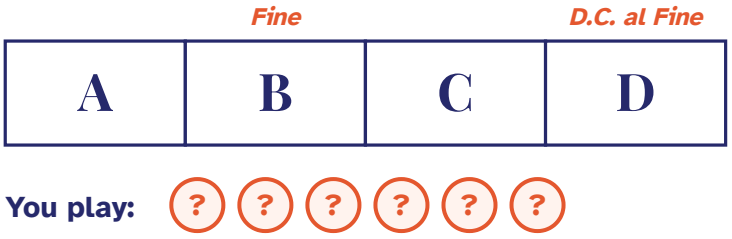
Each letter is one bar. Follow the signs and count every bar you play.

Answer: 1 four (A B A B) · 2 six (A B C B C D) · 3 five (A B C A B).

PART II · TIME · 8 · CONTINUED

EXAMPLE 3

D.C. al Fine

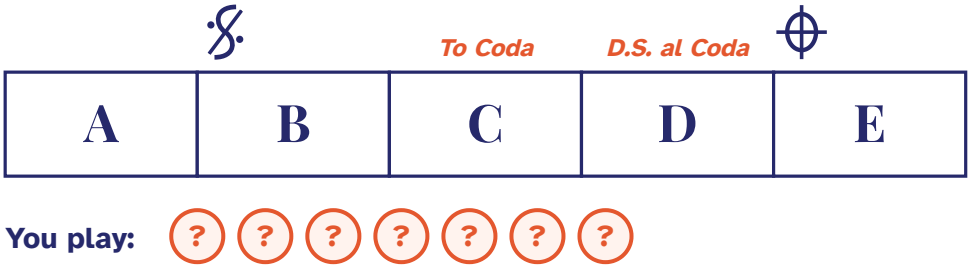


1. **Play straight through** to D.
2. **D.C.** means go back: to where?
3. **al Fine:** where do you stop the second time?

A B C D, then from the head to Fine: **A B C D A B**.

EXAMPLE 4

D.S. al Coda



1. **Play through** to D. (Ignore "To Coda" the first time.)
2. **D.S.** sends you back to the sign: which bar has it?
3. **Play on to "To Coda",** then jump. Where?

A B C D, back to the sign, B C, jump to the coda: **A B C D B C E**.

TRY IT · 10 SECONDS



Clap it, then play it.

Clap A (four beats), clap B (four beats), back to A, B again. Say the letters as you go.

TRAP

- × Taking the repeat again after a D.C.: on the way back, skip the repeats (unless the music says otherwise).
- × Missing "To Coda" the first time through: it only works after the D.S. or D.C.

SEE ALSO

Pulse and the bar p. 46 · Time signatures p. 55

PRACTISE IT

Online: Library › Signs

PART II · TIME · TIPS

When the rhythm slips

A rhythm that slips is nearly always a beat that was not counted. Find which beat, count it out loud, and the rest of the bar usually falls into place.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
It speeds up or slows down	**the pulse**: count one bar aloud before you start	p. 46
A rest gets skipped	**the rest**: it is counted just like a note	p. 49
Long notes come out short	**the hold**: count every beat of a minim, dot or tie	p. 52
The bar feels too long or too short	**the time signature**: how many counts, which note is one	p. 55
An off-beat note lands on the beat	**the "&"**: keep the beat in your foot and count it	p. 64

Four ways to steady a rhythm

1. **Count out loud**
   Say the counts while you play, not in your head: a slip you can hear is a slip you can fix.

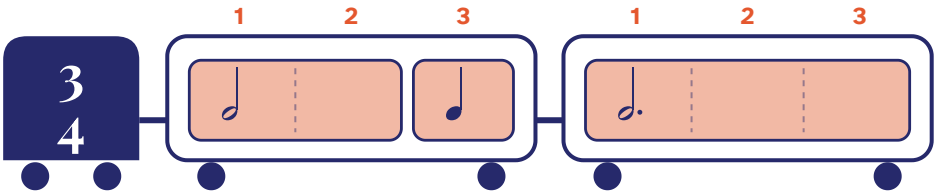
2. **Tap, then play**
   Tap the rhythm on the lid and say the counts; play only when the tapping is steady.

3. **Walk the beat**
   Step the pulse while you clap the rhythm, so the beat lives in your feet.

4. **Split the beat**
   For dots and quavers, count "1 & 2 &": the extra "&" shows exactly where each note goes.

EXAMPLE

Count the long notes



1. **Say 1 2 3** once through before you play, to feel the beat.
2. **The minim books seats 1 and 2:** keep saying "2" while it sounds.
3. **The dotted minim books all three:** let go on the next "1", not before.

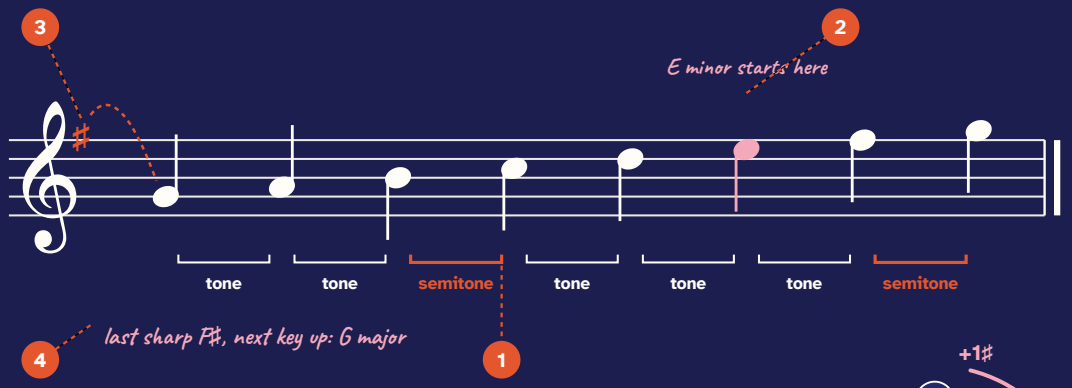
Long notes are where counting slips: **say every seat** they book.

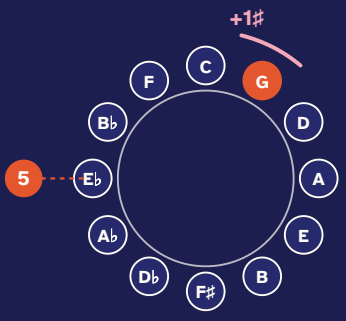
THINK OF IT Like a dance partner: the beat leads, the rhythm follows.

PART III · NOTES IN FAMILIES

Scales and keys

Notes travel in families. Build the scales that whole pieces are made from, then learn to read the key a piece lives in: its signature, its home note, its place on the circle. All of it is in one scale.





1 **Major scales** p. 72
T T S T T T S: the two short steps

2 **Minor scales** p. 75
the major with 3, 6 and 7 one key lower

3 **Keys and key signatures** p. 78
the sharps at the start apply all the way

4 **Finding the key** p. 81
last sharp, next key up; second-last flat

5 **The circle of fifths** p. 84
go right: one more sharp each step

★ **Tips: when a note sounds wrong** p. 87

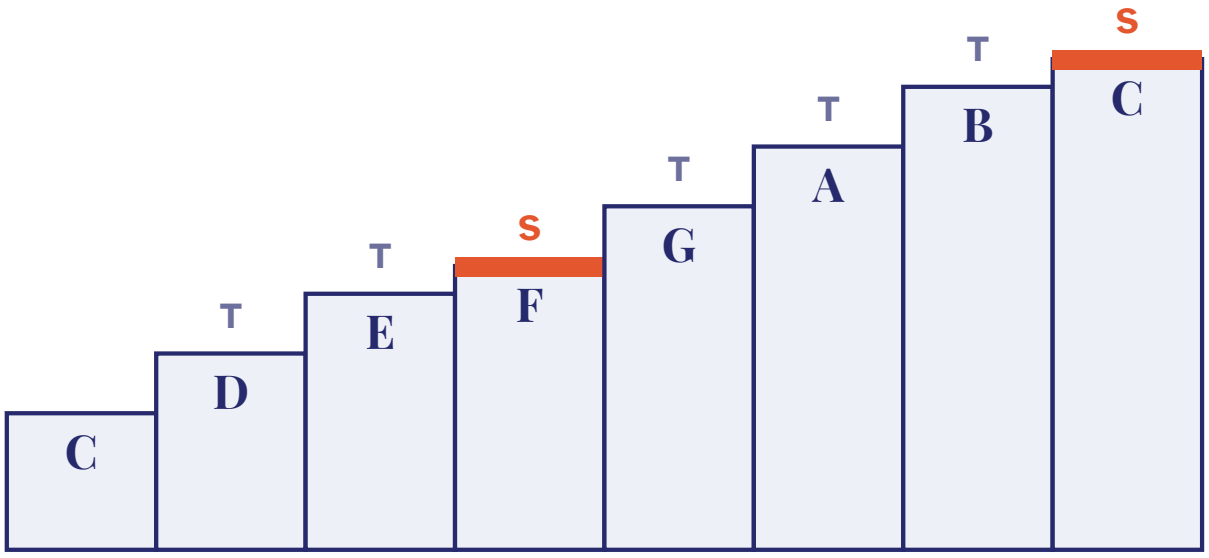
PART III · SCALES AND KEYS · 1

Major scales

A **scale** climbs from a home note, the **key note**, to the next note with the same name, using every letter once. A **major** scale always takes the same steps, whatever note it starts on.

THE SECRET

Every major scale: tone, tone, semitone, tone, tone, tone, semitone.



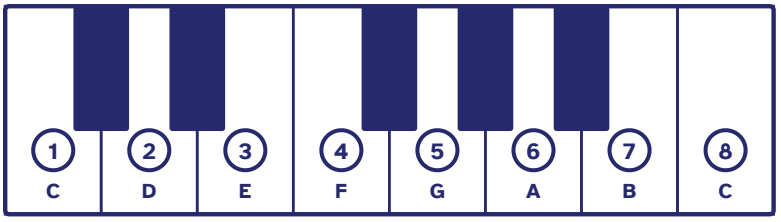
Tone, tone, semitone, tone, tone, tone, semitone

THE RULE

T = tone = one key between. S = semitone = no key between: the very next key, best friends. Every major scale: T T S T T T S.

EXAMPLE 1

Where are the short steps?



?

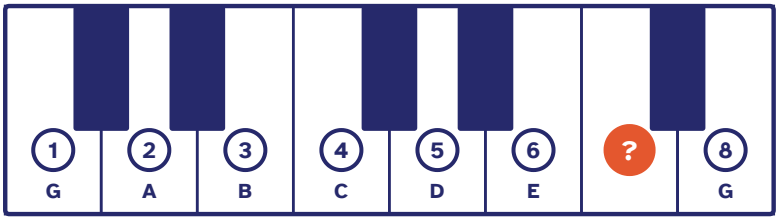
1. **Play C up to C** on the white keys, numbering the notes 1 to 8.
2. **Between each pair,** look for a black key. A key between means a tone.
3. **Which pairs have no black key** between them?

Only **3–4** (E–F) and **7–8** (B–C): nothing between. Those are the two short steps of every major scale.

PART III · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 2

Build G major

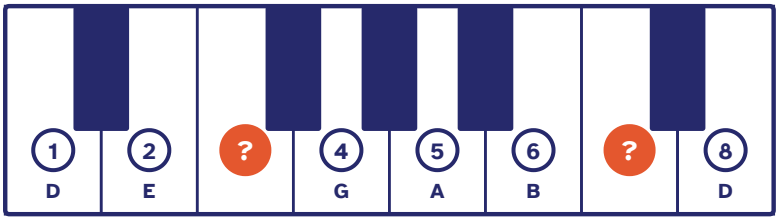


1. **Write the letters** G to G: G A B C D E F G.
2. **Check the top:** 6 to 7 must be T (one key between), 7 to 8 must be S (none between). E to F has nothing between.
3. **Move the ? one key right.** What is it called now?

The **?** becomes **F♯**. G major: G A B C D E F♯ G, one sharp.

EXAMPLE 3

Build D major

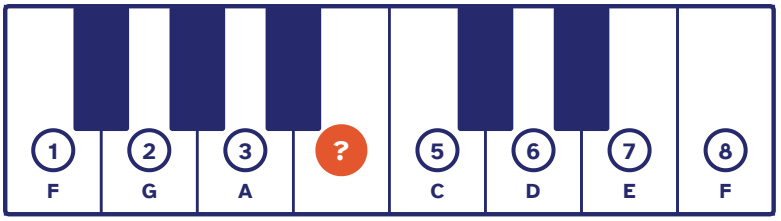


1. **Write the letters** D to D.
2. **Check each step** against T T S T T T S: is there a key between?
3. **Two steps are too short.** Move each ? one key right.

D major: D E **F♯** G A B **C♯** D, two sharps.

EXAMPLE 4

Build F major



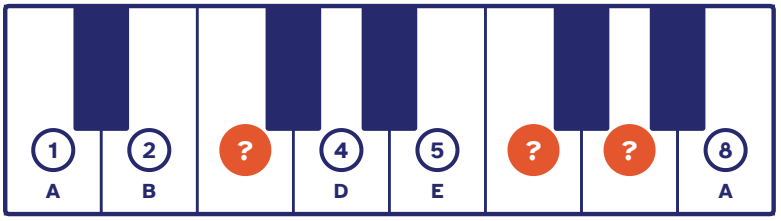
1. **Write the letters** F to F: F G A B C D E F.
2. **Check 3 to 4:** it must be S, nothing between. A to B has a key between: too big.
3. **Move the ? one key left.** What is it called now?

The **?** becomes **B♭**. F major: F G A B♭ C D E F, one flat.

PART III · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 5

Build A major



1 **Write the letters** A to A.

2 **Check each step:** three of them have nothing between where they need a key.

3 **Move each ? one key right.**

A major: A B **C♯** D E **F♯** **G♯** A, three sharps.

GAME

Spot the wrong note

1 G A B C D E F G

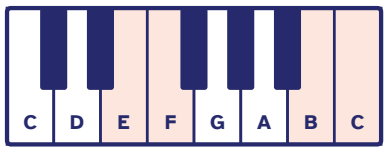
2 D E F♯ G A B C D

3 F G A B C D E F

Each of these major scales has one note that breaks T T S T T T S. Which one, and what should it be?

Answer: 1 F should be F♯ · 2 C should be C♯ · 3 B should be B♭.

TRY IT · 10 SECONDS
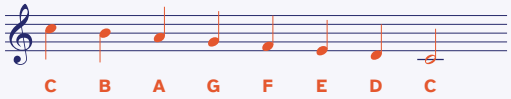


One finger, C to C.

Play the white keys from C up to C and say “tone, tone, semitone, tone, tone, tone, semitone”. Feel the two short steps.

YOU KNOW THIS

Joy to the World



The first line is the **C major scale**, top to bottom: C B A G F E D C.

Try it. Play C major down from the top C and sing “Joy to the world” over it.

TRAP

- × Using the same letter twice: every letter appears once (F♯ in G major, never G♭).
- × Forgetting the short step at the top: 7 to 8 never has a key between.

SEE ALSO

Minor scales p. 75 · Keys and key signatures p. 78

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Scales

PART III · SCALES AND KEYS · 2

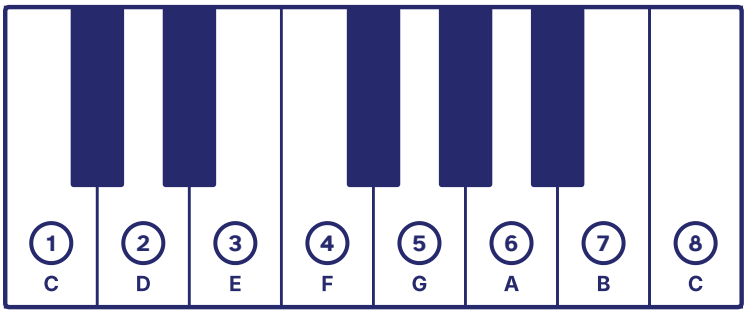
Minor scales

Start from the major scale on the same key note and move three notes, the **3rd**, **6th** and **7th**, one key down. That is the **natural minor**: same home, darker colour.

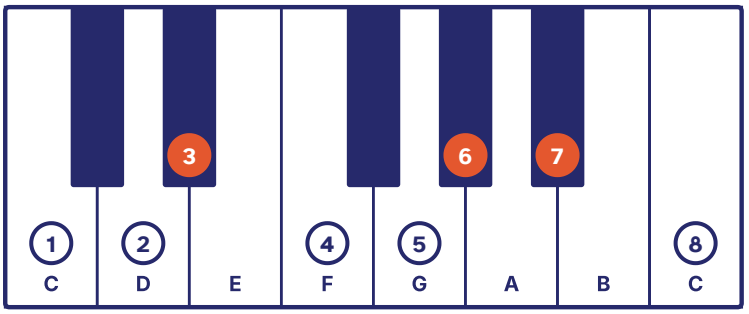
THE SECRET

Minor = the major on the same note with 3, 6 and 7 one key lower.

C major



C minor: 3, 6 and 7 move one key left
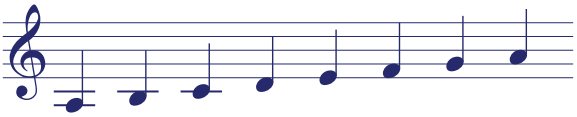


The same home note · three notes one key lower

THE RULE
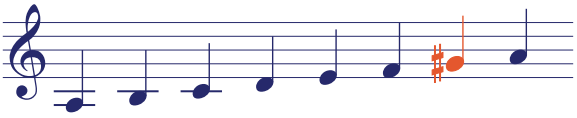
Natural minor = the major with 3, 6 and 7 lowered one key. Harmonic minor = the natural minor with 7 raised again. Melodic minor = going up, 6 and 7 raised; coming down, the natural minor.

A minor in three forms

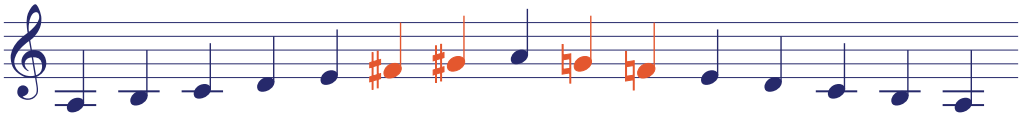


Natural: A major (F♯ C♯ G♯) with 3, 6 and 7 lowered: C, F and G. All white keys.

PART III · SCALES AND KEYS · 2 · CONTINUED



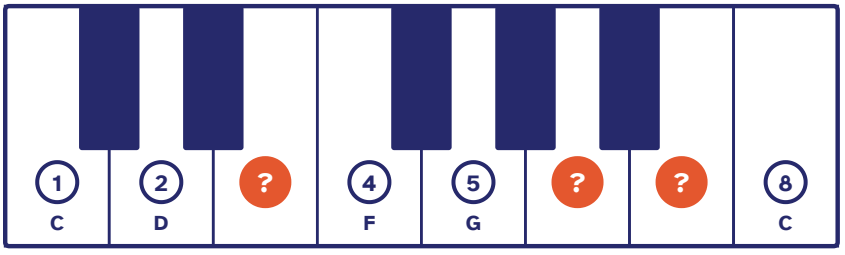
Harmonic: the 7th raised again, one key up: **G♯**.



Melodic: going up, 6 and 7 raised (**F♯ G♯**); coming down, back to **G F**.

EXAMPLE 1

Build C natural minor
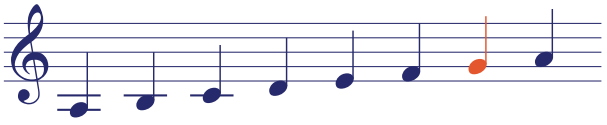


1. **Start from C major:** C D E F G A B C.
2. **The ? keys are 3, 6 and 7.** Move each one key left.
3. **Name the keys you land on.**

C natural minor: C D **E♭** F G **A♭** **B♭** C.

EXAMPLE 2

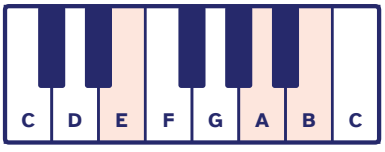
Build A harmonic minor



1. **Start from A natural minor:** A B C D E F G A, all white.
2. **Take note 7** (the ?) and raise it one key.
3. **Name it.**

Note 7 becomes **G♯**: A harmonic minor is A B C D E F G♯ A.

TRY IT · 10 SECONDS



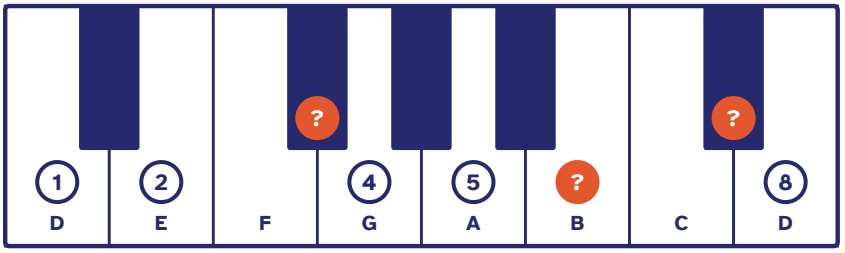
Major, then minor.

Play C major, then play it again with E, A and B one key lower. Same home, new colour: bright, then dark.

PART III · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 3

Build D natural minor



1 **This is D major.** The ? keys are notes 3, 6 and 7.

2 **Move 3, 6 and 7** one key left each.

3 **Name the new keys.**

D natural minor: D E **F** G A **B♭** **C** D.

GAME

Which minor?

1

nothing raised

2

the 7th raised

3
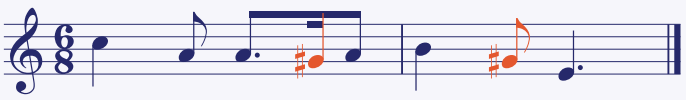
6th and 7th raised going up, lowered coming down

Three forms of the minor scale. Name each one.

Answer: 1 natural · 2 harmonic · 3 melodic.

YOU KNOW THIS

Greensleeves



A minor, with a **G♯**: the raised 7th of the harmonic minor, leaning hard up to A.

Try it. Play it once with G♯, once with G: hear the pull disappear.

TRAP

- × Lowering the wrong notes: only 3, 6 and 7 move.
- × Playing the melodic minor's raised notes on the way down.

SEE ALSO

Major scales p. 72 · Keys and key signatures p. 78

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Scales

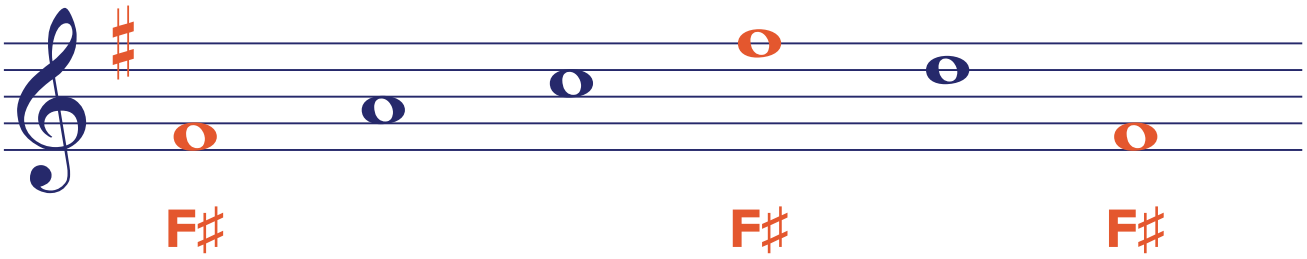
PART III · SCALES AND KEYS · 3

Keys and key signatures

Why write F♯ fifty times? The **key signature** at the start of each line says it once, for the whole piece. Its sharps or flats always line up in the same order.

THE SECRET

The key signature says once, at the start, which notes are always sharp or flat.



One sharp at the start: every F plays F♯



Sharps always come in this order: F C G D A E B.



Flats: B E A D G C F, the sharps backwards.

To remember the order: **F**ather **C**harles **G**oes **D**own **A**nd **E**nds **B**attle; backwards for the flats: **B**attle **E**nds **A**nd **D**own **G**oes **C**harles' **F**ather.

KEY YOU BUILT	KEY SIGNATURE
C major	none
G major	1♯: F♯
D major	2♯: F♯ C♯
F major	1♭: B♭

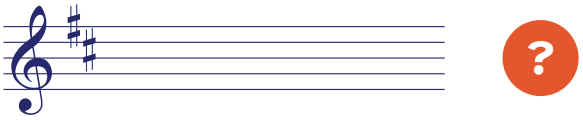
One signature, two keys

Every key signature also belongs to a **minor** key: the **relative minor**. To find it, go **down a minor 3rd**: 2 keys between, then land. C major and A minor share every note: all white keys.

PART III · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 1

Which notes are sharp?

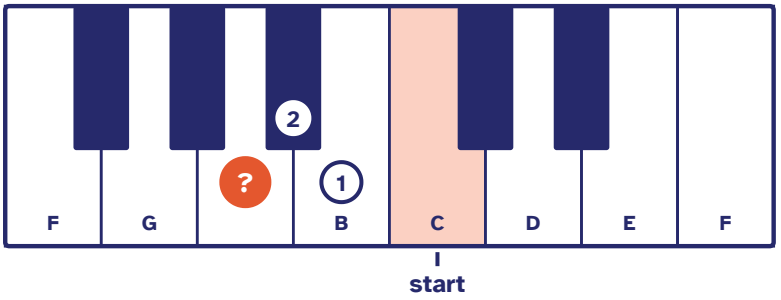


1. **Count the sharps.**
2. **Name them in order:** the first sharp is always F, then ...
3. **Which keys turn black** for the whole piece?

Two sharps, **F♯ and C♯**: every F and every C in the piece, high or low.

EXAMPLE 2

Find the relative minor of C major

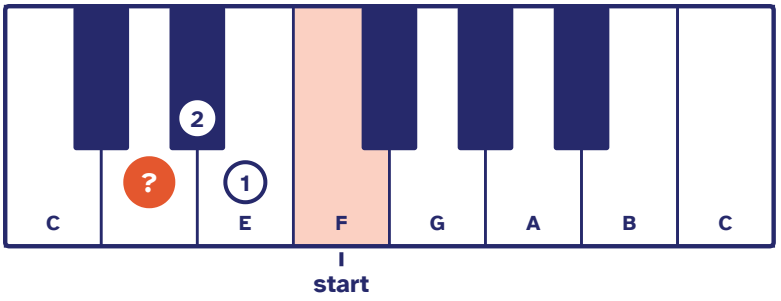


1. **Start on C**, the key note.
2. **Count 2 keys between, going down:** B is 1, B♭ is 2 ...
3. **The next key is the ?.**

You land on **A**: the relative minor of C major is **A minor**.

EXAMPLE 3

Find the relative minor of F major



1. **Start on F.**
2. **Count 2 keys between, going down:** E is 1, E♭ is 2 ...
3. **The next key is the ?.**

The **?** is **D**: D minor, which shares F major's one flat, B♭.

PART III · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 4

Which notes are flat?



1 **Count the flats.**

2 **Name them in order:** the first flat is always B, then …

3 **Which keys turn black?**

Three flats, **B♭ E♭ A♭**: every B, E and A in the piece.

GAME

Which comes next?

1 — sharps: F C G …

2 — flats: B E …

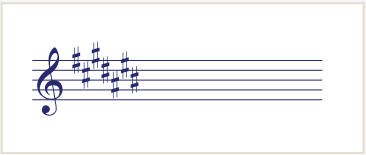
3 — the 1st flat

4 — the 2nd sharp

Use the orders of sharps and flats.

Answer: 1 D · 2 A · 3 B♭ · 4 C♯.

TRY IT · 10 SECONDS
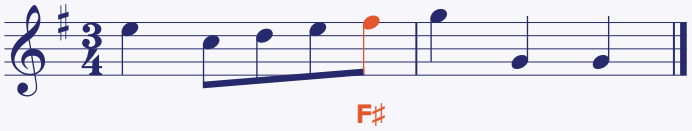


Say the sharps in order.

"Father Charles Goes Down And Ends Battle": F C G D A E B. Then backwards for the flats: B E A D G C F.

YOU KNOW THIS

Minuet in G



One sharp in the key signature: every F is **F♯**, like the one just before the top G. One sharp, and the piece ends on G: **G major**.

Try it. Find every F in the piece and play it as F♯.

THINK OF IT A dress code on a party invitation: written once at the top, it applies to every guest all night. Only an accidental, a word to one guest, overrules it, and only until the bar line.

TRAP

- × Writing a key signature in the wrong order: sharps always F C G D A E B.
- × Reading it once and forgetting it: glance at the key signature at the start of every line.

SEE ALSO

Major scales p. 72 · Finding the key p. 81 · The circle of fifths p. 84

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Keys

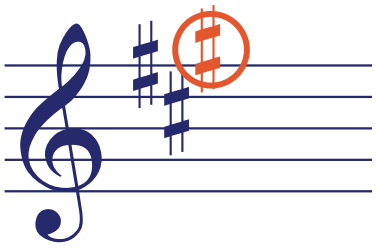
PART III · SCALES AND KEYS · 4

Finding the key

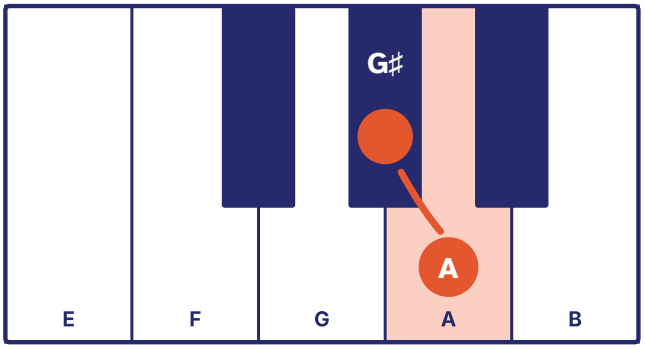
A key signature names two keys, a major and its relative minor. Two quick rules find the major; a minor 3rd down finds the minor; then the music itself tells you which of the two it is.

THE SECRET

Last sharp, then the next key up. Second-last flat: that is the key.



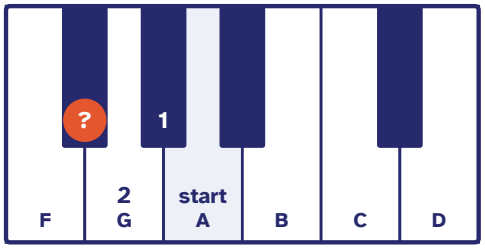
last sharp: G♯



Sharps: the last sharp, then the very next key up. Three sharps: A major



Flats: the second-last flat is the key.
B♭ E♭ A♭: the second-last is E♭, so **E♭ major**. One flat alone is F major.



Major to minor: down a minor 3rd.
From A: 2 keys between, G♯ and G, then land on **F♯**. Three sharps is A major **or** F♯ minor.

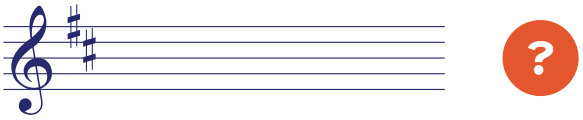
Minor keys and their clue notes

MINOR KEY	SIGNATURE	CLUE NOTE (THE RAISED 7TH)
A minor	none	G♯
E minor	1♯	D♯
D minor	1♭	C♯
B minor	2♯	A♯
G minor	2♭	F♯

PART III · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 1

Name the major key: sharps



1. **Find the last sharp,** the one furthest right.
2. **Name it** (F first, then C ...).
3. **Go to the very next key up.** That is the key.

The last sharp is C♯; the next key up is D: **D major**.

EXAMPLE 2

Name the major key: flats
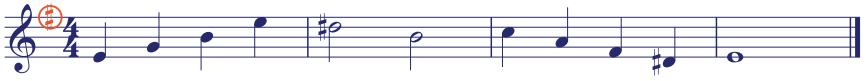


1. **Find the second-last flat.**
2. **Name it** (B first, then E ...).
3. **That flat names the key.**

B♭ E♭: the second-last is B♭: **B♭ major**.

EXAMPLE 3

Major or minor?



1. **The key signature:** one sharp. Which major, and which minor a minor 3rd below it?
2. **Look for a sharp that is not in the signature.** A minor key raises its 7th: whose 7th is it?
3. **Look at the last note.**

One sharp, D♯ clues, ending on E: **E minor**, not G major.

GAME

Name the major key

1	2	3	4
2 sharps	4 sharps	1 flat	3 flats

Sharps: the last sharp, the very next key up. Flats: the second-last flat (one flat alone is F).

Answer: 1 D · 2 E · 3 F · 4 E♭.

PART III · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 4

Find a minor key from its clue



?

1. **Two flats:** B♭ major, or the minor a minor 3rd below it.
2. **Count 2 keys between going down** from B♭: A, A♭ … and land. Which minor key is it, and what is its 7th note?
3. **Raise that 7th**: if the piece is full of it, the key is minor.

Two flats with F♯ clues: **G minor** (its 7th, F, raised to F♯).

TRY IT · 10 SECONDS



Three sharps, stop, step up.

Say “F C G”, stop: the last one is G♯. The next key up is A. Three sharps is A major. Try it with two and four.
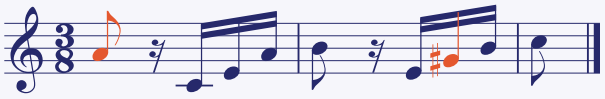
HOW TO

Find the key of a new piece

1. **The key signature:** name the major key, and its relative minor a minor 3rd below.
2. **The clue notes:** look for the minor key’s raised 7th.
3. **The last bass note:** pieces nearly always end on the key note.

YOU KNOW THIS

Für Elise (Beethoven)



No key signature, a G♯ as the clue, and the music keeps coming home to A: **A minor**.

Try it. Play A, then G♯ A: hear the G♯ lean into home.

THINK OF IT Detective work: the key signature narrows it to two suspects; the clue note and the last note name the culprit.

SEE ALSO
Keys and key signatures p. 78 · The circle of fifths p. 84

PRACTISE IT
Workbook Scales and keys 1 · Online: Library › Keys

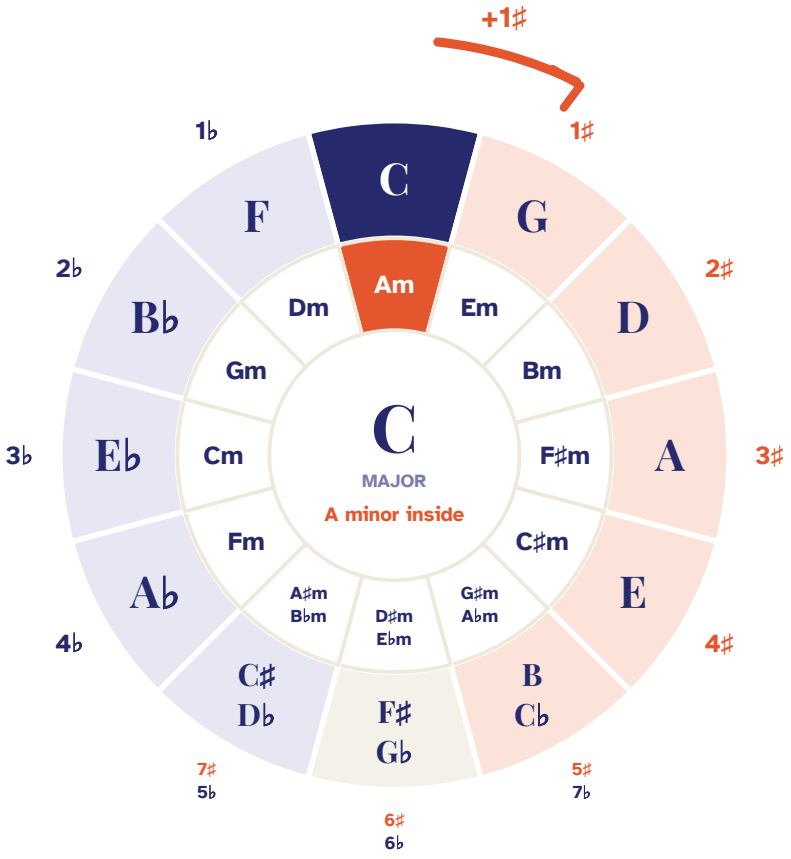
PART III · SCALES AND KEYS · 5

The circle of fifths

All the keys on one clock face. Go **right** round the circle and each key has **one more sharp**. Go **left** and each key has **one more flat**.

THE SECRET

Go right: one more sharp. Go left: one more flat.



Right: one more sharp · left: one more flat

How to read the clock

Outer ring: the major keys. **Inner ring:** their relative minors. **Outside the circle:** how many sharps or flats.

Neighbours are close

Keys side by side share all but one note, so music moves between them easily: from C major to G major or F major, or to the relative minor.

THE RULE

Each step right is a 5th up: count the notes, C D E F G, and G is the next key round. That is why it is called the circle of fifths.

PART III · SCALES AND KEYS · 5 · CONTINUED

The first steps round the circle

STEPS	RIGHT (SHARPS)	LEFT (FLATS)
1	G major: F♯	F major: B♭
2	D major: F♯ C♯	B♭ major: B♭ E♭
3	A major: F♯ C♯ G♯	E♭ major: B♭ E♭ A♭

EXAMPLE 1

How many sharps in A major?

C (none) › G (1♯) › D (2♯) › A (?)

1. **Start at C**, at the top: no sharps.
2. **Go right** one step at a time: G, D …
3. **Count the steps** to A.

Three steps right: A major has **three sharps**, F♯ C♯ G♯.

EXAMPLE 2

How many flats in E♭ major?

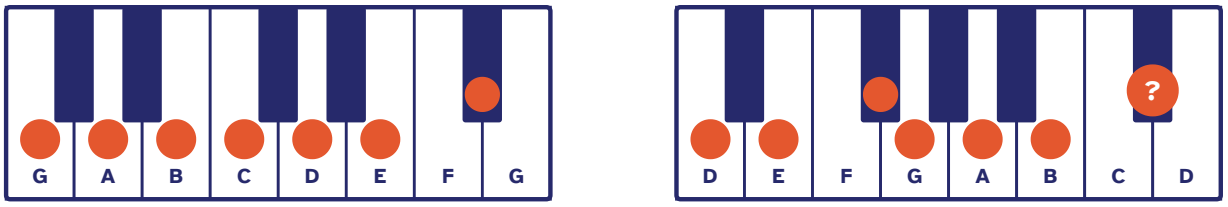
C (none) › F (1♭) › B♭ (2♭) › E♭ (?)

1. **Start at C** again.
2. **Go left:** F, B♭ …
3. **Count the steps** to E♭.

Three steps left: E♭ major has **three flats**, B♭ E♭ A♭.

EXAMPLE 3

Why do G and D sound so close?



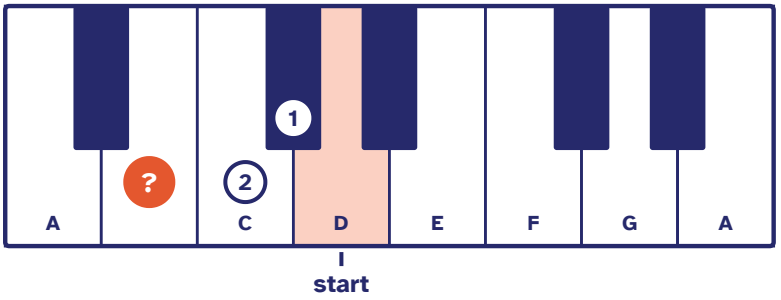
1. **G major** (left) and **D major** (right) are neighbours on the clock.
2. **Compare their notes:** which letters are the same?
3. **Which one letter changes?**

Six notes are the same; only C becomes **C♯**. One step round the circle changes just one note.

PART III · SCALES AND KEYS · 5 · CONTINUED

EXAMPLE 4

Find the minor inside D



1. **Start on D**, on the outer ring.
2. **Count 2 keys between, going down:** C♯ is 1, C is 2 …
3. **The next key is the ?.** It sits on the inner ring, just inside D.

The **?** is **B**: B minor, D major's relative minor. Both have two sharps.

GAME

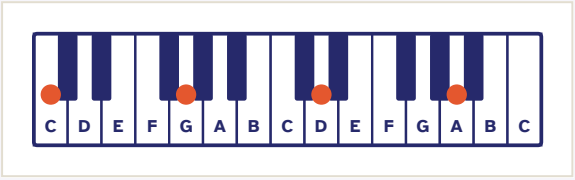
Neighbours on the clock

1	2	3	4
right of D	left of C	opposite C	inside G

Right adds a sharp. Inside each major sits its relative minor.

Answer: 1 A · 2 F · 3 F♯ / G♭ · 4 E minor.

TRY IT · 10 SECONDS



Walk the circle: C G D A.

Play each key note with one finger, a 5th up each time. You just walked three steps right round the circle.

HOW TO

Learn the circle by heart

1. **Right, out loud:** C G D A E B F♯.
2. **Left:** C F B♭ E♭ A♭ D♭ G♭.
3. **Check yourself:** the number of steps from C is the number of sharps or flats.

TRAP

- × Going the wrong way round: right adds sharps, left adds flats.
- × Forgetting the relative minors on the inner ring.

SEE ALSO

Keys and key signatures p. 78 · Finding the key p. 81 · Chords in a key p. 95

PRACTISE IT

Workbook Scales and keys 1 · Online: Library › Related keys

PART III · SCALES AND KEYS · TIPS

When a note sounds wrong

You read the note right, yet it sounds sour. The slip is usually in the key, not the note. Name the slip first, then fix only that note.
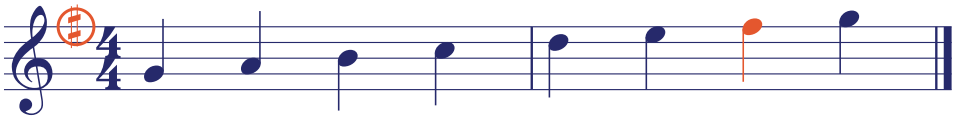
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
A note sounds sour, though you read it right	**the key signature**: is this note sharp or flat all the way through?	p. 78
A key with many sharps or flats feels lost	**the circle**: go right for a sharp, left for a flat	p. 84
A minor tune sounds wrong near the end	**the raised 7th**: harmonic and melodic minor raise it	p. 75
A scale sounds wrong in the middle	**the short steps** (nothing between): 3–4 and 7–8 in every major scale	p. 72
A sharp is missing from the key signature	**the order of sharps**: F C G D A E B	p. 78
You are not sure which major key it is	**the last sharp**: the key is the very next key up	p. 81

EXAMPLE

Find the slip



1. **Read the key signature:** one sharp, on the top line, F (ringed).
2. **The red note is an F**, so the sharp belongs to it too: every F, high or low.
3. **No natural cancels it**, so this F is F♯, here and all through the piece.

Play **F♯**, the black key. Plain F is the most common slip in G major.

HOW TO

Three checks before you play

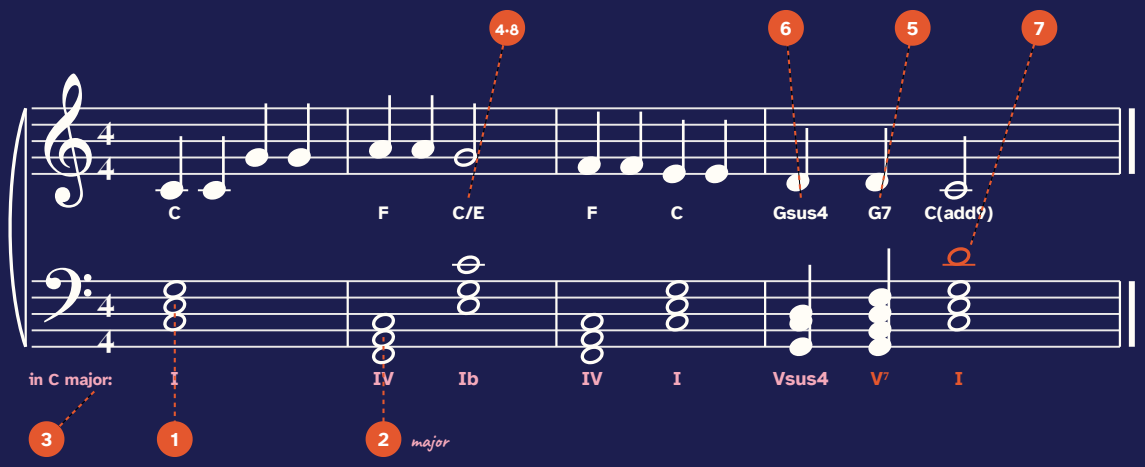
1. **The key signature:** say its sharps or flats aloud.
2. **The key:** name it from the last sharp or the second-last flat, then check the last note.
3. **The scale:** play it once, so your fingers know which black keys belong.

THINK OF IT Like proofreading: a spelling mistake jumps out once you know which words you usually get wrong. Keep a short list of your own slips, and check those first.

PART IV · BUILD THEM

Chords

Press three notes at once and you have a chord. Stack them, name them, turn them upside down, add layers on top. Every chapter of this part is in these four bars of Twinkle.



1 **Chords and triads** p. 89
three notes stacked like a layer cake

2 **Four kinds of triad** p. 92
major 3 + 2, minor 2 + 3

3 **Chords in a key** p. 95
Roman numerals: I, IV and V do most of the work

4 **Inversions** p. 98
the same notes, a different one at the bottom

5 **Seventh chords** p. 101
one more layer on top: G7

6 **Colour chords** p. 104
sus and add: a note moved or added

7 **The tower: 9, 11, 13** p. 107
keep skipping a letter

8 **Slash chords** p. 110
C/E: the chord, then the bass note

★ **Tips: when a chord sounds wrong** p. 113

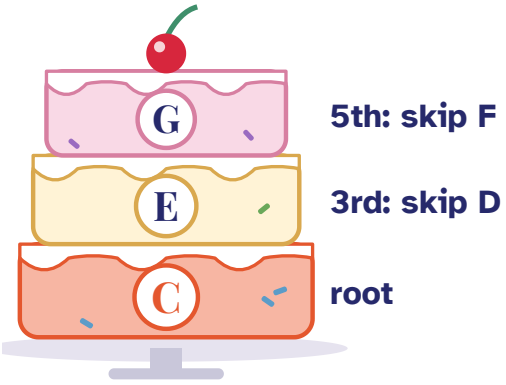
PART IV · CHORDS · 1

Chords and triads

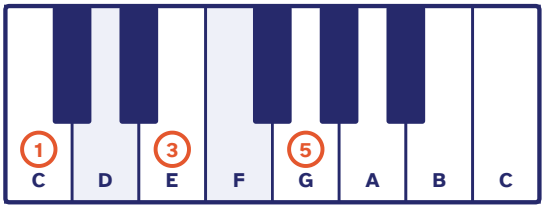
Press three or more notes at once and you have a **chord**. The favourite of all is the **triad**: three notes stacked like a layer cake.

THE SECRET

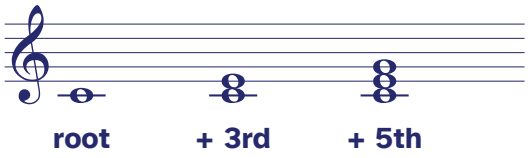
Play one, skip one, play one: the cake rule.



1. Pick a note: the **root**, the big layer at the bottom. It holds the chord up and gives it its name.
2. Skip one letter, take the next: **E**, the **3rd**.
3. Skip again: **G**, the **5th**. C E G is the triad of C.



On the keys: play one, skip one (shaded), play one, skip one, play one.



The cake on the staff: three lines in a row (or three spaces).

On the staff, each layer sits on the **next line up** (or the next space up): the root, a **3rd** above it, a **5th** above the root.

First met: Level ½ (the tonic triad of C, G and F major).

All lines or all spaces

A stacked triad sits on **three lines in a row** or in **three spaces in a row**, in a neat stack. If one note breaks the pattern, it is not a stacked triad.



Lines, spaces, lines, spaces: every one a triad.

PART IV · CHORDS · 1 · CONTINUED

Solid and broken

Played together, a chord is **solid** (or blocked). Played one note at a time, it is **broken**. Spread up and down the keyboard, a broken chord becomes an **arpeggio**. Left-hand accompaniments, the arpeggios in technical work and the chord names over a song are all triads.

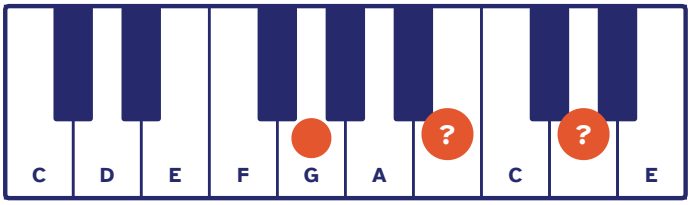
First met: Level 0; arpeggios from Level 1.



The same three notes, solid then broken.

EXAMPLE 1

Build a triad on G

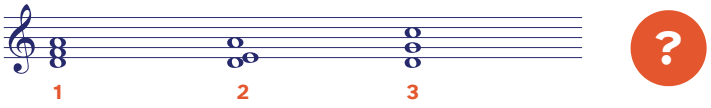


1. **Start on G** (the red dot): the root, the bottom layer.
2. **Skip one white key,** take the next: the first ?.
3. **Skip again,** take the next: the second ?.

G **B D**: the triad of **G**. Same shape as C E G, just moved up.

EXAMPLE 2

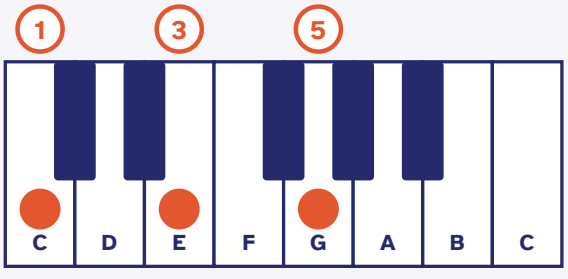
Which one is a triad?



1. **Look at each chord:** are the three notes all on lines, or all in spaces?
2. **Count the notes** from each note to the next one up: is every gap a 3rd?
3. **Only one chord** passes both tests. Which?

Chord 1 (D F A, three spaces in a row). Chord 2 has a 2nd (D–E); chord 3 is stacked in 4ths.

TRY IT · 10 SECONDS



Thumb on C. Fingers 1, 3, 5 down together.

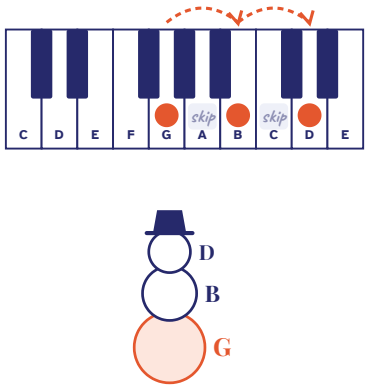
That is the C chord. Slide your hand to any white key and do it again: the cake shape always works.

PART IV · CHORDS · 1 · CONTINUED

GAME

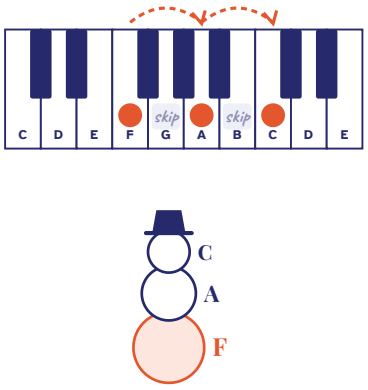
Build one yourself

G



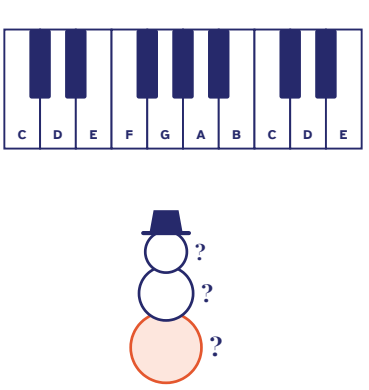
Start on G: **G B D**.

F



Start on F: **F A C**.
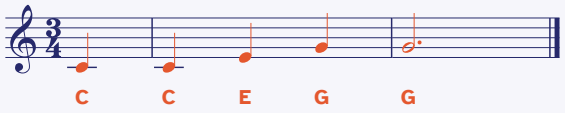
?



Your turn: start on **A**. Which three keys?

Build the cake on A: play one, skip one, play one, skip one, play one.

Answer: A C E: the triad on A.

YOU KNOW THIS

The Blue Danube (Johann Strauss II), here in C



The most famous waltz tune simply climbs the **C triad**: **C, C E G, G**. Root, 3rd, 5th, then it holds the top: that is why it sounds so open.

Try it. Play C E G one at a time, then hum the waltz over it.

THINK OF IT Solid: three friends shout “Hello!” together. Broken: each says it in turn.

TRAP

- × Stacking 2nds or 4ths: a triad skips exactly one letter each time.
- × Naming a chord from its top note: the root names it (at the bottom when the notes are stacked in 3rds).

SEE ALSO
Counting intervals p. 38 · Major scales p. 72

PRACTISE IT
Workbook Harmony 1 · Online: Library › Chords

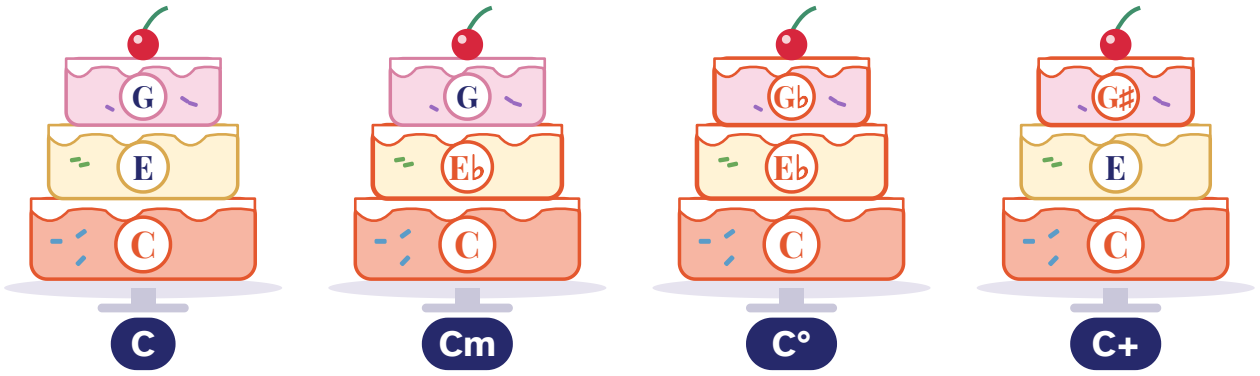
PART IV · CHORDS · 2

Four kinds of triad

Every triad is two 3rds stacked. To tell the kinds apart, count the keys **in between**, black and white, that you do not play: root to 3rd, then 3rd to 5th. Those two numbers name the kind, and its sound.

THE SECRET

Count the keys between: 3 + 2 is major, 2 + 3 is minor.

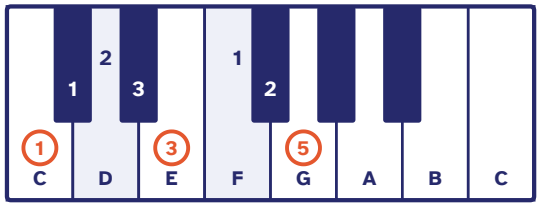


Count the keys between: 3 + 2 is major, 2 + 3 is minor

KIND	KEYS BETWEEN	ON C	CHORD NAME	FIRST MET
major	**3 + 2**	C E G	C	Level ½
minor	**2 + 3**	C E♭ G	Cm	Level 2
diminished	**2 + 2**	C E♭ G♭	C° or Cdim	Level 5
augmented	**3 + 3**	C E G♯	C+ or Caug	Level 6

Chord names: a letter alone means major (C); m means minor (Cm). You see them above the melody in songbooks and lead sheets: C | Am | F | G. In interval names, 3 keys between is a major 3rd and 2 is a minor 3rd (p. 41).

Count the keys in between



Play C and E and count the keys you skip, black and white: C♯, D, D♯ is **3**. Then E to G: F, F♯ is **2**. **3 + 2 is major.** Any root works the same way: from D, skip 3 to F♯, then 2 to A: D F♯ A, D major.

PART IV · CHORDS · 2 · CONTINUED

The seven white-key triads, sorted



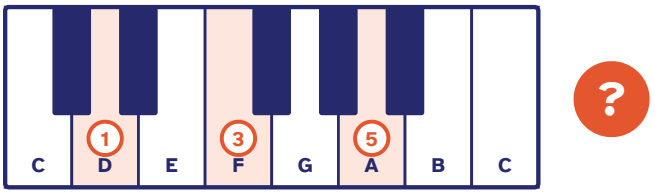
Always the same on the white keys: **three major** (C, F, G), **three minor** (D, E, A), **one diminished** (B), no augmented. Which ones belong to a key is the next chapter (p. 95).

How they sound

KIND	SOUNDS	HEARD IN
major	bright, settled, open	most cheerful tunes, fanfares
minor	darker, softer, sadder	lullabies, laments, film tension
diminished	tense, unstable, wants to move	suspense, spooky scenes
augmented	floating, dreamy, unsettled	dream scenes, a passing colour

EXAMPLE 1

Major or minor? D F A



1. **Root to 3rd:** count the keys between D and F, black and white.
2. **3rd to 5th:** count the keys between F and A.
3. **3 + 2 or 2 + 3?**

D to F has 2 between, F to A has 3: **2 + 3**, so D F A is **D minor** (Dm).

GAME

Which kind?

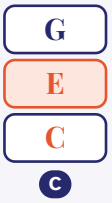
1	2	3
G♭	A	B♯
E♭	F♯	G♯
C	D	E

Count the keys between: 3 + 2 major, 2 + 3 minor, 2 + 2 diminished, 3 + 3 augmented.

Answer: 1 diminished · 2 major · 3 augmented.

PART IV · CHORDS · 2 · CONTINUED

TRY IT · 10 SECONDS

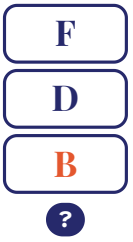


C E G, then C E♭ G, then C E♭ G♭.

Bright, dark, tense. Only the middle layer moves first, then the top one. Listen for the change each time.

EXAMPLE 2

What kind is the triad on B?
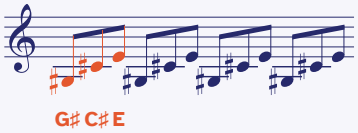


1. **Skip a letter:** B, D, F.
2. **Count the keys between** B and D, then D and F.
3. **Which kind has those two numbers?**

2 + 2: B D F is **B diminished** (B°), the only diminished triad on the white keys.

YOU KNOW THIS

Beethoven, Moonlight Sonata (first movement)



The famous slow triplets are the notes of a **C♯ minor triad**: C♯, E and G♯, rocking gently. Minor is why it sounds so dark and sad.

THINK OF IT Two gaps in the cake: a wide one (3 keys between) and a narrow one (2). Major: wide, then narrow, 3 + 2. Minor: narrow, then wide, 2 + 3. Diminished: two narrow, 2 + 2. Augmented: two wide, 3 + 3.

TRAP

- × Counting the keys you play: count only the keys in between, black and white.
- × Forgetting that in diminished and augmented triads the 5th changes too.
- × Spelling with the wrong letter: a triad skips one letter, so C minor is C E♭ G, never C D♯ G.

SEE ALSO

Chords and triads p. 89 · Major, minor and perfect p. 41

PRACTISE IT

Workbook Harmony 2 · Online: Library › Chords

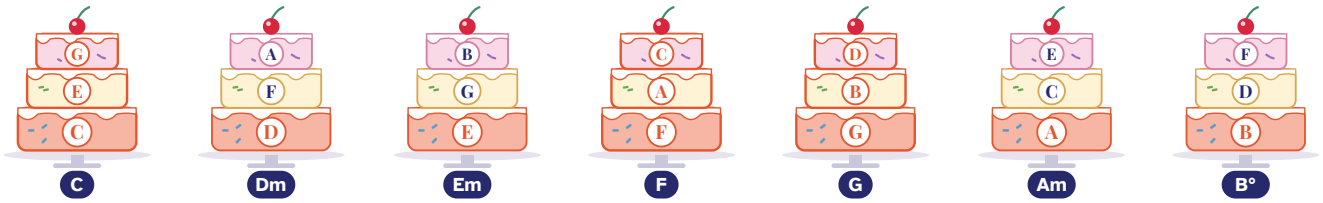
PART IV · CHORDS · 3

Chords in a key
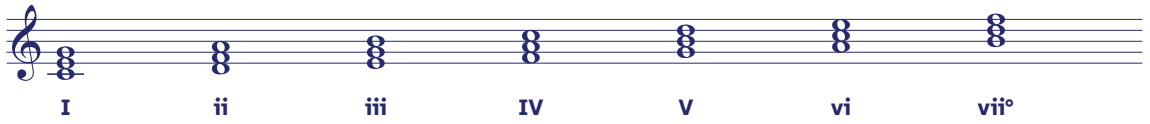
Build a triad on every note of a scale and you have the chords of that key. **Roman numerals** name each one by its place in the scale: I on the 1st note, IV on the 4th, V on the 5th. Capitals are major, small numerals are minor, and small with ° is diminished.

THE SECRET

Build a cake on every note of the scale: I, IV and V are the strongest.



One cake on every note: I, IV and V do most of the work



The seven chords of C major. Only white keys, so the kinds come out as on p. 93.

NUMERAL	NAME	KIND	KEYS BETWEEN	IN C	IN G
I	tonic	major	3 + 2	C	G
ii	supertonic	minor	2 + 3	Dm	Am
iii	mediant	minor	2 + 3	Em	Bm
IV	subdominant	major	3 + 2	F	C
V	dominant	major	3 + 2	G	D
vi	submediant	minor	2 + 3	Am	Em
vii°	leading note	diminished	2 + 2	B°	F♯°

The same in every major key: I, IV and V are major; ii, iii and vi are minor; vii° is diminished. First met: I, IV and V at Levels 1–2; the other chords from about Level 4.

The three primary chords
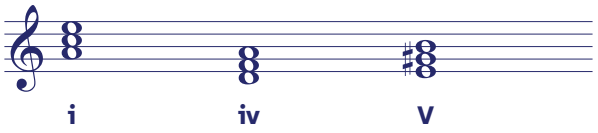
I, IV and V are the strongest chords of a key. Between them they hold **every note of the scale**: C E G, F A C, G B D. That is why so many tunes need only these three.

THINK OF IT I is the key's own chord. IV and V are its two big neighbours. Between the three of them, every note of the scale has a chord to sit in.

PART IV · CHORDS · 3 · CONTINUED

In a minor key

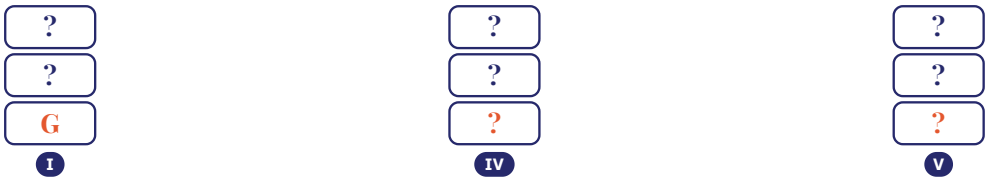
In A minor the three main chords are **i** (A minor), **iv** (D minor) and **V**: E G♯ B, made major by the **raised 7th** of the harmonic minor (p. 75), G♯. Check: E G♯ B is 3 + 2, major.



A minor: i, iv and V. The G♯ makes V major.

EXAMPLE 1

Find I, IV and V in G major



1. **Count up the scale** from G: which notes are 1st, 4th and 5th?
2. **Build a cake on each,** using the key signature (one sharp: F♯).
3. **Check the keys between:** are all three 3 + 2?

In G major: **I = G** (G B D), **IV = C** (C E G), **V = D** (D F♯ A), all major.

GAME

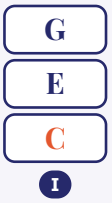
Which numeral?



All in C major. Count up the scale from C to find each chord's number.

Answer: 1 IV · 2 V · 3 vi · 4 ii.

TRY IT · 10 SECONDS



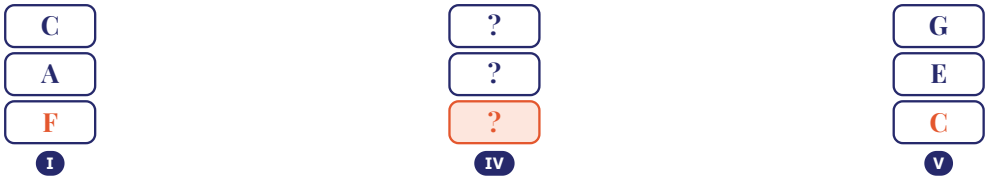
Left hand: C, F, G.

Play the cakes of I, IV and V in C major, then sing any nursery tune over them: these three fit almost every note.

PART IV · CHORDS · 3 · CONTINUED

EXAMPLE 2

Find IV in F major



1. **Count up the scale** from F to the 4th note. Watch the key signature: one flat.
2. **Build the cake** on that note.
3. **Check the keys between.**

IV in F major is **B♭**: B♭ D F, 3 + 2, major.

EXAMPLE 3

Find V in D minor



1. **The 5th note** of D minor is A: build A C E.
2. **The 7th note** of the D minor scale is C. In the harmonic minor it is raised: to what?
3. **Put it in the middle layer.** What kind is the chord now?

C becomes **C♯**: V in D minor is A C♯ E, **A major**.

EXAMPLE 4

The chord on note 6 of C major



1. **Build the cake on A:** skip one, take the next, twice.
2. **Count the keys between:** major or minor?
3. **Note 6 of the scale:** which numeral, capital or small?

A C E, 2 + 3: **A minor**, numeral **vi** (small, because it is minor).

TRAP

- × Writing every numeral big: ii, iii and vi are minor, so they are written small.
- × Forgetting the raised 7th in a minor key: V in A minor is E G♯ B, not E G B.

SEE ALSO

Four kinds of triad p. 92 · Major scales p. 72 · Inversions p. 98

PRACTISE IT

Workbook Harmony 3 · Online: Library › Chords

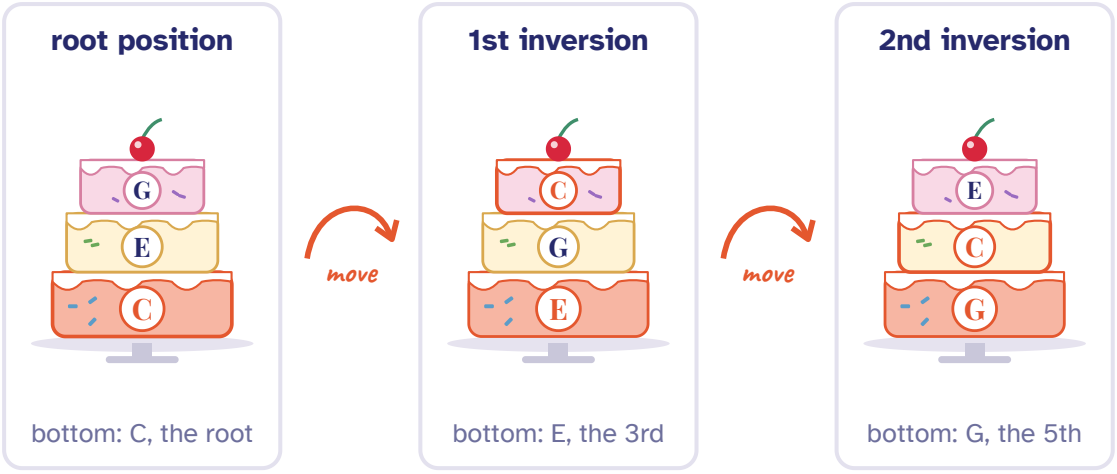
PART IV · CHORDS · 4

Inversions
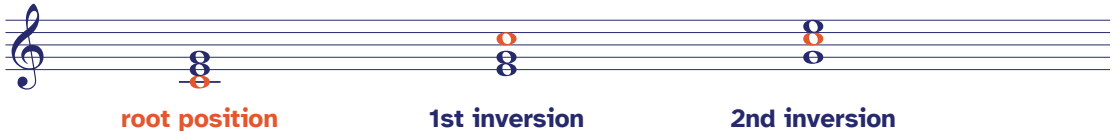
Same three notes, a different one at the bottom. The chord keeps its name; only the bass changes. The red layer is the root, C: watch it travel.

THE SECRET

Move the bottom note to the top: same chord, new bass.



1. Start with the cake **C E G**: root at the bottom.
2. Move the bottom layer to the top: **E G C**. That is the 1st inversion.
3. Move it again: **G C E**, the 2nd inversion. Still **C major**.



The same three chords on the staff; the red note is the root, C. To find the root, restack the notes every other letter (like C E G): then the bottom layer is the root.

Why inversions?

They let your hand **move as little as possible**: keep any note two chords share, and move the others to the nearest note of the next chord. Left-hand accompaniments and hymn parts are written this way.

GAME

Which position?

1	2	3
C E G	E G C	G C E

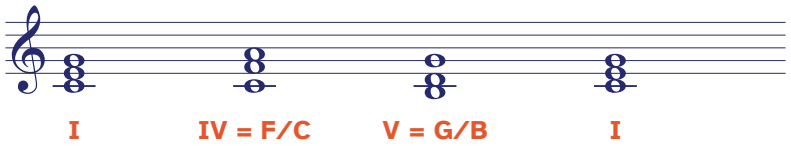
Same three notes. Look at the bottom one: root, 3rd or 5th?

Answer: 1 root position · 2 first inversion (E at the bottom) · 3 second inversion (G at the bottom).

PART IV · CHORDS · 4 · CONTINUED

EXAMPLE 1

Move as little as possible: C, F, G, C



1. **I:** C E G. **IV:** keep C, move E up to F and G up to A: C F A.
2. **V:** no shared note. C steps down to B, F down to D, A down to G: B D G.
3. **I again:** B up to C, D up to E, G stays: C E G.

F stands on C, G stands on B: **inversions** keep the hand in one place.

TRY IT · 10 SECONDS



Climb the chord.

Play C E G, then E G C, then G C E, then C E G an octave up. Same cake, changing its bottom layer as it climbs.

EXAMPLE 2

Write F major in first inversion



1. **Build F major** in root position first.
2. **First inversion:** which layer goes to the bottom, the 3rd or the 5th?
3. **Stack the other two** above it, in order.

The 3rd, A, goes to the bottom: **A C F**.

EXAMPLE 3

Find the second inversion of G



1. **Build G major:** skip one, take the next, from G.
2. **Second inversion:** the 5th goes to the bottom.
3. **Stack the other two** above it.

The 5th, D, at the bottom: **D G B**.

PART IV · CHORDS · 4 · CONTINUED

EXAMPLE 4

Find the root of E G C



1. **Restack the notes** so that each skips one letter: try starting on each note.
2. **Which order works?** C E G, E G C or G C E?
3. **The bottom layer** of that stack is the root.

C E G works: the root is **C**. E G C is C major, 1st inversion.

EXAMPLE 5

Which position? A D F
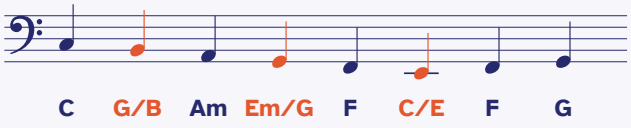


1. **Restack it** so each note skips one letter.
2. **Find the root** (the bottom layer of the stack) and the kind.
3. **Which layer is at the bottom** of A D F: root, 3rd or 5th?

Restacked: D F A, **D minor**. A, the 5th, is at the bottom: **2nd inversion**.

YOU KNOW THIS

Pachelbel's Canon, the bass (here in C)



In the popular version the bass walks **straight down the scale**: C B A G F E. It can, because some chords are **inversions**: G stands on B, Em on G, C on E.

Try it. Play the bass slowly and hum the famous violin tune over it.

TRAP

- × Naming a chord from its bass note: in an inversion the bass is not the root. Restack it in 3rds first.
- × Thinking an inversion is a new chord: E G C is still C major.

SEE ALSO

Chords and triads p. 89 · Chords in a key p. 95 · Slash chords p. 110

PRACTISE IT

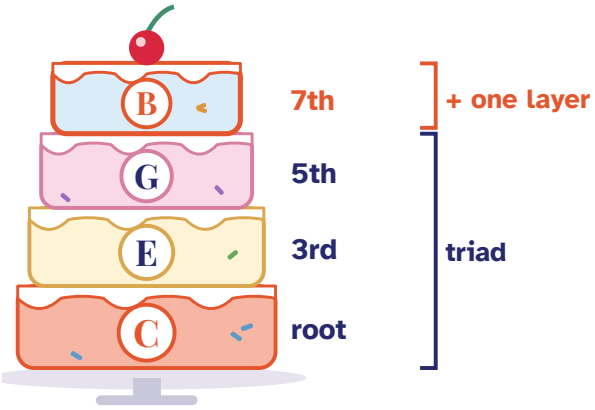
Workbook Harmony 4 · Online: Library › Chords

PART IV · CHORDS · 5

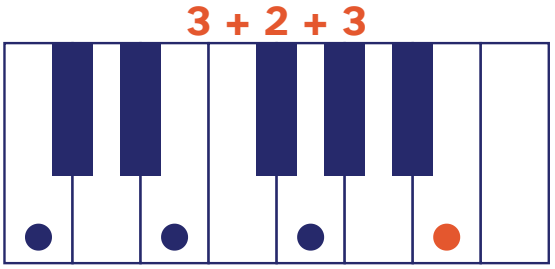
Seventh chords

Add one more layer to the cake and the chord changes colour: dreamy, bluesy, spooky. Same rule as before: skip one note, stack the next. The new note is the **7th**.

THE SECRET
One more layer on the cake and the chord changes colour.



1. Build the triad: **C E G**.
2. Skip one, add the next: **B**.
3. Count the keys between: **3 + 2 + 3**. That is **Cmaj7**.
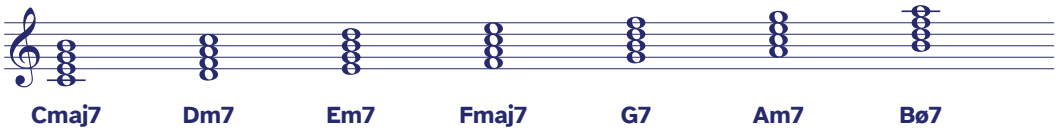


Five flavours, all on C

Cmaj7	C7	Cm7	Cø7	C°7
3 + 2 + 3	3 + 2 + 2	2 + 3 + 2	2 + 2 + 3	2 + 2 + 2
C E G B	C E G B♭	C E♭ G B♭	C E♭ G♭ B♭	C E♭ G♭ B𝄫
dreamy	**bluesy**	**mellow**	**searching**	**spooky**
bossa nova, film	the blues	soul, jazz ballads	jazz, film noir	old film villains

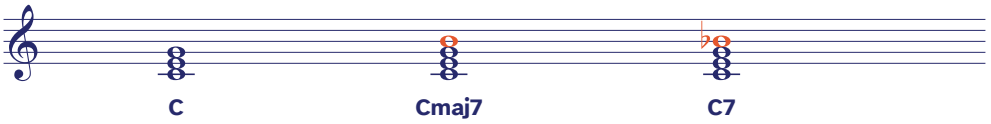
How the names work: major triad + major 7th = **maj7** · major triad + minor 7th = **7** · minor triad + minor 7th = **m7** · diminished triad + minor 7th = **ø7** · diminished triad + diminished 7th = **°7**.

A seventh on every note of C major



Only white keys: count the keys between to name each one.

PART IV · CHORDS · 5 · CONTINUED

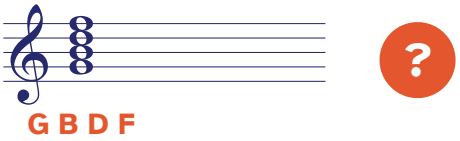


One note decides: B makes Cmaj7, B♭ makes C7. Everything below stays the same.

Spot the difference

EXAMPLE 1

Which seventh is it? G B D F



1. **Count the keys between** each layer: G to B, B to D, D to F.
2. **Which of the five flavours** has those three numbers?
3. **The root is G.** Name the chord.

3 + 2 + 2, the bluesy one, on G: **G7**.

GAME

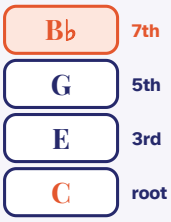
Which seventh?

1	2	3
F A C E	C E G B♭	D F A C

A cake with one more layer. Name each chord.

Answer: 1 Fmaj7 · 2 C7 · 3 Dm7.

TRY IT · 10 SECONDS



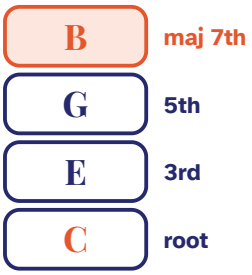
Play C7, then Cmaj7.

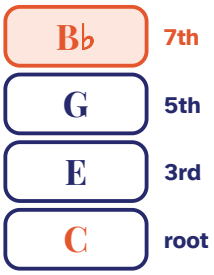
C E G B♭, then C E G B. One key changes the top layer: bluesy, then dreamy.

PART IV · CHORDS · 5 · CONTINUED

EXAMPLE 2

Cmaj7 or C7: which note changes?



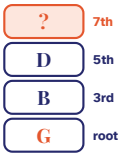


1. **Both start with C major:** C E G.
2. **Compare the top layers.** Which one note is different?
3. **Count the keys between** G and the top layer in each.

Only the top layer changes: **B** (3 between) is Cmaj7, **B♭** (2 between) is C7.

EXAMPLE 3

Build G7

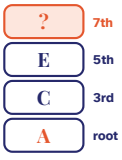


1. **Build G major:** G B D.
2. **Skip one more letter** above D: which?
3. **Count the keys between** D and it: 2 (the bluesy 7) or 3 (maj7)?

G B D **F**: 3 + 2 + 2, so it is **G7**.

EXAMPLE 4

Build Am7



1. **Build A minor:** A C E (2 + 3).
2. **Add the next letter** but one above E.
3. **Count the keys between** E and the new layer.

A C E **G**: 2 + 3 + 2, so it is **Am7**.

TRAP

- × Writing C7 when you mean Cmaj7: "7" alone means the bluesy one, with B♭.
- × Stacking the 7th from the wrong note: count a 7th up from the root, not from the 5th.

SEE ALSO
Four kinds of triad p. 92 · Chords in a key p. 95

PRACTISE IT
Workbook Harmony 5 · Online: Library › Chords

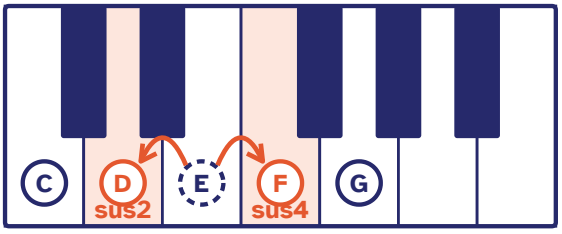
PART IV · CHORDS · 6

Colour chords

The 3rd is the chord's mood: major or minor, happy or sad. Slide it aside, take it away, or add one note, and a plain chord takes on a new colour. Five colours, all from C E G.

THE SECRET

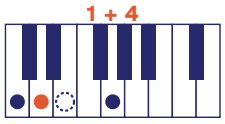
Move the 3rd, take it away, or add one note: five new colours.



1. Start from **C E G**. The **E** (the 3rd) makes it major.
2. Slide the E up to the next letter, **F**: **Csus4**.
3. Or down to the letter below, **D**: **Csus2**. With no 3rd, the chord is neither major nor minor: it **hangs in the air**.

Five colours, all on C

Csus2

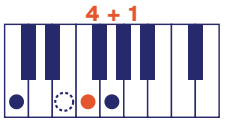


C D G

open

the 3rd steps down

Csus4

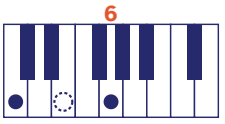


C F G

floating

the 3rd steps up

C5

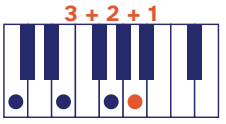


C G

strong

no 3rd at all

C6

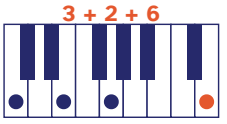


C E G A

warm

add the 6th

Cadd9

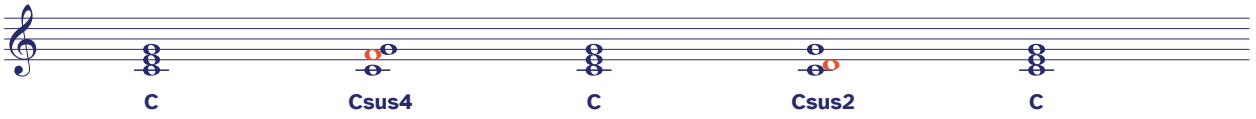


C E G D

sparkly

add the 9th on top

Lean, then slide back



The red note leans, then slides back to E: F steps down, D steps up. Play it slowly: C, Csus4, C, Csus2, C. Guitarists love this shimmer.

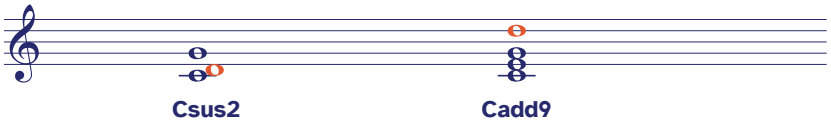
On any root, the counts stay the same

COLOUR	KEYS BETWEEN	ON C	ON G	ON F	ON D
sus2	1 + 4	C D G	G A D	F G C	D E A
sus4	4 + 1	C F G	G C D	F B♭ C	D G A
5	6	C G	G D	F C	D A
6	3 + 2 + 1	C E G **A**	G B D **E**	F A C **D**	D F♯ A **B**
add9	3 + 2 + 6	C E G **D**	G B D **A**	F A C **G**	D F♯ A **E**

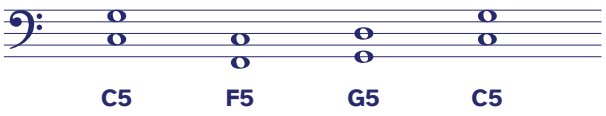
PART IV · CHORDS · 6 · CONTINUED

sus: the 3rd held up or down · **5**: no 3rd at all · **6** and **add9**: one note added on top. Csus means Csus4.

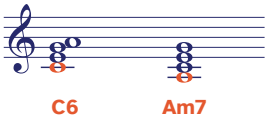
Spot the difference



The same red D, two jobs. Csus2 throws the E out; Cadd9 keeps the E and puts the D on top.



The power chord: just the root and the 5th, played low in the left hand. With no 3rd it is neither happy nor sad, only **strong**: the sound of rock guitar. Play C5, F5, G5, C5 and drum the rhythm.



C6 or Am7? The same four notes, C E G A. The red bottom note decides: C at the bottom is **C6**; A at the bottom is **Am7**. More on the bass note in Slash chords, p. 110.

EXAMPLE 1

Which colour is it? G C D



1. **Count the keys between:** G to C, then C to D.
2. **Find the card** with the same two numbers.
3. **Is there a 3rd (B)?** What has taken its place?

4 + 1, no B: G C D is **Gsus4**. Swap the C for **B** and you have G major.

GAME

sus2, sus4 or add9?

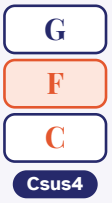
1	2	3
C D G	C F G	C E G D

In a sus chord the 3rd has moved; in an add chord nothing moved and a note was added.

Answer: 1 Csus2 · 2 Csus4 · 3 Cadd9.

PART IV · CHORDS · 6 · CONTINUED

TRY IT · 10 SECONDS

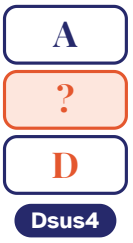


Csus4, then C.

Play C F G, then C E G. No 3rd, then a 3rd: floating, then bright.

EXAMPLE 2

Build Dsus4

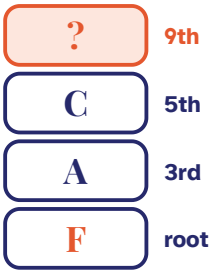


1. **Start with D major:** D F♯ A.
2. **sus4:** move the middle layer up to the next letter.
3. **Which key is it?**

F♯ moves up to **G**: Dsus4 is **D G A**.

EXAMPLE 3

Build Fadd9



1. **Start with F major:** F A C, and keep every note.
2. **The 9th** is the 2nd, an octave up: which letter comes just after F?
3. **Put it on top.**

Fadd9: F A C **G**. Nothing moved; one note was added.

THINK OF IT Decorating the cake: a ribbon (the 6th) or a candle (the 9th) and it is still the same cake. Sus moves its middle layer; a power chord takes the middle layer out.

TRAP

× Calling Cadd9 a ninth chord: C9 also has the 7th (B♭). Add9 is just the triad plus the 9th (D on C), nothing else (The tower, p. 107).

SEE ALSO
Four kinds of triad p. 92 · Seventh chords p. 101

PRACTISE IT
Workbook Harmony 6 · Online: Library › Chords

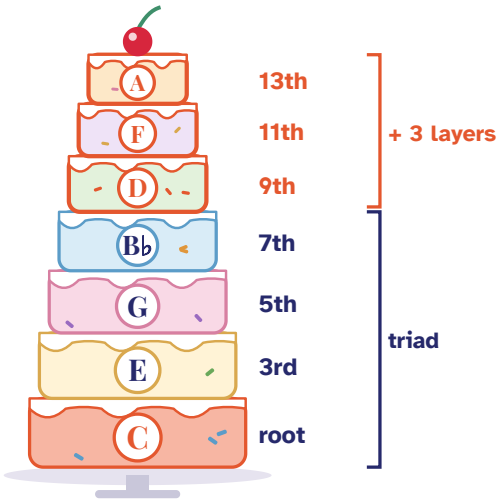
PART IV · CHORDS · 7

The tower: 9, 11, 13

Why stop at the 7th? Keep stacking 3rds and the cake grows three more layers: the 9th, the 11th and the 13th. At the top you have used **all seven letters, A to G**.

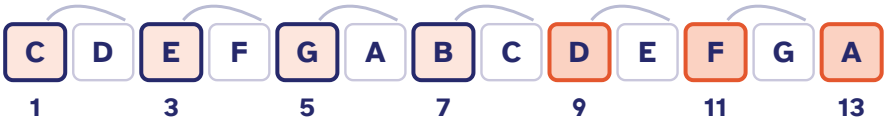
THE SECRET

Keep skipping letters above the 7th: 9, 11, 13 are 2, 4, 6 an octave up.



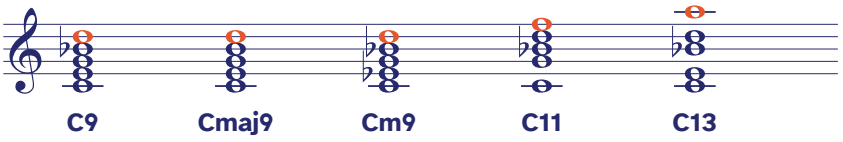
1. Build the seventh chord: **C E G B♭** (C7).
2. Keep skipping one letter: **D**, **F**, **A**. Those layers are the **9th, 11th and 13th**.
3. Take 7 away to find the letter: 9 → 2nd (D), 11 → 4th (F), 13 → 6th (A). Same letters, an octave up.

Skip a letter, all the way up



Start on C and take every other letter: C E G B D F A (letters only: in C7 the B is B♭). After the 13th comes C again: the tower is full.

Five tall chords on C

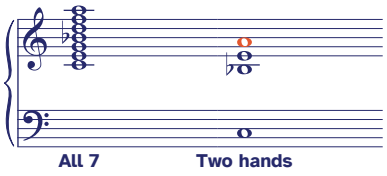


The red note is the top layer that gives each chord its name. C11 and C13 leave some notes out: see p. 108.

CHORD	BUILT AS	NOTES	SOUNDS	HEARD IN
C9	C7 + 9th	C E G B♭ **D**	soulful	funk, soul
Cmaj9	Cmaj7 + 9th	C E G B **D**	dreamy	film, ballads
Cm9	Cm7 + 9th	C E♭ G B♭ **D**	smooth	neo-soul, R&B
C11	C9 + 11th, no 3rd	C G B♭ D **F**	airy	gospel, pop
C13	C7 + 13th	C E B♭ (D) **A**	rich	jazz, blues endings

PART IV · CHORDS · 7 · CONTINUED

Too many notes? Leave some out



All seven notes sound muddy. Two hands play C13 with four: C in the left hand, B♭ E A in the right.

Ten fingers, seven layers: you choose. **Keep** the 3rd and the 7th (they say what kind of chord it is) and the red top note (its colour). **Leave out** the 5th first. The right hand can drop the root too, when the left hand or a bass player plays it.

With a major 3rd, the 11th (F) rubs against E. So C13 leaves out the F. C11 does the opposite: it keeps the F and leaves out the E.

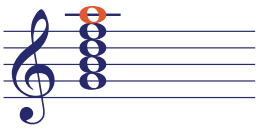
6 or 13? add9 or 9?

SAME TOP NOTE	WITHOUT THE 7TH	WITH THE 7TH (B♭)
A on C	**C6**: C E G A	**C13**: C E B♭ A
D on C	**Cadd9**: C E G D	**C9**: C E G B♭ D

Look for the 7th. No 7th: a colour chord (p. 104). A 7th: a tower chord.

EXAMPLE 1

Name this tall chord: G B D F A



G B D F A

?

1. **Find the root:** each note skips one letter, so the bottom note is the root.
2. **The first four layers:** count the keys between. Which seventh?
3. **The red top layer:** which number is it above G?

G7 (3 + 2 + 2) with A, the 9th, on top: **G9**.

GAME

Which floor?

1 — the 9th of C

2 — the 11th of C

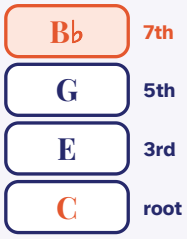
3 — the 13th of C

Build C7 (C E G B♭), then keep skipping one letter above the B♭.

Answer: 1 D · 2 F · 3 A.

PART IV · CHORDS · 7 · CONTINUED

TRY IT · 10 SECONDS

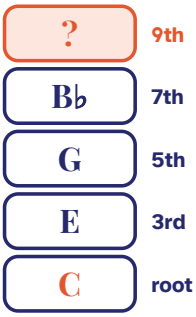


Play C7, then add one more.

Left hand C E G, right hand B♭, then add the next letter but one above it. You just built a tall chord.

EXAMPLE 2

Build C9



1. **Build C7:** C E G B♭.
2. **Skip one more letter** above B♭: which letter?
3. **Five notes:** share them between the hands if you need to.

C9: C E G B♭ **D**, one layer above the 7th.

HOW TO

Build any tall chord

1. **Start on the root** and skip one letter each time: 1 3 5 7 9 11 13.
2. **Stop** at the number in the name.
3. **Leave out** the 5th first; in a 13th chord, the 11th too, when it is too many notes for your hands.

THINK OF IT You don't need every layer. Keep the 3rd and the 7th, which say what kind of chord it is, and the top layer, which gives it its colour.

TRAP

- × Hunting far up for the 9th: it is just the 2nd, an octave higher (D on C).
- × Calling C13 a C6: the 13th chord has the 7th (B♭) as well.
- × Playing F and E together in C13 or C11: they clash. C13 drops the F; C11 drops the E.

SEE ALSO
Seventh chords p. 101 · Colour chords p. 104

PRACTISE IT
Workbook Harmony 7 · Online: Library › Chords

PART IV · CHORDS · 8

Slash chords

C/E means one chord with a chosen bass note: play a C chord, with E as the lowest note. It is not a fraction, and not two chords.

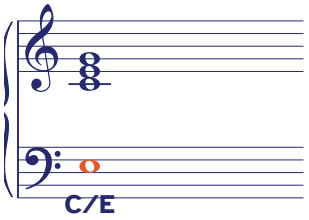
THE SECRET
Chord before the slash, bass note after it.

the chord
right hand

C/E

the bass
left hand, lowest note

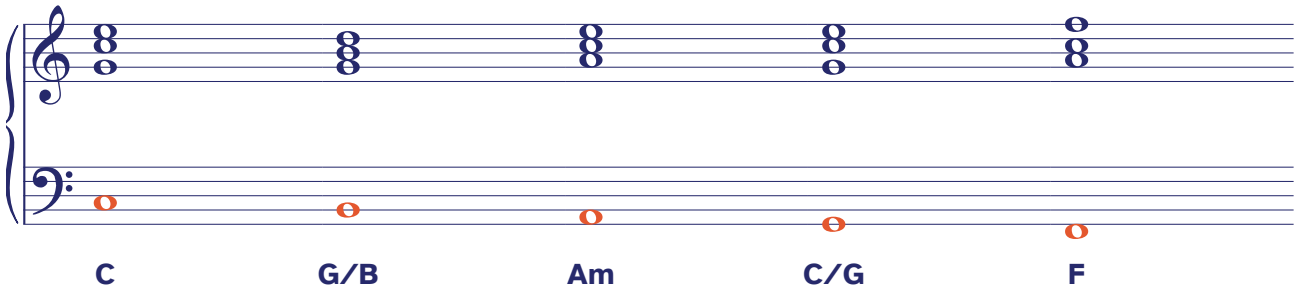
1. Before the slash: the **chord**. C means C E G, in the right hand.
2. After the slash: the **bass**. E goes in the left hand, below everything.
3. Play them together: **C/E**, “C over E”.



Four jobs for a slash chord

JOB	EXAMPLE	THE BASS NOTE IS	WHAT YOU HEAR
Inversion	C/E · C/G	a note of the chord	smoother changes (p. 98)
Walk-down	C · G/B · Am	one step down each time	a bass line that walks down
Pedal	F/C · G/C	the same note, held	the chords change above it
Colour	F/G · D/C	not in the chord	open, modern: pop, gospel

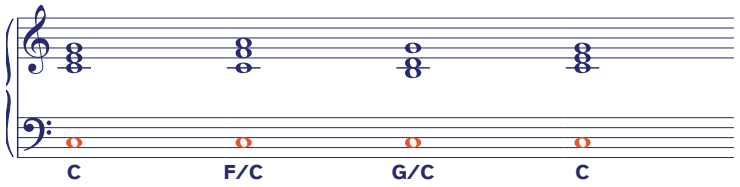
Job 2: a bass that walks down



The red bass walks straight down, C B A G F, while the right hand moves a step at most. G/B and C/G make the steps possible.

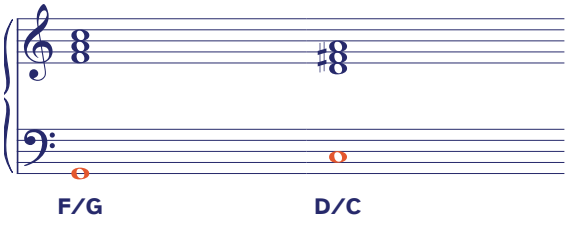
PART IV · CHORDS · 8 · CONTINUED

Job 3: a pedal, one bass under everything



The bass holds C while the chords change above it: C, F/C, G/C, C. The chords change while the C stays. (It is called a pedal after the organ's foot pedals, which hold long low notes.)

Job 4: a bass from outside the chord

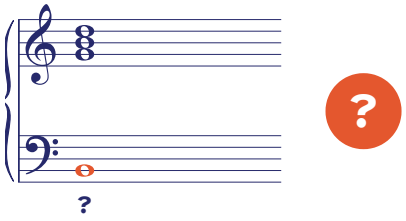


F/G: F A C over a G. G is not in the F chord, so it floats, open and unfinished. Pop and gospel players love it.

D/C: D F♯ A over a C. The C is not in the D chord, so it rubs against the chord.

EXAMPLE 1

Name the slash chord



1. **The right hand:** which chord is G B D?
2. **The lowest note:** which note is in the left hand?
3. **Write chord, slash, bass.** Is the bass in the chord?

It is **G/B**, "G over B": B is in the G chord, so it is the 1st inversion.

GAME

Read the slash

1	2	3	4
G/B	C/G	F/A	Am/C

Before the slash: the chord. After it: the bass note. Which note goes in the left hand?

Answer: 1 B · 2 G · 3 A · 4 C.

PART IV · CHORDS · 8 · CONTINUED

TRY IT · 10 SECONDS

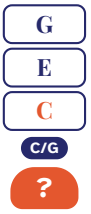


Right hand C, left hand E.

Play C E G in the right hand and a low E in the left: C/E. Now try a low A under it: what would you call that?

EXAMPLE 2

C/G: which note goes in the left hand?

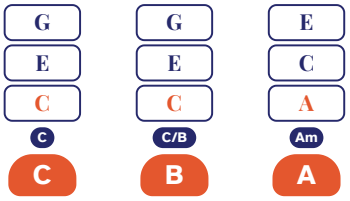


1. **Before the slash:** the chord. Which three notes go in the right hand?
2. **After the slash:** the bass. Which note goes low in the left hand?
3. **Play them together.**

Right hand C E G, left hand a low **G**: the C chord standing on G.

EXAMPLE 3

Walk the bass down: C, C/B, Am



1. **Right hand** holds C E G for two chords.
2. **Left hand** walks down: C, then B.
3. **Then A**, under A C E: Am.

The bass walks **C B A**, one step at a time, while the chords hardly move.

THINK OF IT A cake and the plate it stands on: the slash says which cake (the chord) and which plate (the bass). Same cake, new plate, new view.

TRAP

- × Writing it back to front: the chord comes first, the bass second. E/C would be an E chord over C.
- × Letting the right hand dip below the bass: the letter after the slash must be the lowest note you play.
- × Forgetting the chord itself: in C/E the right hand still plays the whole C chord, C E G, not just C.

SEE ALSO
Inversions p. 98 · Colour chords p. 104

PRACTISE IT
Workbook Harmony 8 · Online: Library › Chords

PART IV · CHORDS · TIPS

When a chord sounds wrong

A chord that sounds wrong usually has one small slip. Name the slip, fix that one note, and the chord comes right.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
A chord sounds dark when it should be bright	**the kind**: keys between, 3 + 2 or 2 + 3	p. 92
Your hand jumps all over	**an inversion**: keep the shared note still	p. 98
A big chord sounds muddy	**leave notes out**: drop the 5th first	p. 107
A seventh chord sounds wrong	**the 7th**: G7 adds F, a 7th above G (9 keys between)	p. 101
A sus chord sounds unfinished	**the 4th**: the middle layer moved up a letter	p. 104
The bass note sounds odd	**the slash**: the letter after it is the lowest note	p. 110

Four ways to learn chords fast

1. **Build, then play**
   Say the cake (C E G) before your fingers go down.

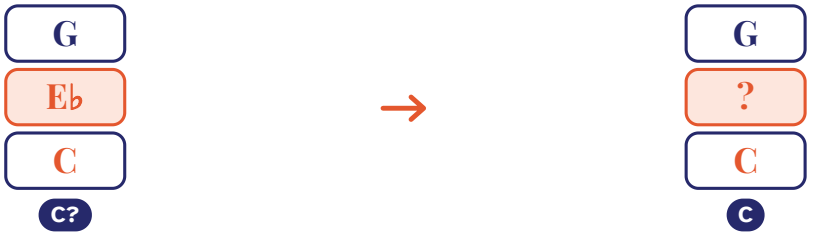
2. **Hands alone first**
   Left hand chords alone, then the tune alone, then together.

3. **One cake, three keys**
   Build the same kind of chord on C, G and F.

4. **Listen for the kind**
   Play major and minor side by side: bright, then dark.

EXAMPLE

Fix a dark chord



1. **It sounds dark**, but the music wants a bright C chord.
2. **Count the keys between:** C to E♭, then E♭ to G. Which kind is that?
3. **Move one note** so it becomes 3 + 2. Which layer, and to which key?

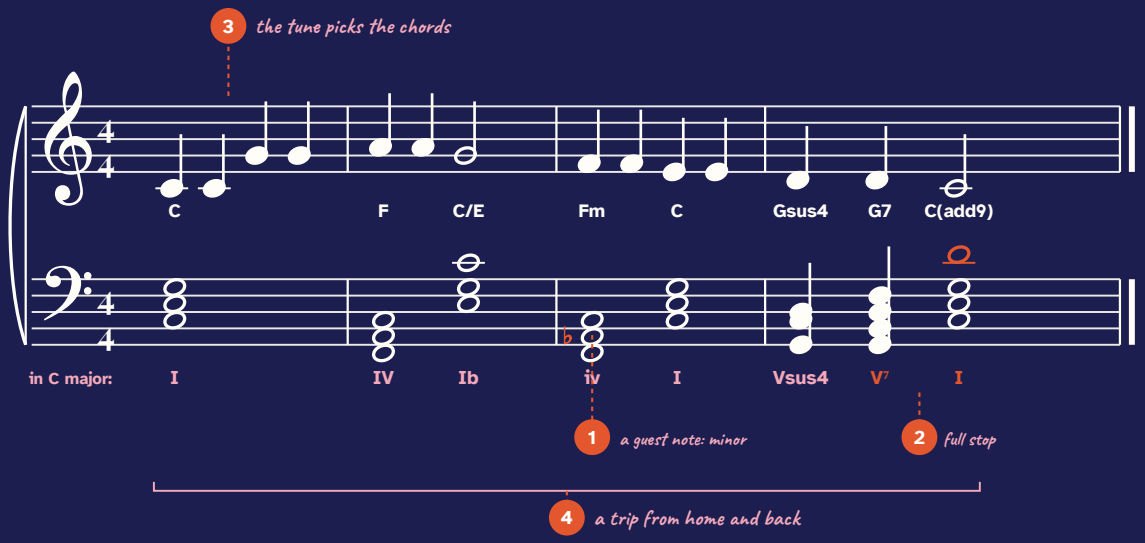
Move the middle layer from E♭ up to **E**: 3 + 2, bright C major again.

THINK OF IT Like a mechanic: one odd sound points to one part. Fix that part and the engine runs smoothly.

PART V · PUT THEM TO WORK

Harmony in music

A melody is the singer; harmony is the band behind it. Here the chords start moving: guests from outside the key, endings, chords under a tune, and the trips every song takes. All of it is in the same four bars.



1 **Chords from outside the key** p. 115
a guest note: A♭ in C major

2 **Cadences** p. 118
V to I ends a phrase like a full stop

3 **Chords under a melody** p. 121
the tune's main notes pick the chord

4 **Progressions everywhere** p. 123
the same few trips, in every song

★ **Tips: when harmony goes wrong** p. 125

PART V · HARMONY IN MUSIC · 1

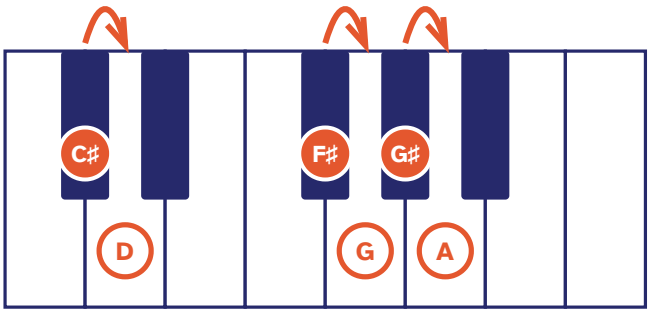
Chords from outside the key

In C major, home is the white keys, so a black key in a chord is a **guest**. Guests never stay: each one moves to the **very next key** (no key between), and shows you where the music goes next.

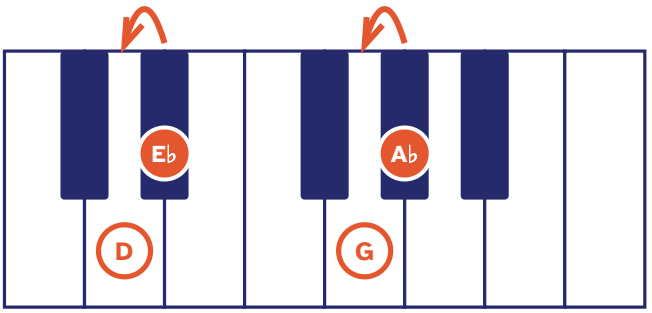
THE SECRET

Sharp guests push up, flat guests sigh down: follow them one key and you find the next chord.

Sharp guests push up

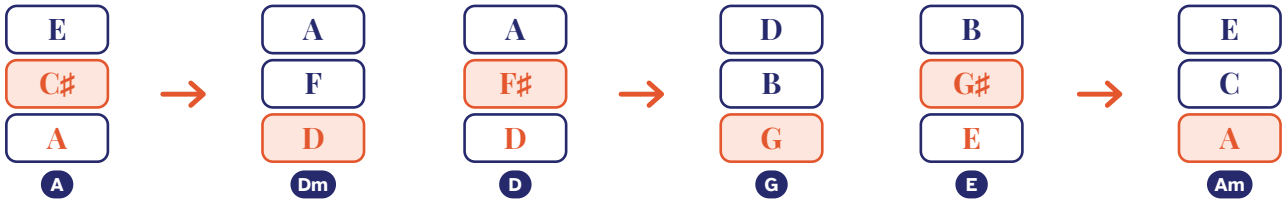


Flat guests sigh down

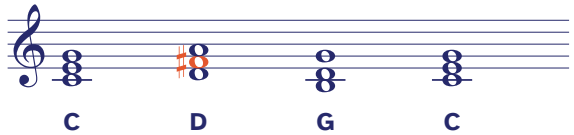


1. Find the **black key** in the chord: that is the guest.
2. A **sharp**? Go **up** to the very next key. A **flat**? Usually **down** to the very next key.
3. You have found a note of the **next chord**. Play it and hear it land.

Sharp guests: up one key names the next chord

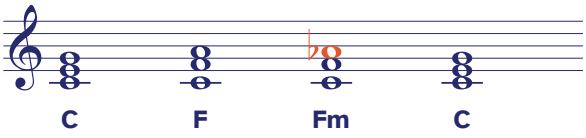


C♯ goes up to **D**, so the next chord is D minor. F♯ goes up to **G**: G. G♯ goes up to **A**: A minor.



Try it in a song: C, D, G, C. The **F♯** pushes up into G, then G leads home to C.

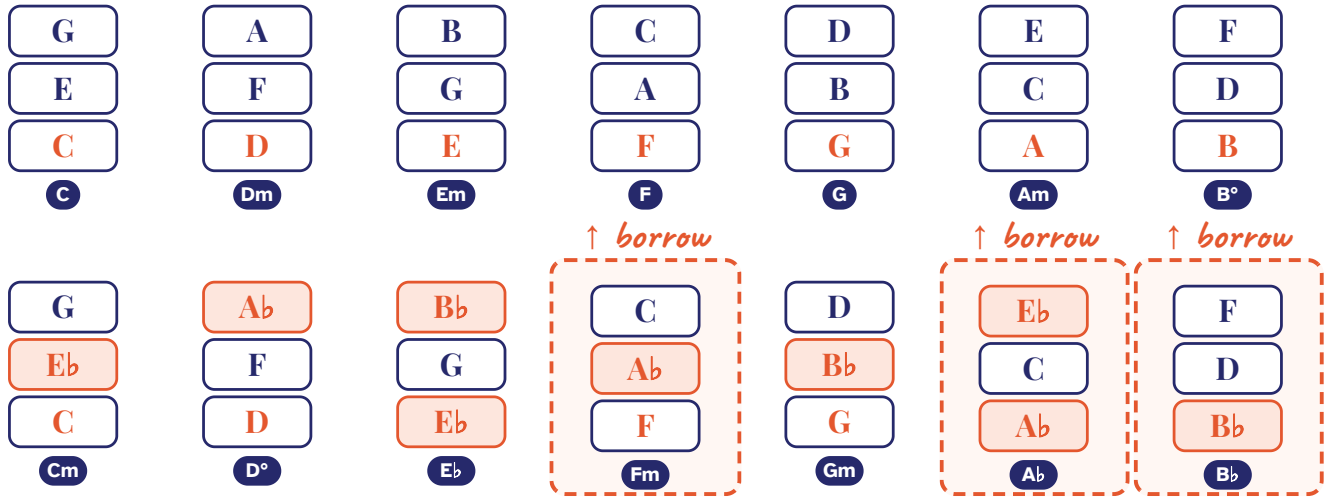
Flat guests: borrowed from the twin, they sigh down



C major and C minor are **twins** (the parallel minor): the same home note, C, in different clothes. A flat guest is one piece of the twin's outfit. In C, F, Fm, C the **A♭** sighs down to G.

PART V · HARMONY IN MUSIC · 1 · CONTINUED

C major *at home*



C minor *the twin: same home note, darker clothes*

Their names, for later

GUEST	NOTES	MOVES	NEXT CHORD	ITS NAME
sharp guest	D F♯ A	F♯ up to G	G	V of V
sharp guest	A C♯ E	C♯ up to D	Dm	V of ii
sharp guest	E G♯ B	G♯ up to A	Am	V of vi
flat guest	F A♭ C	A♭ down to G	C	borrowed iv
flat guest	A♭ C E♭	A♭ down to G	G or C	borrowed ♭VI
flat guest	B♭ D F	B♭ on to C	C	borrowed ♭VII

B♭ is the flat guest that walks **up** instead, on to C (one key between: a tone): the rock and film ending B♭ → C.

EXAMPLE

Where do the guests go? B D♯ F♯ in C major



1. **Find the black keys:** there are two guests here.
2. **Both are sharps:** move each up to the very next key. Where do they land?
3. **Which cake in C major** holds both landing notes?

D♯ goes up to E, F♯ up to G: both live in **E minor** (E G B). B D♯ F♯ leads to Em.

PART V · HARMONY IN MUSIC · 1 · CONTINUED

GAME

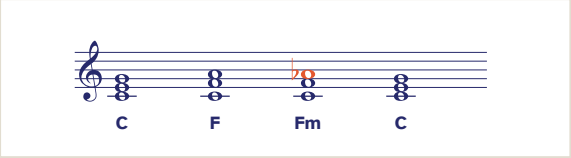
Where does the guest go?

1	2	3
E	E♭	A
C♯	C	F♯
A	A♭	D
A	A♭	D

In C major: find each guest, follow it one key, and write the chord that comes next.

Answer: 1 C♯ up to D: D minor · 2 A♭ down to G, E♭ down to D: G · 3 F♯ up to G: G.

TRY IT · 10 SECONDS



Play C, F, Fm, C slowly.

Hear the A♭ sigh down to G? Now play C, D, G, C: the F♯ pushes up instead. One sighs, one pushes.

THINK OF IT A surprise guest at dinner: one new face changes the whole conversation, and then everyone goes home.

TRAP

- × Following a sharp downwards: sharps push **up**, flats usually sigh **down**.
- × Calling every black key a guest: in G major F♯ is at home. A guest is a note from outside the key you are in.
- × Learning the names first: play it, hear where the guest goes, then name it.
- × Mixing up the relatives: C minor is C major's twin (same home note). A minor is its neighbour (same key signature). We borrow from the twin.

SEE ALSO

Chords in a key p. 95 · Slash chords p. 110 · Cadences p. 118

PRACTISE IT

Workbook Harmony 9 · Online: Library › Chromatic chords

PART V · HARMONY IN MUSIC · 2

Cadences

A phrase is a musical sentence, and it needs punctuation. The last two chords of a phrase are its **cadence**. They tell you: finished, taking a breath, or surprise!

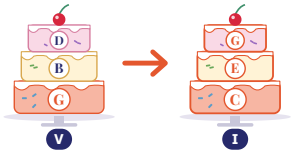
THE SECRET

Music has full stops and commas too: listen to the last two chords of a phrase.

.

Perfect

V → I

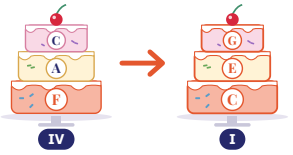


Home. The end.

Amen

Plagal

IV → I

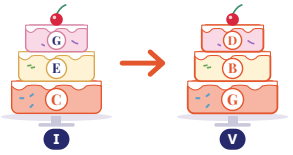


A soft, gentle end.

,

Imperfect

… → V

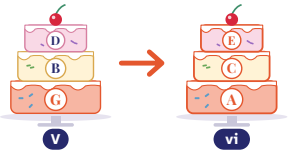


To be continued…

?!

Interrupted

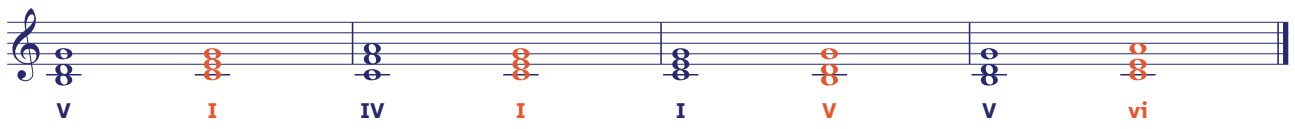
V → vi



Surprise! Not home.

1. Find the end of a phrase: often a long note, or a breath.
2. Name its **last two chords** with numerals: V I, IV I, anything then V, or V vi.
3. Match the punctuation: full stop (.), Amen, comma (,) or ?!

The four cadences in C

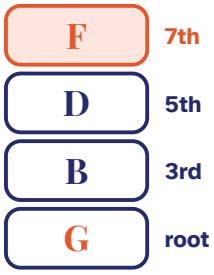


Perfect, plagal, imperfect, interrupted. The red chord is where each one lands.

CADENCE	CHORDS	FEELS LIKE	ALSO CALLED
Perfect	V → I	a full stop: finished	authentic, full close
Plagal	IV → I	the “A-men” sung at the end of a song in church	Amen cadence
Imperfect	any → V	a comma: waiting	half close, half cadence
Interrupted	V → vi	a surprise: not home yet	deceptive

PART V · HARMONY IN MUSIC · 2 · CONTINUED

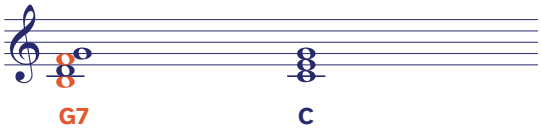
Why G7 pulls home



Add the 7th to V and the full stop gets stronger.

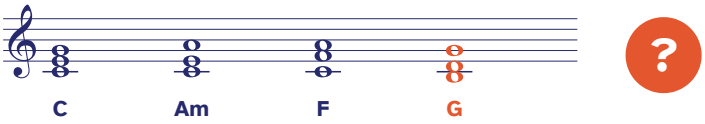
In **G7** two notes lean towards C major. The **B** rises to **C**, the **F** falls to **E**: each to the very next key, no key between. Two tiny steps, and you are home (Seventh chords, p. 101).

It works like a guest: B pushes up, F sighs down (p. 115).



EXAMPLE 1

How does this phrase end? C Am F G

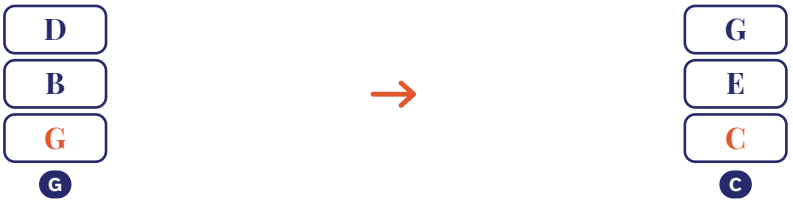


1. **Find the last two chords** of the phrase.
2. **Turn them into numerals** in C major.
3. **Which punctuation card** ends that way?

F then G is IV → V: it lands on V, the comma. An **imperfect cadence**.

EXAMPLE 2

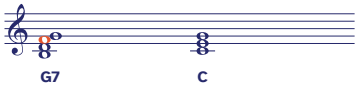
How does Twinkle end?



1. **The last two chords** of Twinkle in C are G, then C.
2. **Turn them into numerals.**
3. **Which card** goes from that chord to that chord?

V → I: a **perfect cadence**, the full stop. That is why Twinkle sounds finished.

TRY IT · 10 SECONDS



Play G7, then C. The end!

Now play G7, then A minor: “huh?!” That is the interrupted cadence. Then play F, then C, and sing “A-men”.

PART V · HARMONY IN MUSIC · 2 · CONTINUED

EXAMPLE 3

Surprise! G7, then A minor

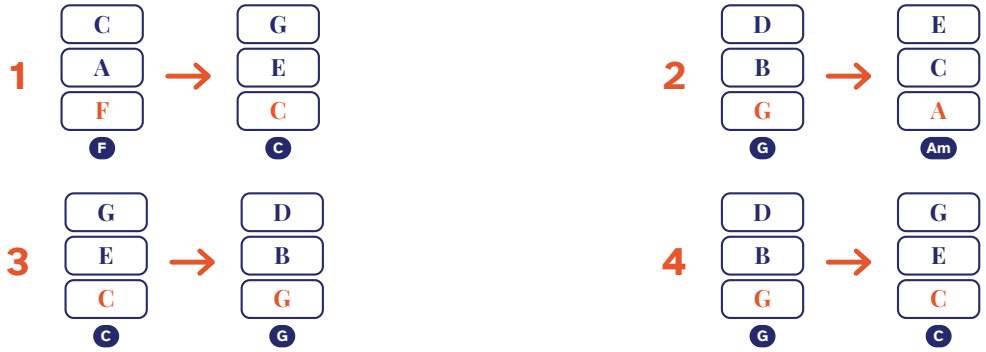


1. **G7 leans towards home**: you expect C.
2. **But the next chord is A minor.** Which numeral is it in C?
3. **V → that numeral:** which card?

V → vi: an **interrupted cadence**. The music was heading home, then turned the corner.

GAME

Punctuate the phrases



Each pair ends a phrase in C major. Turn the letters into numerals first (C is I, F is IV, G is V, Am is vi), then write full stop, comma, Amen or ?! beside each.

Answer: 1 Amen (plagal), 2 ?! (interrupted), 3 comma (imperfect), 4 full stop (perfect).

THINK OF IT Punctuation in a sentence: a full stop ends it, a comma makes you wait, and a surprise makes you gasp. Music ends its phrases the same way.

TRAP

- × Calling every ending on I (C) perfect: a perfect cadence comes **from V**. F to C is plagal.
- × Mixing up imperfect and perfect: imperfect **ends on** V; perfect **starts on** V and lands on I.

SEE ALSO

Chords in a key p. 95 · Seventh chords p. 101 · Chords under a melody p. 121

PRACTISE IT

Workbook Harmony 10 · Online: Library › Cadences

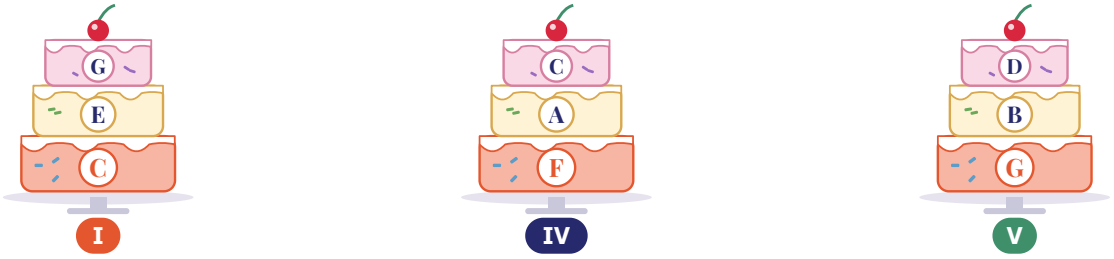
PART V · HARMONY IN MUSIC · 3

Chords under a melody

The melody is the singer; the chords are the band. To choose a chord for each bar, look at the melody's main notes and pick the cake holding them.

THE SECRET

Look at the melody's main notes, then pick the cake that holds them.



C	D	E	F	G	A	B
I, IV	V	I	IV	I, V	IV	V

Which cake holds the tune's main notes?

Every note of C major lives in I, IV or V. C and G live in two: let your ear choose.
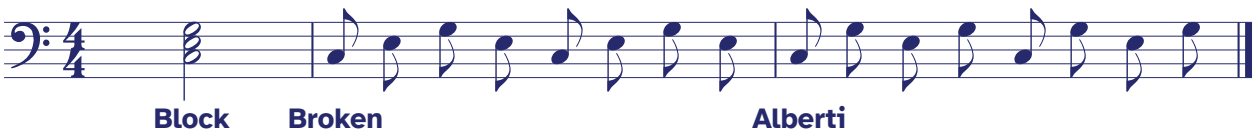
1. Look at the **main notes** of the bar: the long ones, and the ones on beats 1 and 3.
2. Find the cake that holds them: **I, IV or V**.
3. Play that chord in the left hand while the right hand plays the tune.

Twinkle, Twinkle, with its chords



C C G G sit in I (C); A A in IV (F), then G in I; F F in IV, E E in I; D D in V (G), and the last C in I. The chord changes halfway through bars 2, 3 and 4.

Three ways to play the chord in the left hand



Block: all three at once. Broken: one at a time, up and down. Alberti: low, high, middle, high, named after Domenico Alberti, who loved it.

PART V · HARMONY IN MUSIC · 3 · CONTINUED

EXAMPLE

Choose the chord: E D C D | E E E

E D C D | E E E

red = beats 1 and 3

1. **Bar 1:** which notes fall on beats 1 and 3 (red)?
2. **Which cake** holds both of them: I, IV or V?
3. **Bar 2:** E E E. Which cake holds E?

E and C both live in **I**, and so does E: both bars take **I** (C). That is the start of Mary Had a Little Lamb.

GAME

Pick the cake
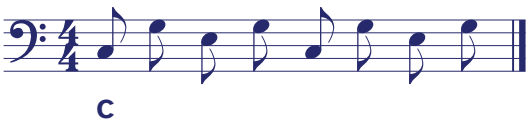
F A F A D B G B G E C E

1 2 3

Each bar is in C major. Which cake holds its notes: I, IV or V?

Answer: 1 IV (F A C) · 2 V (G B D) · 3 I (C E G).

TRY IT · 10 SECONDS



Left hand: C G E G, over and over.

That is the Alberti bass on the C chord. Keep it going while you hum Twinkle, then try it on F (F C A C) and G (G D B D).

YOU KNOW THIS

Mozart, Sonata in C, K. 545: the left hand



The opening left hand is an **Alberti bass**: C G E G is just the C chord, broken into low, high, middle, high; D G F G is G7 (G B D F, without the B). Mozart wrote it an octave higher, starting on middle C.

Try it. Play this left hand alone, then add the famous tune: C, E G, B, C D C.

THINK OF IT Like picking a coat for the weather: when the tune changes, change the chord.

TRAP

- × Choosing a chord for a passing note: use the main notes (strong beats, long notes).
- × Playing the left hand louder than the tune: the melody is the star, the chords are the stage.

SEE ALSO

Chords in a key p. 95 · Cadences p. 118

PRACTISE IT

Workbook Harmony 11 · Online: Library › Chords

PART V · HARMONY IN MUSIC · 4

Progressions everywhere

A **progression** is a trip from chord to chord. A handful of trips turn up in song after song. Learn them by their numbers and you can play them in any key.

THE SECRET
Most songs take the same few trips: home, away, a pull, and home again.

I home · vi iii home's cousins · IV ii away · V the pull back home

The pop four
countless pop songs and ballads

I C → V G → vi Am → IV F

The fifties
old rock 'n' roll and doo-wop

I C → vi Am → IV F → V G

The jazz three-step
jazz standards (as ii7, V7, Imaj7)

ii Dm → V G → I C

The canon
Pachelbel's Canon and its many children

I C → V G → vi Am → iii Em → IV F → I C → IV F → V G

1. Learn the trip by **numbers**: I V vi IV.
2. Pick a key and turn the numbers into chords: in C, **C G Am F**.
3. Play it round and round. Change key: the numbers stay the same.

Same trip, any key

The pop four	I	V	vi	IV
in C	C	G	Am	F
in G	G	D	Em	C
in F	F	C	Dm	B♭
in D	D	A	Bm	G

The twelve-bar blues

I C	I C	I C	I C
IV F	IV F	I C	I C
V G	IV F	I C	V G

The most famous trip of all is twelve bars long and uses only **I, IV and V**. Almost every blues, and a lot of rock 'n' roll, follows it. Play it in C with one chord per bar, four beats each.

Make it bluesy: turn every chord into a seventh (C7, F7, G7, p. 101).

PART V · HARMONY IN MUSIC · 4 · CONTINUED

EXAMPLE

Play the pop four in A major

I ? → V ? → vi ? → IV ?

1. **Write the A major scale** (three sharps) and number the notes.
2. **Find notes 1, 5, 6 and 4.**
3. **Build each cake:** which one is minor?

A, E, F♯m, D: **the pop four in A.** Same trip, new key.

GAME

Name the trip

1 C · Am · F · G

2 Dm · G · C

3 C · G · Am · F

All in C major. Turn each into numbers, then name the trip: the pop four, the fifties or the jazz three-step.

Answer: 1 the fifties (I vi IV V) · 2 the jazz three-step (ii V I) · 3 the pop four (I V vi IV).

TRY IT · 10 SECONDS

I C → V G → vi Am → IV F

Play C, G, Am, F, four beats each.

Go round four times without stopping, left hand plays the chord's name note (C for C, A for Am), right hand the whole cake. Hum any tune you like over it.

TRAP

× Mixing up the pop four and the fifties: same four chords, different order. Listen to which chord comes second.

SEE ALSO

Chords in a key p. 95 · Cadences p. 118 · Chords under a melody p. 121

PRACTISE IT

Workbook Harmony 12 · Online: Library › Chords

PART V · HARMONY IN MUSIC · TIPS

When the harmony goes wrong

Every chord can be right and the music still sound wrong: the ending will not close, or the left hand fights the tune. The slip is in how the chords move. Find which move it is, then fix that one.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
The end does not feel finished	**the cadence**: end with V → I	p. 118
A sharp or flat surprises you	**a guest**: sharps go up, flats go down, one key	p. 115
The left hand fights the tune	**the main notes**: pick the cake holding them	p. 121
A new song feels hard to learn	**the trip**: find I, IV, V and vi by number	p. 123

Four ways to hear the harmony

1. **Hum the bass**
   Sing the lowest notes: they show where the chords move.

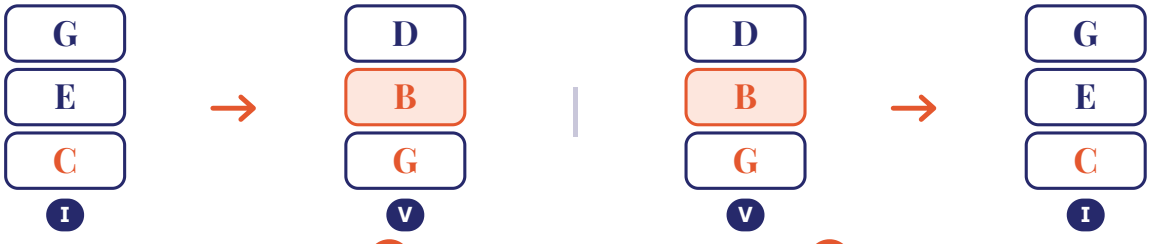
2. **Tick the full stops**
   Mark every V → I in your piece with a red tick.

3. **One trip, three keys**
   Play I–V–vi–IV in C, G and F: same numbers, new letters.

4. **Chord a tune you know**
   Pick I, IV or V for each bar of Twinkle or Happy Birthday.

EXAMPLE

Fix the ending



1. **The piece stops, but it does not feel finished.** Look at the last two chords.
2. **Name them:** C then G is I → V. That is a comma, not a full stop.
3. **Swap the order:** G then C is V → I. The red B climbs one key to C, and the music lands.

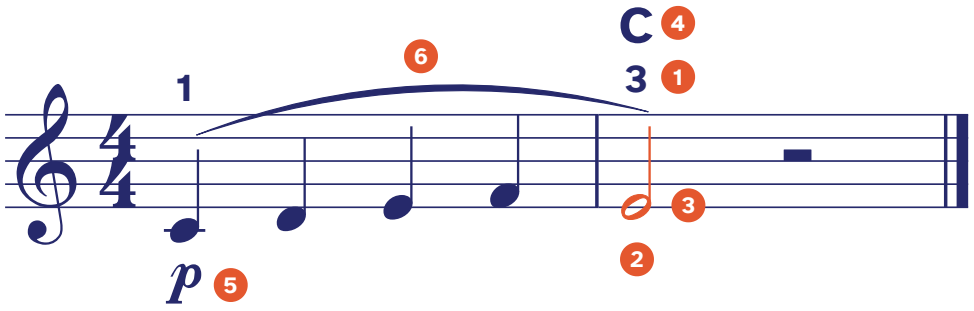
A strong ending needs **V → I** (IV → I for a soft Amen): the last two chords are the full stop.

THINK OF IT Like a road trip: once you know where home is, every detour makes sense, and every road leads back to I.

PUTTING IT TOGETHER · 1

Everything one note tells you

Look at the red note. It carries six pieces of information at once, and each one can be read straight off the page. Read all six and the note is right; miss one and it is only nearly right.



1 The finger number

3: the middle finger plays it. Fingering is planned, not guessed.

PLAYING AND LISTENING (LATER)

2 The beat

Beat 1 of bar 2, and it is held for **two beats**.

PART II · TIME · P. 46

3 The note name

E: the white key just right of the two black keys.

PART I · PITCH · P. 22

4 The chord

The **C** above the staff: this E belongs to the C major chord.

PART IV · CHORDS · P. 89

5 The marks

p and the slur: play it softly and smoothly.

EXPRESSION (LATER)

6 The phrase

The slur ends here, so the note lands gently, like the end of a sentence.

SHAPE AND STYLE (LATER)

THE SAME SIX, IN ANY MUSIC

A CLASSICAL SCORE
Most of it is written in: finger numbers, dynamics, slurs. Your job is to read every one.

A POP LEAD SHEET
A melody and chord names only. You choose the fingers, the marks and the shape yourself.

A JAZZ CHART
Chords and a style word such as Swing: the beat and the shape are yours to feel.

Try it on any piece: point to one note and find all six. Finger numbers count as much as the note name.

THINK OF IT A note is like a word said aloud: its spelling, its timing, its tone of voice and its place in the sentence all change what it means.

PUTTING IT TOGETHER · 2

When a passage won't come right

Name the problem first, then fix that one thing. Most hard bars are hard for one reason, and the six things a note tells you point straight to it.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
You stumble in the same place every time	**the finger number**	Playing and listening
It rushes, drags or the counting slips	**the beat**	Part II · Time
A note sounds wrong	**the note name** and the key	Part I · Pitch, Part III · Scales and ke
A chord sounds odd or thin	**the chord**	Part IV · Harmony
It is correct but flat and dull	**the marks**	Expression
It does not seem to go anywhere	**the phrase**	Shape and style

Six ways to fix a hard bar

1. **Slow it right down**
   Slow enough to be right three times in a row, then a little faster.

2. **Hands separately**
   Each hand on its own until it is easy, then together, slowly.

3. **Start from the end**
   Play the last beat, then the last two, then three: you always finish on what you know best.

4. **Practise the join**
   The last note of one bar and the first of the next: most slips live there.

5. **Climb the metronome**
   Up four clicks each time it is clean; down eight if it falls apart.

6. **Put it back**
   Play two bars before and two after, so the fixed bar lives inside the music again.

Before you play a new piece: a thirty-second look

Key signature
Which sharps or flats, all the way through?

Time signature
How many beats in a bar, and which note is one beat?

Tempo and mood
The words at the top: how fast, what character?

Patterns
Bars that repeat or nearly repeat: learn them once.

THINK OF IT Like a doctor: name the symptom first, then treat that one thing. Practising everything harder rarely fixes the one bar that is wrong.